Estimating a Rational Inattention Discrete Choice Model: A Semiparametric Approach

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Abstract. People's choices depend on both the payoffs of the available options and what they learn about those payoffs before deciding. This paper develops methods that use aggregate choice data to distinguish unobserved payoff heterogeneity from costly information acquisition in a rational inattention discrete choice model. Under Shannon information costs, I establish conditions for semiparametric point identification using exogenous payoff and information-cost shifters, leaving the taste shock distribution and the information-cost function nonparametric. With two alternatives, I show that the probability of choosing one option, as a function of an exogenous payoff shifter, can be interpreted as a cumulative distribution function whose moments identify the payoff coefficients and the information-cost function. A standard deconvolution argument then identifies the taste shock distribution. I extend identification to multiple alternatives. I then propose an estimator based on observed market shares and establish its consistency. Because binding parameter constraints can lead to nonnormal limiting distributions, I construct confidence regions by test inversion without requiring asymptotic normality of the estimator. An application to Amazon Mechanical Turk illustrates the implications of information costs for workers' responses to wages.

Keywords. Rational inattention; discrete choice; semiparametric identification; payoff heterogeneity; sieve estimation; test inversion.

JEL classification. C12, C14, C25, D83, J22.

1 Introduction

Following McFadden (1973), econometricians use random utility models to explain heterogeneity in observed choices. These models typically assume that individuals know their taste shocks, the idiosyncratic payoff components unobserved by the econometrician. In real-world situations, however, individuals may be uncertain about these components, and learning about them may require time and effort. The rational inattention (RI) framework developed by Sims (2003) incorporates this learning decision into the choice problem. In this framework, individuals optimally choose what to learn by trading off the expected payoff gains from learning against the cost of acquiring information.

Suppose an analyst observes the choices of rationally inattentive individuals who are initially uncertain about their taste shocks. These choices reflect both their underlying preferences and the information they choose to acquire. Suppose further that the analyst seeks to recover both the distribution of taste shocks and how the unit cost of information varies with observed characteristics in order to evaluate policy counterfactuals. To help the analyst do so, this paper establishes conditions under which exogenous variation in choice data identifies the taste-shock distribution and the unit cost of information as a function of observed characteristics, *without imposing parametric forms on either*.

Parametric specifications have made RI models tractable for empirical analysis. For example, Brown and Jeon (2024) estimate an insurance-demand model using a Cardell distribution (Cardell, 1997) for taste shocks and a parametric specification of how the unit cost of information varies with observed characteristics. Their distributional assumptions yield closed-form choice probabilities, allowing them to analyze identification directly from explicit formulas. Like Brown and Jeon (2024), I exploit exogenous variation in observed characteristics that affect payoffs and information costs to identify the taste-shock distribution and the information-cost function nonparametrically. The required exogenous variation comes from two observed characteristics with distinct roles. A utility shifter enters the payoff component known to the individual linearly. An information-cost shifter changes the unit cost of information without changing that known payoff component. The taste shocks are independent of these two shifters.

I first establish identification in the RI model for choices between two alternatives by extending Lewbel's (1997) method of recovering payoff parameters from moments of an unobserved choice threshold. In the RI model, I construct a similar threshold, which I call a switch point; when the utility shifter is a wage, it can be interpreted as an implied reservation wage. Holding the information-cost shifter fixed, its cumulative distribution function is the choice probability as a function of the utility shifter. The moments of this switch point identify the

payoff coefficients and the information-cost function. A deconvolution argument then recovers the taste-shock distribution. To extend identification to multiple alternatives, I recover the choice curves between each alternative and a reference alternative as the known payoffs of the remaining alternatives become arbitrarily low, for example through lower wages. Applying the moment identities to these curves identifies the payoff coefficients and the common information-cost function. I then use the probability of choosing the reference alternative and multivariate deconvolution to recover the joint distribution of taste shocks.

I propose a sieve minimum-distance estimator based on market shares. It uses finite-dimensional approximations to the taste-shock distribution and information-cost function. The RI model gives an equation that the true parameters and population shares must satisfy. I estimate the parameters by minimizing the average squared difference between the equation's two sides after replacing population shares with observed shares. I establish consistency even though the criterion can be discontinuous at zero information cost. Inference must accommodate estimates at the limits imposed by nonnegativity: information costs cannot be negative, so an estimated cost can reach its lower bound of zero. Similarly, when the taste-shock distribution is approximated by nonnegative probability weights on a grid extending beyond its true support, some estimated weights can be zero. These constraints can make the estimator's limiting distribution nonnormal. I therefore construct confidence regions by test inversion and obtain intervals for functions of the model parameters by taking their ranges over parameter values that the test does not reject.

I apply these methods to workers' decisions to accept tasks on Amazon Mechanical Turk, an online platform for short paid tasks. Estimated information costs are lowest and close to zero when descriptions specify both what workers must do and how much work is required. For a doubling of posted rewards, the wage elasticity of the average task-taking probability falls from 0.287 at estimated information costs to 0.068 when costs are halved and 0.037 under full information. At the original rewards, the average task-taking probability rises from 2.62% at estimated costs to 8.90% when costs are halved and 12.87% under full information. These counterfactual results suggest that cheaper task assessment can increase workers' willingness to take jobs while reducing their responsiveness to higher wages in this online labour market.

Related Literature My identification and estimation results use the RI framework of Sims (2003) and the choice-probability formula derived by Matějka and McKay (2015). Sims (2003) models individuals who choose how to use a limited capacity to process information about changing economic conditions, with the limit measured using Shannon information. He shows that this constraint can generate gradual responses to new information even without costs of adjusting actions. Under Shannon information costs, Matějka and McKay show that optimal choice probabilities conditional on payoffs take a generalized multinomial logit form. I use

this formula in my identification and estimation arguments. Fosgerau et al. (2020) extend the connection between RI and additive random utility models through a broader class of information costs. Caplin et al. (2022) characterize Shannon costs and broader information-cost functions through behavioral axioms.

RI research also develops revealed-preference tests and studies information costs using experimental data. These approaches use known prior beliefs[1] and known payoffs for each possible realization of the uncertainty, together with choice probabilities conditional on those realizations. Caplin and Dean (2015) provide revealed-preference tests of rational inattention and bounds on relative information costs. Dean and Neligh (2023) use laboratory experiments to show that participants' choices are broadly consistent with optimal information acquisition under a general rational-inattention model. In my setting, the researcher observes neither individual taste shocks nor their distribution, and choice probabilities average over these unobserved realizations. Under Shannon information costs, I establish conditions for identifying both the taste-shock distribution and the information-cost function from these aggregate choice probabilities.

The empirical RI literature studies how costly information acquisition affects demand and consumers' responses to prices and product characteristics. The closest empirical comparison is Brown and Jeon (2024), who estimate an insurance-demand model using Cardell distributions and parametric information costs. In Brown and Jeon's model, consumers initially know their individual taste shocks. In their application, the econometrician observes expected out-of-pocket costs and other plan characteristics that individuals must acquire information to learn about. By contrast, I study identification when individuals learn about taste shocks that are unobserved by the econometrician. Joo (2023) develops an empirical RI demand model with parametric information costs and choice probabilities, while leaving individuals' subjective prior beliefs unspecified.

A related identification literature studies which preferences and prior beliefs are consistent with observed choices when the researcher does not observe the information individuals use to make their decisions. Liao (2024) studies learning about shocks shared by individuals in the same market, while individuals know their own taste shocks, and characterizes the prior beliefs consistent with observed choices. Liao also shows that exogenous choice-set variation can distinguish full-information and positive-cost models. Gualdani and Sinha (2024) allow unrestricted information structures and derive sharp identified sets for preferences and counterfactuals using Bayes correlated equilibrium. I obtain point identification under Shannon costs and the stated exclusion and support conditions.

[1] An individual's probability distribution over possible payoffs before acquiring information is called a prior belief. A prior is correct when it matches the true payoff distribution. Subjective priors reflect individuals' own assessments and may differ from that distribution.

The semiparametric discrete-choice literature develops methods for identifying payoff coefficients and distributions of unobserved heterogeneity without imposing parametric forms on those distributions. Lewbel (1997) shows how moments of unobserved payoff components can be recovered from discrete choices, an approach I extend to RI. Fox et al. (2012) identify the random-coefficient distribution in mixed logit using local variation in product characteristics. Wang (2023) develops identification and sieve estimation of mixed-logit demand with a nonparametric random-coefficient distribution. A related question is how properties of choice probabilities allow identification with limited covariate support. Iaria and Wang (2025) establish real analyticity of mixed-logit and mixed-probit demand in mean utility indices and study its implications for identification. In my RI model, at positive information costs, real analyticity allows the choice curve to be recovered from an observed interval where its probability lies between zero and one. Continuity determines its endpoints, so full support of the utility shifter is unnecessary for recovering the complete choice-probability function.

RI provides a theory of the endogenous formation of consideration sets, the subsets of available alternatives from which individuals actually choose. Caplin et al. (2019) characterize these sets under Shannon information costs and show how payoffs, prior beliefs, and information costs determine which alternatives receive positive choice probability. This connects RI to the consideration set literature. Masatlioglu et al. (2012) show how changes in available alternatives can reveal preferences and which alternatives attract attention. Manzini and Mariotti (2014) identify preferences and consideration probabilities when alternatives are considered independently with probabilities that do not vary across choice sets. Cattaneo et al. (2020) allow more general random consideration and develop partial identification and inference for preference rankings under a monotonicity restriction on attention: removing an option outside a particular consideration set cannot reduce the probability of considering exactly that set.

Beyond demand estimation and identification, RI has implications for market outcomes and welfare. Afrouzi (2024) develops a model in which firms with fewer competitors pay less attention to aggregate conditions, a prediction supported by survey evidence; strategic inattention amplifies monetary nonneutrality in the calibrated model. Angeletos and Sastry (2025) show that a version of the first welfare theorem holds with Shannon costs in otherwise frictionless inattentive economies, while other information costs can create gains from simplifying markets.

These broader implications have also motivated empirical studies of attention to macroeconomic news and information treatments. Song and Stern (2025) construct a text-based measure of firms' attention to macroeconomic news and relate it to monetary-policy responses. Weber et al. (2025) find that households and firms become better informed and less responsive to supplied information as inflation rises. Complementing this evidence, Maćkowiak and

Wiederholt (2026) show why information-treatment effects depend on learning before and during an experiment.

2 The rational inattention model

2.1 The rational inattention problem

I consider a binary RI model in which individuals choose between alternatives 0 and 1. Before choosing, they can acquire information about an initially unknown component of the payoff from alternative 1. Each market m, such as a job posting, has observed characteristics (Z_m, X_m). For individual i in this market, normalize the payoff of alternative 0 to zero and specify

$$U_{i0m} = 0, \qquad U_{i1m} = \alpha + \delta Z_m + \Omega_{im}, \qquad \lambda_m = \lambda(X_m). \tag{2.1}$$

Both the individual and the analyst observe (Z_m, X_m). Before acquiring information, the individual knows the payoff component $\alpha + \delta Z_m$, the unit cost $\lambda(X_m)$, and the taste-shock distribution G, but not the realization Ω_{im}. The analyst observes market characteristics and choice shares but does not know (α, δ), λ, or G, and observes neither Ω_{im} nor the information the individual acquires.[2]

The taste shocks Ω_{im} are independent and identically distributed across individuals i and markets m, with common distribution G. Individuals share the correct prior G about their initially unknown taste shocks. I assume

$$\Omega_{im} \perp (Z_m, X_m).$$

The utility shifter changes known payoffs while leaving the unit cost fixed; the cost shifter changes the unit cost while leaving known payoffs fixed.

For example, an online job posting offers four dollars to "Transcribe this MP3 of a 5 minute interview." The reward is known, but the worker may not know how demanding accurate transcription would be for her. The same recording can require different amounts of concentration and repeated listening from different workers because its pronunciation and vocabulary match their experience differently. These differences in personal difficulty contribute to Ω_{im}.

To assess this uncertain payoff before choosing, the individual can acquire a *signal* T. The

[2] Additional covariates can be considered by conditioning the binary identification argument in Subsection 2.4 on utility characteristics H_U and cost characteristics H_λ. For the specification $\alpha(H_U) + \delta Z + \Omega$ with cost $\lambda(X, H_\lambda)$, the recovery formulas apply at conditioning values where the stated exclusion, support, and low-cost conditions hold. This gives a partially linear payoff specification with a flexible intercept. A common slope or taste distribution across conditioning values is an additional restriction when imposed.

signal is the information produced by a learning procedure. For the transcription task, it can be an internal assessment formed by reading the requirements, recalling comparable work, and considering which parts of that experience apply to the assignment. One worker may conclude that the work is manageable; another may judge that recovering every word would require exhausting repeated checking. These assessments help workers decide whether to take the task, even when they leave some uncertainty about the task's difficulty.

To describe what such a learning procedure reveals, I use an *information structure*, which specifies how likely each signal is at each possible payoff. Within a fixed market, I suppress the individual and market indices and write $U = (U_0, U_1)$. Let P_U be the prior over U induced by the known payoff component and G. The individual chooses a joint distribution $F_{T,U}$ satisfying $F_U = P_U$. This constraint holds the payoff distribution fixed while allowing the individual to choose how informative the signal will be.

In the example, an informative assessment procedure is more likely to produce a favorable judgment when the assignment is a good personal match and an unfavorable judgment when it is a poor match. A procedure producing the same generic judgment regardless of personal difficulty conveys no information: its signal is independent of the payoff. The information structure describes the procedure before its outcome is known; the signal is the particular assessment it produces. Even with the same procedure, individuals with different payoff draws can receive different signals.

The signal matters for choice because it changes the individual's assessment of the alternatives. After receiving $T = t$, she chooses an alternative maximizing

$$\mathbb{E}_F[U_j \mid T = t]. \tag{2.2}$$

The conditional expectations in (2.2) use the payoff distribution implied by $F_{T,U}$ after observing signal t, called the individual's posterior distribution. A worker can therefore take an assignment with a positive conditional expected payoff and subsequently discover that it is unsuitable. This outcome is compatible with a correct prior because the signal need not reveal the payoff perfectly.

The individual chooses the information structure before seeing its signal. She compares the expected payoff from the decisions it will support with the cost of acquiring that information. Following Matějka and McKay (2015), I take this cost to be proportional to Shannon mutual information. The individual's problem is

$$\max_{F_{T,U}:\, F_U = P_U} \left\{ \mathbb{E}_T \left[\max_{j \in \{0,1\}} \mathbb{E}_F[U_j \mid T] \right] - \lambda I_F(U;T) \right\}, \tag{2.3}$$

where

$$I_F(U;T) = \int \log\left(\frac{dF_{T,U}}{d(F_T \otimes P_U)}\right) dF_{T,U}. \tag{2.4}$$

The first term in (2.3) is the expected payoff from choosing after receiving the signal; the second is the cost of the information obtained. The integral in (2.4) is interpreted as relative entropy and can be infinite. Mutual information measures the average information the signal conveys about the payoffs and equals zero when the signal and payoffs are independent. For the worker, it measures what the internal assessment reveals about personal difficulty, rather than the number of words read or sources consulted.

The coefficient $\lambda \geq 0$ is the unit cost of information, measured in payoff units.[3] A lower coefficient makes any given amount of information cheaper. At positive cost, the individual may choose a partially informative procedure because resolving the remaining uncertainty would not justify its cost. At zero cost, she can make a full-information choice; it is enough to learn which alternative yields the higher payoff. This decision-relevant signal takes only two values and avoids any need to reveal a continuously distributed payoff with infinite mutual information.

Gathering evidence and assessing its implications require cognitive effort. The model specifies the cost of information without describing how particular reading or search activities produce it. The interpretation of λ is the cost of achieving a given improvement in the individual's information about the payoff.

This choice problem depends on $\theta = (\alpha, \delta, \lambda, G)$. The payoff coefficients (α, δ) are finite-dimensional, while the information-cost function λ and taste-shock distribution G are infinite-dimensional. Identifying these parameters from observed choices is therefore a semiparametric problem, with nonparametric identification of λ and G.

2.2 Choice probabilities of rationally inattentive agents

I write $b(z, x; \theta)$ for the probability of choosing alternative 1 before the signal is received. The analyst does not observe the signals and uses this probability averaged over their realizations. At $\lambda(x) > 0$, Matějka and McKay (2015) characterize this probability as the maximizer of

$$b(z, x; \theta) = \arg\max_{q \in [0,1]} \int \log\left(1 - q + q e^{(\alpha + \delta z + \omega)/\lambda(x)}\right) dG(\omega). \tag{2.5}$$

Equation (2.5) makes it possible to characterize the choice probability without solving directly for the entire information structure. Multiplying its objective by $\lambda(x)$ gives the optimized value

[3] I use natural logarithms, so information is measured in nats. Fully resolving a payoff-relevant yes-or-no question whose answers are initially equally likely provides one bit. One bit equals $\log 2$ nats and costs $\lambda \log 2$.

in payoff units; omitting that positive multiplier leaves the maximizing probability unchanged. A nondegenerate G makes the objective strictly concave in q, so the optimal share is unique.

Matějka and McKay (2015) also give the choice probability conditional on the taste shock. I call the optimal share *interior* when $0 < b(z,x;\theta) < 1$, so both alternatives have positive choice probability. When $\lambda(x) > 0$ and $0 < b(z,x;\theta) < 1$, the probability of choosing alternative 1 conditional on $\Omega = \omega$ is

$$p_\theta(\omega;z,x) = \frac{b(z,x;\theta)e^{(\alpha+\delta z+\omega)/\lambda(x)}}{1-b(z,x;\theta)+b(z,x;\theta)e^{(\alpha+\delta z+\omega)/\lambda(x)}}. \tag{2.6}$$

At a given payoff, the optimized learning procedure can produce different signals and therefore different choices. Averaging the conditional choice probability over G gives the unconditional probability:

$$b(z,x;\theta) = \int p_\theta(\omega;z,x)\,dG(\omega). \tag{2.7}$$

Since the unconditional share also appears in (2.6), equation (2.7) is a fixed-point restriction: the average of the conditional probabilities must equal the share used in their formula.

If the share $b(z,x;\theta)$ maximizing (2.5) satisfies $0 < b(z,x;\theta) < 1$, the derivative of the objective with respect to q, evaluated at $q = b(z,x;\theta)$, must be zero. This first-order condition is equivalent to (2.7). At shares zero and one, (2.7) is automatically satisfied, so this equation alone does not establish optimality. The maximization problem (2.5) determines whether an endpoint is optimal. At zero cost, choice follows the full-information payoff comparison:

$$b(z,x;\theta) = \mathbb{P}_G\{\alpha + \delta z + \Omega > 0\}. \tag{2.8}$$

With a nonzero slope and a continuous distribution of Z conditional on X, an exact payoff tie has probability zero across markets. Choosing either alternative at a tie therefore gives the same choice probabilities for almost every market. For identification, a right-continuous version of the full-information curve will be used.

2.3 Observations

I use market shares to connect these population probabilities to the data. The sample contains M markets, with a deterministic number $N_{m,M}$ of sampled individuals in market m. The analyst observes market characteristics $W_m = (Z_m, X_m)$ and the share

$$\widehat{s}_m = \frac{1}{N_{m,M}} \sum_{i=1}^{N_{m,M}} Y_{im}, \tag{2.9}$$

where Y_{im} is the indicator for choosing alternative 1. Conditional on market characteristics, choices are independent across individuals and markets, with

$$Y_{im} \mid W_m \sim \text{Bernoulli}(s_m), \qquad s_m = \pi(Z_m, X_m).$$

The population choice probability satisfies

$$\pi(z,x) = \mathbb{P}(Y = 1 \mid Z = z, X = x) = b(z,x;\theta_0). \tag{2.10}$$

Here θ_0 denotes the true parameter. Individual payoffs and signals remain unobserved; the identification and estimation arguments use variation in the market characteristics and their associated choice probabilities.

2.4 Identification

I now ask whether the choice probabilities described above determine the model's unknown parameters. For identification, I work with the family of population choice curves; the estimation section explains how market shares can replace these probabilities.

Write $\mathcal{X} = \text{supp}(X)$ and $\mathcal{Z}_x = \text{supp}(Z \mid X = x)$.

Definition 1 (Identification)**.** The parameter θ_0 is *identified* if there exists a mapping $\mathcal{T}$, known to the analyst, that recovers it from the observed population choice probabilities:

$$\theta_0 = \mathcal{T}\Big(\big\{\{\pi(z,x)\}_{z\in\mathcal{Z}_x}\big\}_{x\in\mathcal{X}}\Big). \tag{2.11}$$

Individual signals and payoff components are not observed. The identification argument uses how aggregate choice probabilities change with the two observed shifters.

A choice curve describes how the probability of choosing alternative 1 changes with the utility shifter at a fixed value of the cost shifter. Comparing curves at different cost-shifter values shows how choice probabilities change with information cost. The following assumptions specify the payoff normalizations, independence of taste shocks from the shifters, and the support needed to make these comparisons.

Assumption ID1.

(a) The taste shocks are independent of the shifters:

$$\Omega_{im} \perp (Z_m, X_m). \tag{2.12}$$

(b) $\mathbb{E}_{G_0}\Omega = 0$, $\mathrm{Var}_{G_0}(\Omega) = 1$.

(c) $\delta_0 \neq 0$.

(d) $\lambda_0 : \mathcal{X} \to [0, \infty)$ is continuous.

(e) For every $x \in \mathcal{X}$ with $\lambda_0(x) = 0$, the function $z \mapsto b(z, x; \theta_0)$ is continuous on $\mathcal{Z}_x$.

Under ID1(a), the taste-shock distribution is the same at every value of the observed shifters.

ID1(b) imposes location and scale normalizations. Whether the choice probability rises or falls as z increases identifies the sign of δ_0. Because replacing Z by $-Z$ and δ_0 by $-\delta_0$ leaves the payoff unchanged, I assume $\delta_0 > 0$ without loss of generality.

ID1(e) imposes continuity only in the utility shifter at zero cost. Together with the common taste distribution, continuity of λ_0, and the zero-cost full-support condition in ID3(b), the RI problem then implies joint continuity in (z, x).

This joint continuity makes the population choice curve uniquely determined on the observed support. Conditional choice probabilities are initially determined by the distribution of the data only up to sets of covariate values with probability zero. Two continuous choice functions that agree almost everywhere must agree at every point in the support of (Z, X): a difference at any support point would persist on a neighborhood with positive probability. Thus the choice function used in Definition 1 is uniquely determined there.

The next assumption requires the cost shifter to include values at which obtaining useful information becomes nearly costless.

Assumption ID2: arbitrarily low information costs.

$$\inf_{x \in \mathcal{X}} \lambda_0(x) = 0. \tag{2.13}$$

When information is clearly organized and easy to interpret, learning which option is better may require very little cognitive effort. ID2 assumes that some values of the cost shifter make this learning nearly effortless.[4] With finitely many cost categories, the assumption requires one category to have zero cost. With a continuous shifter, costs can approach zero without reaching it.

The next assumption concerns variation in the utility shifter conditional on the cost shifter. In the job example, posted wages must vary enough within a given information-cost category to reveal how the task-taking probability changes with pay. The support requirements differ between positive and zero information cost.

Assumption ID3: Support of the utility shifter.

(a) For each x, $Z \mid X = x$ has a continuous distribution with interval support $\mathcal{Z}_x$.

(b) Define

$$\mathcal{X}_0 = \{x \in \mathcal{X} : \lambda_0(x) = 0\}.$$

For every $x \in \mathcal{X}_0$, $\mathcal{Z}_x = \mathbb{R}$. For every $x \notin \mathcal{X}_0$, there exists

$$z_x^* \in \text{int}(\mathcal{Z}_x), \qquad 0 < \pi(z_x^*, x) < 1.$$

ID3(a) allows continuous variation in the utility shifter over an interval while holding the information cost fixed. At positive cost, ID3(b) requires an observed neighborhood of a point where both alternatives have positive choice probability. The RI model makes the choice curve real analytic wherever its values lie strictly between zero and one. Analyticity, together with continuity and monotonicity, determines the complete curve from its values on that neighborhood. Thus the observed utility shifter can have bounded support at positive cost.

At zero cost, the choice curve is determined directly by the otherwise unrestricted taste distribution. Its behavior outside the observed interval need not be determined by its values inside that interval. ID3(b) therefore requires full conditional support at cost-shifter values where information is free. If $\lambda_0(x) > 0$ for every $x \in \mathcal{X}$, this full-support requirement does not apply.

[4] Both Brown and Jeon (2024, equation (15)) and Joo (2023, equation (16)) use exponential information-cost specifications. For a specification $\lambda_0(x) = \exp(x'\beta_0)$ with coefficient vector β_0, ID2 holds if $\text{supp}(X'\beta_0) = \mathbb{R}$, since the index can then become arbitrarily negative.

Together with the payoff normalizations and low-cost variation in ID1–ID2, these support conditions yield the main identification result.

Theorem 1 (Identification)**.** *Under Assumptions ID1–ID3, the true parameter*

$$\theta_0 = (\alpha_0, \delta_0, \lambda_0, G_0)$$

is identified.

The complete proof is in Section C.2.

To establish the theorem, I start with the connection between a choice curve and a latent threshold in Lewbel (1997). Consider the full-information binary choice rule

$$Y = \mathbf{1}\{\alpha_0 + \delta_0 Z \geq -\Omega\}, \qquad \Omega \perp Z, \quad \delta_0 > 0.$$

Define the threshold

$$S = -\frac{\alpha_0 + \Omega}{\delta_0}, \qquad \alpha_0 + \delta_0 S = -\Omega.$$

Rearranging the payoff comparison gives $Y = \mathbf{1}\{S \leq Z\}$. Since S depends only on Ω, it is independent of Z, and hence

$$\mathbb{P}(Y = 1 \mid Z = z) = \mathbb{P}(S \leq z).$$

Thus the choice probability as a function of z is the CDF of S. When Z is a wage, S is a reservation wage: alternative 1 is chosen when the wage reaches this threshold. Under the mean-zero and unit-variance normalizations for Ω in ID1(b),

$$\mathbb{E}S = -\frac{\alpha_0}{\delta_0}, \qquad \mathrm{Var}(S) = \frac{1}{\delta_0^2}.$$

The choice curve therefore determines the moments that recover the payoff coefficients.

To construct an analogous threshold in the RI model, I first express its choice equation as a probability statement. For this construction, $\pi(z, x) = b(z, x; \theta_0)$ denotes the model's choice curve for every $z \in \mathbb{R}$, including values outside the observed support. Let $\ell(p) = \log\{p/(1-p)\}$ and let ε be standard logistic, independent of Ω, with CDF $\sigma(t) = (1 + e^{-t})^{-1}$. I introduce this logistic random variable to write the generalized logit formula as a probability. At positive cost

and an interior share, rewriting (2.6) gives

$$p_{\theta_0}(\omega; z, x) = \sigma\left(\ell(\pi(z,x)) + \frac{\alpha_0 + \delta_0 z + \omega}{\lambda_0(x)}\right)$$
$$= \mathbb{P}\left\{\varepsilon \leq \ell(\pi(z,x)) + \frac{\alpha_0 + \delta_0 z + \omega}{\lambda_0(x)}\right\}.$$

Averaging over $\Omega \sim G_0$, independently of ε, therefore gives

$$\pi(z,x) = \mathbb{P}\{h_x(z) \geq -\Omega + \lambda_0(x)\varepsilon\}, \tag{2.14}$$

where

$$h_x(z) = \alpha_0 + \delta_0 z + \lambda_0(x)\ell(\pi(z,x)). \tag{2.15}$$

For x with $\lambda_0(x) > 0$, define

$$I_x = \{z \in \mathbb{R} : 0 < \pi(z,x) < 1\}.$$

The RI model implies that I_x is a nonempty open interval and that $h_x : I_x \to \mathbb{R}$ is strictly increasing and onto. Its inverse therefore converts the threshold in (2.14) into units of the utility shifter.

Definition 2 (Switch point)**.** For each $x \in \mathcal{X}$, define the *switch point* S_x by

$$S_x = \begin{cases} h_x^{-1}(-\Omega + \lambda_0(x)\varepsilon), & \lambda_0(x) > 0, \\ -\dfrac{\alpha_0 + \Omega}{\delta_0}, & \lambda_0(x) = 0, \end{cases}$$

where $\Omega \sim G_0$ and ε is standard logistic and independent of Ω.

At positive cost, monotonicity of h_x gives, for $z \in I_x$,

$$\mathbb{P}(S_x \leq z) = \mathbb{P}\{-\Omega + \lambda_0(x)\varepsilon \leq h_x(z)\} = \pi(z,x).$$

The same CDF equality holds for every real value of the utility shifter, at both positive and zero cost:[5]

$$\mathbb{P}(S_x \leq z) = \pi(z,x), \qquad z \in \mathbb{R}. \tag{2.16}$$

The choice curve is therefore the CDF of S_x at each x.

[5]At positive cost, $S_x \in I_x$ almost surely. To the left of I_x, both $\mathbb{P}(S_x \leq z)$ and $\pi(z,x)$ equal zero; to its right, both equal one. Continuity gives the same equalities at any finite endpoints. At zero cost, the equality follows from the full-information threshold, using the right-continuous choice curve described in Subsection 2.2.

The moments used below depend on the complete CDF $z \mapsto \pi(z, x)$. ID3 determines this curve from the observed probabilities through the support and analyticity argument described before Theorem 1.

The distribution of S_x is now available from the choice curve. The next step is to relate its moments to the unknown parameters. At positive information cost, applying h_x to the switch point gives the identity

$$-\Omega + \lambda_0(x)\varepsilon = \alpha_0 + \delta_0 S_x + \lambda_0(x)\ell(\pi(S_x, x)). \tag{2.17}$$

Every moment of S_x used below is a function of its identified CDF. I write $\mu(x) = \mathbb{E}S_x$, $V(x) = \mathrm{Var}(S_x)$, and

$$C(x) = \mathrm{Cov}\big(S_x, \ell(\pi(S_x, x))\big). \tag{2.18}$$

This covariance is positive for a nondegenerate switch-point distribution.

The two logistic terms in (2.17) have the same variance. At positive cost, $\pi(S_x, x)$ is uniform on $(0, 1)$, so its log odds have the same mean and variance as ε. Taking means and variances therefore cancels the two logistic variance terms and gives

$$\mu(x) = -\frac{\alpha_0}{\delta_0}, \qquad \delta_0^2 V(x) + 2\delta_0\lambda_0(x)C(x) = 1. \tag{2.19}$$

The full-information threshold gives the same restrictions at zero cost. Equation (2.19) explains what information costs do to switch-point dispersion: at positive cost, the positive covariance term makes the variance smaller than the full-information value $1/\delta_0^2$. That value is attained exactly when information is free, since δ_0 and $C(x)$ are positive.

To recover δ_0, I use the largest switch-point variance across values of the cost shifter. As information cost approaches zero, switch-point variance approaches its full-information value; Cauchy–Schwarz bounds the difference from that value by a constant times $\lambda_0(x)$. Under ID2, these variances can therefore be arbitrarily close to the full-information value. Their supremum identifies the slope, and the common mean then identifies the intercept:

$$\delta_0 = \left(\sup_{x \in \mathcal{X}} V(x)\right)^{-1/2}, \tag{2.20}$$

$$\alpha_0 = -\delta_0 \mu(x). \tag{2.21}$$

Once (α_0, δ_0) are known, solving the same moment identity gives $\lambda_0(x)$ at each x:

$$\lambda_0(x) = \frac{1 - \delta_0^2 V(x)}{2\delta_0 C(x)}. \tag{2.22}$$

For a single value of the cost shifter, (2.19) contains both an unknown payoff slope and an unknown information cost. The variance limit as information cost approaches zero identifies the slope. The information cost can then be identified from each curve separately.

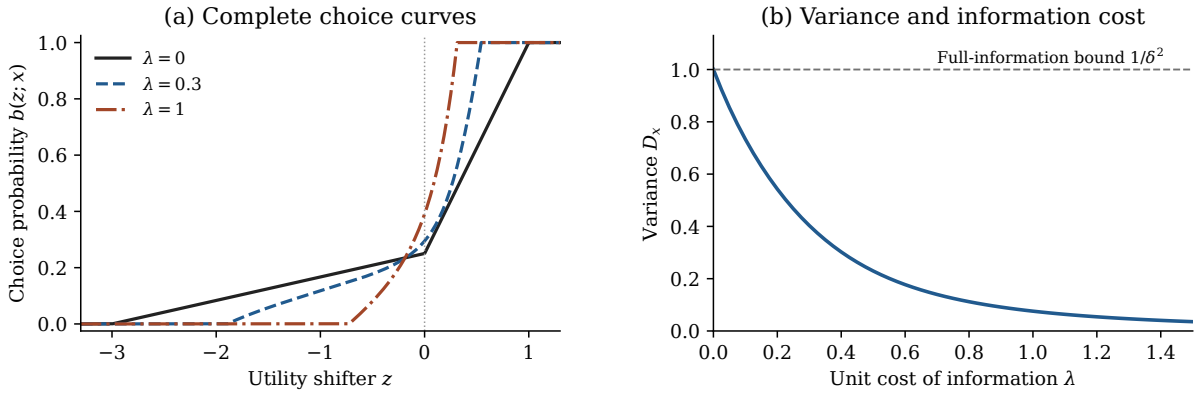


Figure 1: Choice curves and switch-point dispersion. The illustration uses $\alpha = 0$, $\delta = 1$, and an asymmetric taste density equal to $3/4$ on $[-1,0]$ and $1/12$ on $[0,3]$. This distribution has mean zero and variance one. The choice curves define distributions with the same mean; their variances fall as information costs rise. The curves are computed from the RI model by deterministic integration.

Only the taste-shock distribution G_0 remains to be recovered. The identified coefficients (α_0, δ_0) and cost function λ_0 make h_x known, so the observed distribution of S_x also determines the distribution of $h_x(S_x)$. At a value of the cost shifter with positive information cost, (2.17) expresses this variable as the sum of $-\Omega$ and an independent logistic variable with the known scale $\lambda_0(x)$. Writing $\varphi_V(t) = \mathbb{E}e^{itV}$, independence gives

$$\varphi_{-\Omega}(t) = \frac{\varphi_{h_x(S_x)}(t)}{\varphi_\varepsilon(\lambda_0(x)t)}. \tag{2.23}$$

The denominator is known and has no real zeros, so the ratio identifies the characteristic function, and hence the distribution, of $-\Omega$. Changing its sign gives the distribution of Ω, namely G_0. If costs are zero everywhere, no deconvolution is needed: G_0 is directly the distribution of $-\alpha_0 - \delta_0 S_x$. These moment and deconvolution calculations establish identification. For estimation, I use the RI optimality equation directly, so I do not first need to estimate switch-point moments or extend the observed choice curves beyond the observed support.

2.4.1 Extension: multiple alternatives

I extend identification to choices among alternatives $0, 1, \ldots, J$, taking alternative 0 as the default with payoff zero. The argument first recovers the probability of choosing each alternative

j when the only other available option is the default. I then use the probability of choosing the default when all alternatives are available to recover the dependence among taste shocks. Appendix A states the formal conditions and identification theorem; Section C.5 gives the complete proof. The remaining payoffs are

$$U_0 = 0, \qquad U_j = \alpha_j + \delta_j z_j + \Omega_j, \qquad j = 1, \ldots, J. \tag{2.24}$$

Here $z = (z_1, \ldots, z_J)$ is the vector of utility shifters. The unit information cost $\lambda(x)$ is common across the alternatives. The individual acquires information about the vector $\Omega = (\Omega_1, \ldots, \Omega_J)$, whose joint distribution G is independent of the shifters. The parameter is now $\theta = (\alpha, \delta, \lambda, G)$, where $\alpha = (\alpha_1, \ldots, \alpha_J)'$ and $\delta = (\delta_1, \ldots, \delta_J)'$ collect the payoff coefficients, λ is the common cost function, and G is the joint taste distribution. Dependence among taste components allows a worker who finds one kind of task attractive to find another kind attractive as well.

To fix the common payoff scale, I normalize $\mathbb{E}\Omega_j = 0$ for every j and $\mathrm{Var}(\Omega_1) = 1$. Only the first taste variance is normalized; the other marginal variances will be recovered. Changing the sign of each utility shifter when necessary allows me to assume $\delta_j > 0$ without loss of generality. I let $b(z, x; \theta)$ collect the $J + 1$ probabilities of choosing each alternative. At positive cost, the generalized logit representation gives

$$b(z, x; \theta) = \arg\max_{q \in \Delta_J} \lambda(x) \mathbb{E}_G \log\left(q_0 + \sum_{j=1}^{J} q_j e^{U_j/\lambda(x)} \right), \tag{2.25}$$

where $\Delta_J = \{q : q_j \geq 0, \ \sum_{j=0}^{J} q_j = 1\}$. Non-collinearity of the exponentiated payoff vectors ensures uniqueness of the share vector. This excludes taste distributions for which two different share vectors give the same expression inside the logarithm almost surely.

I first consider the choice between alternative j and the default. If the remaining alternatives have zero shares, their terms in (2.25) are zero, leaving

$$\max_{p \in [0,1]} \lambda(x) \mathbb{E}_G \log\left(1 - p + p e^{U_j/\lambda(x)}\right).$$

The remaining problem is exactly the binary RI problem with the marginal taste distribution of Ω_j. I can therefore use its choice curve to construct the switch-point moments $\mu_j(x)$, $V_j(x)$, and $C_j(x)$ needed for the binary recovery formulas.

To recover the choice curve for alternative j when only j and the default are available at positive cost, I start in a region where every alternative has a positive share, which I call a full-consideration region. Analyticity determines the choice probabilities throughout this connected region from their values on an observed open subset. I then construct a sequence along which the shares of alternatives other than j and the default approach zero and z_j approaches

the chosen value. Every alternative has a positive share at each finite point of the sequence. The optimality conditions along this sequence determine the limiting binary choice probability. Section C.5 proves this result using finite taste moments. At zero cost, the stated full conditional support condition makes the full-information choice curves directly observed.

Because the first taste variance is normalized, the choice curve for alternative 1 and the default identifies α_1, δ_1, and the common cost function λ by the binary formulas. For alternatives $j = 2,\dots,J$, recovering the payoff coefficients requires one more step because the marginal taste variances have not been normalized. For such an alternative, I write $\sigma_j^2 = \mathrm{Var}(\Omega_j)$ and use the corresponding moment identity

$$\delta_j^2 V_j(x) + 2\delta_j \lambda(x) C_j(x) = \sigma_j^2, \qquad \mu_j(x) = -\frac{\alpha_j}{\delta_j}. \tag{2.26}$$

The right-hand side is an unknown marginal variance, but the cost function λ on the left-hand side has already been recovered from the choice curve for alternative 1 and the default. This makes variation in information costs useful for separating the remaining slope from its taste variance.

First suppose $\lambda(x) > 0$ at some $x \in \mathcal{X}$. I evaluate (2.26) at two cost-shifter values with $V_j(x) \neq V_j(x')$ and subtract the equations. The unknown variance σ_j^2 cancels because it is the same at both values. Since $\delta_j > 0$, the difference identifies

$$\delta_j = \frac{2\{\lambda(x')C_j(x') - \lambda(x)C_j(x)\}}{V_j(x) - V_j(x')}. \tag{2.27}$$

The mean then gives the intercept,

$$\alpha_j = -\delta_j \mu_j(x),$$

and substituting the coefficients into either variance equation gives σ_j^2. For a nondegenerate marginal taste, a full-information limit and a value of the cost shifter with positive information cost give different switch-point variances. Variation in information costs therefore identifies the relative payoff scales once one common scale is normalized.

Applying the binary identification argument to each of these choice curves identifies the marginal taste distributions, but does not identify the dependence between taste shocks for different alternatives. To identify that dependence, I use an observed full-consideration region at positive information cost. I write $b^j(z,x)$ for the population share of alternative j, the jth component of $b(z,x;\theta_0)$. Let $\eta_0,\dots,\eta_J$ be independent standard Gumbel variables, independent

of Ω. The conditional logit probabilities can be represented as choices among

$$U_j + \lambda(x)\log b^j(z,x) + \lambda(x)\eta_j, \qquad j = 0,\dots,J.$$

The default is selected only when its value exceeds that of every other alternative. Its choice probability therefore involves all taste components jointly. I define $\zeta_j = \eta_j - \eta_0$ and the known thresholds

$$c_j(z,x) = -\alpha_j - \delta_j z_j + \lambda(x)\log\frac{b^0(z,x)}{b^j(z,x)}. \tag{2.28}$$

The probability of these simultaneous inequalities is

$$b^0(z,x) = \mathbb{P}\{\Omega_j + \lambda(x)\zeta_j \le c_j(z,x) \text{ for every } j\}. \tag{2.29}$$

Equation (2.29) expresses the probability of choosing the default as the joint CDF of $H_x = \Omega + \lambda(x)\zeta$, evaluated at the thresholds determined by the identified coefficients and observed shares. As the utility shifters vary, these thresholds change, giving values of the joint CDF at different points.

To identify the complete distribution, I use the threshold variation generated by the observed full-consideration region. The map $z \mapsto (c_1(z,x),\dots,c_J(z,x))$ has a nonsingular Jacobian on this region, as shown in Section C.5. The observed region therefore gives an open set of threshold values. At positive cost, adding the independent vector $\lambda(x)\zeta$ to Ω produces a random vector with a real-analytic density. Real analyticity determines this density on its entire domain from its values on that open set, and hence identifies the distribution.

As in the binary case, I use deconvolution to identify the taste distribution from the distribution of this sum. Independence of $\eta_0,\dots,\eta_J$ and the standard Gumbel characteristic function (Janson, 2010, Section 3) give the characteristic function of ζ:

$$\varphi_\zeta(t) = \Gamma\left(1 + i\sum_{j=1}^{J} t_j\right)\prod_{j=1}^{J}\Gamma(1 - it_j), \qquad t \in \mathbb{R}^J.$$

Here Γ is the gamma function. This characteristic function is nonzero at every real t because the gamma function has no zeros and its arguments here have real part one, away from its poles (NIST, 2026, Section 5.2(i)). Independence of the taste vector and the auxiliary vector therefore gives

$$\varphi_G(t) = \frac{\varphi_{H_x}(t)}{\varphi_\zeta(\lambda(x)t)}. \tag{2.30}$$

This ratio identifies the characteristic function, and hence the joint distribution, of the taste

vector. The components of the auxiliary vector are dependent because each subtracts the same Gumbel variable η_0. Their joint distribution is known, including this dependence.

If $\lambda(x) = 0$ for every $x \in \mathcal{X}$, the binary formulas still identify α_1, δ_1, and the zero cost function. Full conditional support then permits comparisons between any two nondefault alternatives. Increasing their utility shifters together identifies the ratio of their slopes: only one relative rate of increase leaves both alternatives with positive limiting choice probabilities. The known value of δ_1 fixes the remaining slopes, and the binary switch-point means identify the intercepts. The default probability directly identifies the joint taste distribution because

$$b^0(z,x) = G(-\alpha_1 - \delta_1 z_1, \ldots, -\alpha_J - \delta_J z_J), \qquad z \in \mathbb{R}^J.$$

The proof in Section C.5 establishes these conclusions without requiring a positive information cost at any cost-shifter value.

Once the payoff coefficients (α, δ), cost function λ, and joint taste distribution G are known, the model determines choices at every covariate value, including where the default is never chosen.

3 Estimation

For ease of exposition, I present the estimator and its consistency argument for binary choice. Section 3.2.1 discusses the estimator for multiple alternatives.

3.1 The estimator

The RI choice-probability equation (2.7) gives a restriction on the parameters and population shares. I construct a minimum-distance estimator by measuring the discrepancy between its two sides. I write $w = (z,x)$ and $s(w) = \pi(z,x)$, and let $\sigma(u) = (1+e^{-u})^{-1}$ be the logistic CDF. At positive cost and an interior input share, Ψ averages the conditional choice probabilities generated by a candidate parameter. The following definition also specifies its value at endpoint shares and zero cost:

$$\Psi(\theta; w, q) = \begin{cases} \displaystyle\int \sigma\left(\ell(q) + \frac{\alpha + \delta z + \omega}{\lambda(x)}\right) dG(\omega), & \lambda(x) > 0, \quad 0 < q < 1, \\ q, & \lambda(x) > 0, \quad q \in \{0,1\}, \\ \mathbb{P}_G\{\alpha + \delta z + \Omega > 0\}, & \lambda(x) = 0. \end{cases} \tag{3.1}$$

At the true parameter and population share, this average must equal the share supplied to the formula. I measure their difference with the RI residual

$$r(\theta; w, q) = \Psi(\theta; w, q) - q. \tag{3.2}$$

At the true parameter and population share,

$$r(\theta_0; W_m, s_m) = 0. \tag{3.3}$$

In the data, the population share s_m is unavailable. I replace it by the observed share $\widehat{s}_m$ and minimize the average squared discrepancy:

$$\widehat{Q}_M(\theta) = \frac{1}{M} \sum_{m=1}^{M} r(\theta; W_m, \widehat{s}_m)^2. \tag{3.4}$$

The cost function λ and taste distribution G are both unknown. I let Θ be the set of parameters satisfying the model restrictions and approximate these functions in a finite-dimensional set $\Theta_M \subseteq \Theta$, called a sieve. A nonnegative basis expansion can approximate a continuous cost function, while probability masses on an increasingly rich grid or a histogram with increasingly many bins can approximate the taste distribution. The probability weights sum to one and satisfy the payoff normalizations. When E2(c)'s analytic alternative is used at zero cost, candidates must generate analytic choice curves; finite Gaussian mixtures provide one such taste approximation. These approximations make the optimization finite-dimensional while allowing the functions to become more flexible as the sample grows.

A measurable estimator is an approximate minimizer if

$$\widehat{Q}_M(\widehat{\theta}_M) \leq \inf_{\theta \in \Theta_M} \widehat{Q}_M(\theta) + \epsilon_M, \qquad \epsilon_M = o_p(1). \tag{3.5}$$

This definition allows a numerical optimization error relative to the infimum over the sieve, provided that error converges to zero in probability. The numerical implementation uses several initial parameter values to check whether they give different optimization results.

A zero residual at an endpoint share does not establish that a candidate is correct. At positive cost, (3.1) makes the residual zero whenever the input share is zero or one, even if that share would not maximize (2.5).

The criterion can be discontinuous when a candidate information cost reaches zero. Fix $w = (z, x)$ and the payoff parameters (α, δ, G), and write $r_t(w, q)$ for the residual when the

unit cost at x is $t \geq 0$. At an input share of zero,

$$r_t(w,0) = \begin{cases} 0, & t > 0, \\ \mathbb{P}_G\{\alpha + \delta z + \Omega > 0\}, & t = 0. \end{cases}$$

If this full-information probability is positive, then

$$\lim_{t \downarrow 0} r_t(w,0)^2 = 0 < r_0(w,0)^2. \tag{3.6}$$

Thus the market contributes no loss at a positive cost but a positive loss at zero cost. Continuity of the true choice curve does not prevent this jump in a candidate's loss. The consistency proof below uses interior population shares to address it.

Positive bounded weights can also be included.[6]

3.2 Consistency

Market characteristics W_m are i.i.d., and within-market choices are conditionally i.i.d. as specified in Section 2. The criterion averages over markets and uses an estimated share within each market. E1 requires both the number of markets and the smallest within-market sample size to grow.

Assumption E1. The market sizes satisfy

$$M \to \infty, \qquad N_M := \min_{m \leq M} N_{m,M} \to \infty.$$

Consistency does not require a particular relative growth rate between M and N_M.

The next condition restricts the candidate parameter space. Besides compactness and the model restrictions, it ensures that matching observed interior shares determines the complete choice curves.

[6]The weighted criterion is

$$\widehat{Q}_M^w(\theta) = \frac{1}{M} \sum_{m=1}^{M} w(\theta; W_m, \widehat{s}_m) r(\theta; W_m, \widehat{s}_m)^2. \tag{3.7}$$

I require $0 < c \leq w(\theta; w, q) \leq C < \infty$, uniformly over candidate parameters, market characteristics, and input shares. These bounds keep the weighted and unweighted losses comparable, even when the weights depend on the candidate. The comparison argument in Section C.3 then gives consistency for a weighted approximate minimizer as well.

Assumption E2: parameter space and regularity.

(a) Θ contains θ_0 and is compact under

$$d(\theta, \widetilde{\theta}) = |\alpha - \widetilde{\alpha}| + |\delta - \widetilde{\delta}| + \|\lambda - \widetilde{\lambda}\|_\infty + d_L(G, \widetilde{G}), \tag{3.8}$$

where d_L is the Lévy metric for weak convergence.

(b) Every $\theta \in \Theta$ satisfies the restrictions in Assumptions ID1(a)–(d) and ID2, with θ replacing θ_0.

(c) For every $\theta = (\alpha, \delta, \lambda, G) \in \Theta$ and $x \in \mathcal{X}$, at least one of the following holds: $\lambda(x) > 0$; $\mathcal{Z}_x = \mathbb{R}$; or the choice curve $z \mapsto b(z, x; \theta)$ is real analytic on $\mathbb{R}$, with

$$0 < b(z, x; \theta) < 1 \qquad \text{for every finite } z.$$

Convergence in d means convergence of (α, δ), uniform convergence of λ, and weak convergence of G. The compactness assumption applies jointly to the coefficients and both unknown functions. It can be satisfied using a uniformly bounded, equicontinuous set of cost functions and a weakly closed set of taste distributions with a common moment bound of order greater than two, provided the restrictions in E2(b)–(c) are preserved under limits.

The first alternative in E2(c) covers a parameter space in which every candidate has $\lambda(x) > 0$ for all $x \in \mathcal{X}$. This pointwise positivity allows costs to approach zero, as required by ID2; it does not impose a positive lower bound uniform in x.[7] E2(c) applies to the entire compact parameter space Θ, including limits of candidate sequences, rather than only to the finite-dimensional sieves.

To establish consistency, I must show that a candidate giving the observed interior shares has the same complete choice curves as θ_0. For a candidate assigning zero information cost at a given x, E2(c) requires either full conditional support or a globally real-analytic choice curve. Full conditional support permits comparison at every z. When support is restricted, real analyticity determines the candidate's complete curve from its values on the observed interval. The requirement that probabilities lie strictly between zero and one at finite utility-shifter values follows from global analyticity: a monotone choice curve that reaches zero or one at a finite point is constant on a half-line, and the identity theorem would then make it constant everywhere, contradicting its limits at the two ends of $\mathbb{R}$. Discrete candidate taste distributions are permitted when E2(c) is satisfied through positive costs or full conditional support. The

[7] If $\mathcal{X}$ is compact, continuity and strict positivity would imply a positive minimum, so this everywhere-positive special case would be incompatible with ID2.

analytic alternative must hold for every zero-cost candidate when neither of the other alternatives applies. ID1(e) requires the true zero-cost curves to be continuous in z; the RI model and ID3(b) then imply joint continuity of the true choice function. No separate requirement of an observed interior share is imposed on positive-cost candidates: at zero population loss, their shares must equal the true observed interior shares.

The finite-dimensional sieve Θ_M must also become flexible enough to approximate λ_0 and G_0 while respecting the model restrictions.

Assumption E3: sieve approximation. A deterministic sequence $\{\theta_M^*\}_{M=1}^{\infty}$ satisfies

$$\theta_M^* \in \Theta_M \subseteq \Theta, \qquad d(\theta_M^*, \theta_0) \to 0 \quad (M \to \infty).$$

E3 requires increasingly accurate approximations to θ_0 without imposing an approximation rate. E1 makes the observed shares increasingly precise, while E2 restricts the candidate parameter space so that matching the observed interior shares determines the true parameter. Together, these conditions give the following consistency result for an approximate minimizer of the sample criterion.

Theorem 2 (Consistency)**.** *Let $\widehat{\theta}_M \in \Theta_M$ satisfy (3.5). Under ID1–ID3 and E1–E3,*

$$d(\widehat{\theta}_M, \theta_0) \to_p 0 \qquad \text{as } M \to \infty.$$

Section C.3 gives the complete proof.

To prove consistency, I address two issues in the minimum-distance argument. First, the criterion uses estimated shares. Second, its zero-cost limit can have the jump shown in (3.6). I first bound the effect of share estimation uniformly over candidate parameters. I then use a continuous lower bound on the original criterion to show that the estimator converges to the true parameter.

I first introduce the population-share criterion $Q_M(\theta) = M^{-1}\sum_m r(\theta; W_m, s_m)^2$. I compare it with the criterion based on estimated shares. The logistic derivative bound makes it possible to control their loss difference without using the candidate parameter in the bound. I define

$$H(t,s) = \begin{cases} \min\{1, \frac{1}{2}|\ell(t)-\ell(s)| + 2|t-s|\}, & t,s \in (0,1), \\ 0, & t = s \in \{0,1\}, \\ 1, & \text{otherwise.} \end{cases} \tag{3.9}$$

The resulting comparison holds simultaneously for every candidate:

$$\sup_{\theta\in\Theta}|\widehat{Q}_M(\theta)-Q_M(\theta)| \le \frac{1}{M}\sum_{m=1}^{M} H(\widehat{s}_m, s_m) = o_p(1). \tag{3.10}$$

The bound goes to zero because the observed shares become accurate as market sizes grow. For each fixed market type with an interior population share, share consistency makes the expected bound vanish; at a true endpoint, the observed share is exact. Since H is bounded by one and the minimum market size grows, bounded convergence and Markov's inequality make the average bound vanish. The bound contains no candidate parameter, which is why the comparison is uniform.

The share-substitution bound controls sampling error. To address the zero-cost jump in (3.6), I retain only interior population shares in an auxiliary criterion:

$$\begin{aligned} Q^{\circ}_M(\theta) &= \frac{1}{M}\sum_{m=1}^{M} \mathbf{1}\{0 < s_m < 1\} r(\theta; W_m, s_m)^2, \\ Q^{\circ}(\theta) &= \mathbb{E}\Big[\mathbf{1}\{0 < s(W) < 1\} r(\theta; W, s(W))^2\Big]. \end{aligned} \tag{3.11}$$

The auxiliary criterion is used only in the proof; the estimator still minimizes the original criterion. Since $0 \le Q^{\circ}_M(\theta) \le Q_M(\theta)$, a small value of the original criterion implies a small value of the auxiliary criterion. Using only interior shares excludes the endpoint discontinuity. At such shares, the zero-cost limit agrees with the full-information probability except at payoff ties. The choice made when payoffs tie can affect the probability only at thresholds where G has a point mass. There are at most countably many such thresholds, and the nonzero slope and continuous conditional distribution of Z make them probability-zero events for each candidate. Thus, at each fixed θ, the auxiliary loss is continuous in the candidate parameter for almost every market characteristic w and is bounded by one. This is the continuity needed for the uniform-convergence argument. Under E2, the sample auxiliary criterion converges uniformly to its continuous population expectation.

It remains to connect this auxiliary criterion to approximate minimization. Under ID1–ID3 and E3, the sieve approximant satisfies $Q_M(\theta^*_M) = o_p(1)$, so the sieve contains an approximant with a vanishing population-share loss. Let ζ_M be the uniform share-substitution error in (3.10). Approximate minimization then gives

$$0 \le Q^{\circ}_M(\widehat{\theta}_M) \le Q_M(\widehat{\theta}_M) \le Q_M(\theta^*_M) + 2\zeta_M + \epsilon_M = o_p(1). \tag{3.12}$$

The estimator's auxiliary criterion therefore approaches zero. Uniform convergence gives $Q^{\circ}(\widehat{\theta}_M) \to_p 0$. At zero population loss, a candidate reproduces the true observed interior shares.

Analyticity and monotonicity extend this agreement when both costs are positive; E2(c) supplies full support or global analyticity when the candidate cost is zero. The complete-curve moment argument then makes θ_0 the criterion's unique zero. By continuity and compactness, parameters a fixed distance from θ_0 have criterion values bounded away from zero. The estimator must approach θ_0, despite the discontinuity of the criterion it actually minimizes.

3.2.1 Extension: multiple alternatives

For the $J+1$ alternatives in Section 2.4.1, let $\mathbf{q} = (q_1, \ldots, q_J)'$ collect the shares for alternatives 1 through J and set $q_0 = 1 - \sum_{j=1}^{J} q_j$. The default share is determined by the others, so only J residual components are needed. Write $v_j(\omega; z) = \alpha_j + \delta_j z_j + \omega_j$ and $v_0(\omega; z) = 0$. At positive cost, the averaged conditional choice probability is

$$\Psi_j(\theta; w, \mathbf{q}) = \int \frac{q_j e^{v_j(\omega;z)/\lambda(x)}}{q_0 + \sum_{k=1}^{J} q_k e^{v_k(\omega;z)/\lambda(x)}} \, dG(\omega), \qquad j = 1, \ldots, J. \tag{3.13}$$

At zero cost, I use the full-information probabilities that each alternative has the highest payoff, with the model's tie convention. I define the column vector

$$\mathbf{r}(\theta; w, \mathbf{q}) = \left(\Psi_j(\theta; w, \mathbf{q}) - q_j\right)_{j=1}^{J}.$$

Each component is the difference between an alternative's conditional choice probability averaged over tastes and the share supplied to the formula, as in the binary model. Summing their squared discrepancies gives

$$\widehat{Q}_M(\theta) = \frac{1}{M} \sum_{m=1}^{M} \|\mathbf{r}(\theta; W_m, \widehat{\mathbf{s}}_m)\|_2^2. \tag{3.14}$$

The sieve now approximates the joint taste distribution. Positive bounded market weights can be included as in (3.7).

The consistency argument can follow the binary proof. For fixed J, bounds on the effect of replacing population shares by estimated shares in each residual component also bound the effect on their sum of squares. An auxiliary criterion using only population share vectors with positive probabilities for every alternative excludes the zero-cost discontinuity at the simplex boundary. Under the corresponding continuity and identification conditions, approximate minimization and compactness imply convergence to the unique population minimizer, as in the binary case.

4 Inference

For ease of exposition, I describe the inference procedure for binary choice. Section 4.1.1 discusses inference for multiple alternatives. Appendix B gives the formal binary conditions and coverage theorem.

4.1 Confidence intervals

Inference needs to accommodate the same constraints used in estimation. Information costs and sieve probability weights can equal zero, placing estimates on the boundary of the parameter space. An interior normal approximation would exclude these cases or require a separate boundary analysis. I instead test candidate parameters through their RI residuals and collect the candidates that are not rejected. This constructs a confidence region without requiring asymptotic normality of the estimator.

If the candidate parameter is the true value, the population residual is zero, but the residual evaluated at the estimated share fluctuates because of sampling error. The test asks whether this sampling variation can explain the observed residuals. For exposition, I take $N_{m,M} = N_M$ in every market and write

$$r_m(\theta, q) = r(\theta; W_m, q), \qquad A_m(\theta, q) = \partial_q r_m(\theta, q).$$

The derivative $A_m(\theta, q)$ measures how much $r_m(\theta, q)$ changes with its input share q. At the true parameter, the residual's leading sampling variance is $A_m(\theta_0, s_m)^2 s_m(1-s_m)/N_M$. Dividing each squared residual by its estimated sampling variance lets me compare residuals across markets without differentiating with respect to parameters subject to inequality constraints.

This standardization becomes unstable when shares approach zero or one or when the residual derivative is close to zero. I therefore retain markets whose shares and derivatives stay away from these values. Fix $\eta \in (0, 1/2)$ and $a_* > 0$. The retained set is

$$\widehat{\mathcal{A}}_M(\theta) = \{m : \eta \le \widehat{s}_m \le 1-\eta,\ |A_m(\theta, \widehat{s}_m)| \ge a_*\}. \tag{4.1}$$

The derivative depends on the candidate, so I recompute the retained set for each null being tested. On these markets, I sum squared residuals relative to their estimated sampling variances:

$$T_M(\theta) = \sum_{m \in \widehat{\mathcal{A}}_M(\theta)} \frac{N_M r_m(\theta, \widehat{s}_m)^2}{A_m(\theta, \widehat{s}_m)^2 \widehat{s}_m(1-\widehat{s}_m)}. \tag{4.2}$$

For a rejection probability $\gamma \in (0,1)$, let $\chi^2_{1-\gamma,k}$ be the $(1-\gamma)$ chi-square quantile with k degrees of freedom, and set $\chi^2_{1-\gamma,0} = 0$. The confidence region contains the sieve parameters not rejected by the test:

$$\mathcal{C}^s_{M,\gamma} = \left\{\theta \in \Theta_M : T_M(\theta) \leq \chi^2_{1-\gamma,|\widehat{\mathcal{A}}_M(\theta)|}\right\}. \tag{4.3}$$

If no market is retained, the statistic supplies no evidence against the candidate and the test accepts it. Under the conditions in Appendix B, the sieve approximant to the truth is accepted with asymptotic probability $1-\gamma$. The conditions account separately for share sampling, changes in trimming decisions, and residual bias from sieve approximation.

The confidence region can then be used to report a parameter or a counterfactual. For a scalar function $\phi : \Theta \to \mathbb{R}$, I take its range over the accepted parameters:

$$L_M = \inf_{\theta \in \mathcal{C}^s_{M,\gamma}} \phi(\theta), \qquad U_M = \sup_{\theta \in \mathcal{C}^s_{M,\gamma}} \phi(\theta). \tag{4.4}$$

For example, $\phi(\theta)$ can equal a payoff coefficient such as δ, the information cost $\lambda(x)$ at a fixed x, or a counterfactual change in task-taking probability. The other parameters are allowed to vary within the confidence region when I minimize and maximize this function. If the data do not reject many different values of the reported function, the resulting interval can be wide. Theorem 3 gives coverage under an additional condition on the sieve approximation error and the interval around the approximant's value.

4.1.1 Extension: multiple alternatives

For $J+1$ alternatives, I use the J shares and residuals for alternatives 1 through J from Section 3.2.1. The main change is that share-estimation errors are correlated: choosing one alternative excludes choosing another. Let $\mathbf{s}_m$ denote the vector of population shares for these alternatives. For conditionally independent multinomial choices, the covariance of $\widehat{\mathbf{s}}_m$ is

$$\frac{\Sigma_m}{N_M}, \qquad \Sigma_m = \operatorname{diag}(\mathbf{s}_m) - \mathbf{s}_m \mathbf{s}_m'.$$

I account for this dependence when computing the covariance of the residual vector. Let $B_m(\theta, \mathbf{q}) = \partial_{\mathbf{q}'} \mathbf{r}(\theta; W_m, \mathbf{q})$ be its Jacobian. At the true parameter, the leading residual covariance is

$$\frac{1}{N_M} B_m(\theta_0, \mathbf{s}_m) \Sigma_m B_m(\theta_0, \mathbf{s}_m)'.$$

I estimate this covariance multiplied by N_M with

$$\widehat{V}_m(\theta) = B_m(\theta, \widehat{\mathbf{s}}_m)\widehat{\Sigma}_m B_m(\theta, \widehat{\mathbf{s}}_m)',$$
$$\widehat{\Sigma}_m = \operatorname{diag}(\widehat{\mathbf{s}}_m) - \widehat{\mathbf{s}}_m \widehat{\mathbf{s}}_m'.$$

For trimming, the shares of every alternative, including the default, must be bounded away from zero, and the smallest eigenvalue of $\widehat{V}_m(\theta)$ must be bounded away from zero. For retained markets, the squared standardized residual in the binary statistic is replaced by

$$N_M\, \mathbf{r}(\theta; W_m, \widehat{\mathbf{s}}_m)' \widehat{V}_m(\theta)^{-1} \mathbf{r}(\theta; W_m, \widehat{\mathbf{s}}_m).$$

The binary proof indicates how to justify this extension. The scalar derivative $A_m(\theta, q)$ is replaced by the Jacobian $B_m(\theta, \mathbf{q})$, and the leading residual covariance multiplied by N_M is $B_m(\theta_0, \mathbf{s}_m)\Sigma_m B_m(\theta_0, \mathbf{s}_m)'$. The corresponding assumptions must control multinomial sampling error, the probability that a market lies close to the trimming cutoffs, and the population residual at the sieve approximant. These bounds control the Taylor remainder and the probability that sample and population trimming retain different markets. Under these conditions, the test uses a chi-square critical value with J degrees of freedom per retained market. Test inversion then gives a confidence region, and minimizing and maximizing the reported function over this region gives its interval. An interval-inclusion condition analogous to F5 is needed to cover the function's true value.

5 Empirical Application: assessing tasks in an online labour market

5.1 Data and specification

I use workers' decisions on Amazon Mechanical Turk, an online market for short assignments, to examine wage responses under different information costs. A worker sees a posting's reward and description but may still be uncertain how demanding the assignment would be for her. This setting connects the model's learning problem to a concrete decision: whether to take a task after assessing its personal difficulty. I relate the estimated cost of that assessment to how concretely the description specifies the work.

The posting data collected by Dubé et al. (2020) record rewards, requester identifiers, titles, descriptions, and repeated observations of available assignments. I retain postings with a positive reward, an identified requester, at least two inventory observations within the first 30 minutes, and an initial batch of at least ten assignments. To compare similar work at different

rewards, I then select groups from the same requester with matching task descriptions that contain at least four postings, at least three distinct rewards, and a highest reward at least twice the lowest. These restrictions yield 174 postings. Requiring a recorded inventory observation at the common three-hour endpoint leaves 137 postings in 35 groups, observed between June 1 and August 20, 2017. The repeated tasks provide within-group reward variation, and the common endpoint gives a comparable observation window for task taking.

The outcome of interest is the probability that a worker takes a task after encountering its posting, which I call the task-taking probability. Since worker visits are not directly observed, I construct a proxy from assignment inventories. I sum positive inventory decreases over the three-hour window; inventory increases do not cancel earlier decreases. Dividing this count by an exposure measure proportional to the initial batch size, using a common arrival-rate calibration, gives the observed task-taking share.

Descriptions vary along two dimensions. *Task specificity* requires a concrete action and its target, such as finding an agent's email address. *Workload specificity* requires a stated duration, a numerical amount of work, or an effort limit, such as inspecting at most three search results. A broad label such as "survey" need not specify a concrete task, and "quick" alone does not specify a numerical workload. Titles and descriptions are read together, with identical text assigned the same code.

I combine these two specificity criteria to form the categorical task assessment difficulty index x_m in Table 1. Its numbers label categories; they do not impose equal distances or rank the two middle categories. In particular, a posting can identify a broad type of work while meeting neither specificity criterion. The last category therefore includes recognizable but insufficiently specific descriptions.

Table 1: Task assessment difficulty categories

x	Category	Postings	Illustrative description
0	Both specific	29	Find the agent's email address; inspect up to three search results.
1	Task only	83	Find the agent's email address.
2	Workload only	9	Complete a two-minute study.
3	Neither specific	16	Complete a survey.

Notes: Examples illustrate the coding criteria. Task specificity requires a concrete action and target; workload specificity requires a numerical duration, quantity, or effort limit. The sample contains 44 distinct posting texts. Category values are labels.

I interpret information acquisition as the cognitive effort required to evaluate a task's uncertain payoff. The signal is an internal assessment of difficulty and personal suitability, formed by reading the posting and considering the worker's knowledge and experience. A description

specifying both the activity and the workload helps the worker judge what the assignment involves and how demanding it would be for her. This can reduce the cognitive effort required to obtain a given amount of information about the payoff. I use the task assessment difficulty index to record these differences in description specificity and allow $\lambda(x)$ to vary across its categories.

To interpret these differences as information costs, I treat description categories as cost shifters and maintain a common distribution of the remaining uncertain payoff component after controlling for task-group attractiveness. This exclusion matters because descriptions could otherwise change workers' prior beliefs about task payoffs. I estimate the four costs separately, without imposing an ordering or setting a category's cost to zero.

Worker i decides whether to take the task in posting m. I normalize the payoff from not taking the task to zero. The payoff from taking it and the unit information cost are

$$v_{im} = a_{c(m)} + \delta z_m + \Omega_{im}, \qquad \lambda_m = \lambda(x_m). \tag{5.1}$$

The index $c(m)$ groups posting m with other postings from the same requester that have matching task descriptions. The intercept $a_{c(m)}$ allows average payoffs to differ across these groups, while the reward coefficient is estimated from reward variation within groups. I define the reward index as

$$z_m = \log r_m - \overline{\log r}_{c(m)},$$

where r_m is the posted reward and the group average is computed from the estimation sample. Within each task group, I treat the posted reward as independent of Ω_{im}.

The taste distribution has mean zero and variance one and is approximated by a 41-bin histogram. I jointly estimate its bin probabilities, the 35 task-group intercepts, the reward coefficient, and four information costs by minimum distance.

5.2 Estimated information costs

Figure 2 shows how the estimated assessment cost varies across descriptions. It is approximately 0.0033 when both task and workload are specific, 0.0164 when only the task is specific, 0.0079 when only the workload is specific, and 0.0134 when neither criterion is met. The both-specific estimate is the smallest and close to zero, consistent with concrete activity and workload descriptions making an informative assessment easier to form.

The distinction between assessment and performance is important for interpreting the figure. These costs describe how difficult it is to evaluate a task. Task-group intercepts allow average attractiveness to differ across assignments, while taste shocks represent uncertain worker-specific

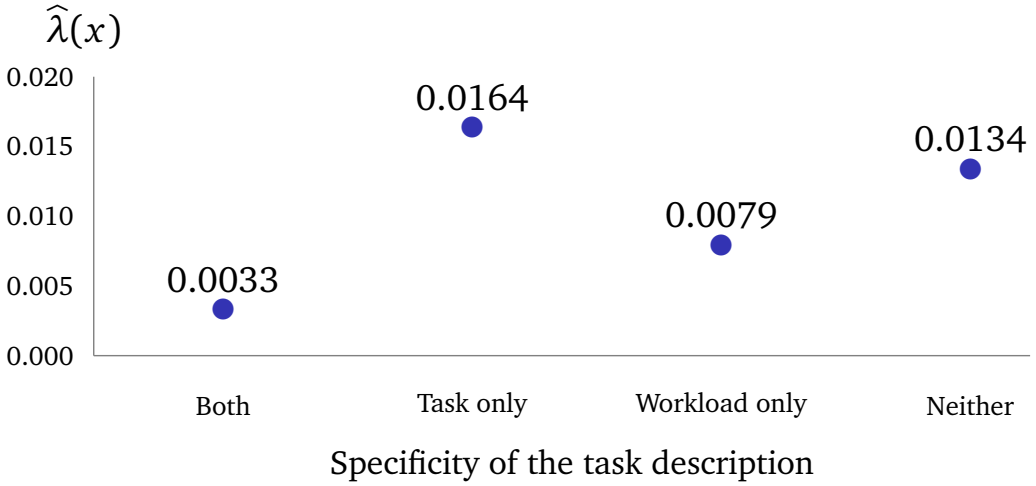


Figure 2: Estimated unit information costs by task-description category.

suitability. The two middle categories need not have a predetermined cost ranking: one specifies the activity, while the other specifies the workload.

The point estimates alone do not establish the ranking statistically. Table 2 reports them together with nominal 95% numerical confidence ranges obtained by simulation-based test inversion for the posting counts. The actual-data ranges are broad and all include zero. The calibrated column reports the corresponding ranges for a larger simulated sample generated at the estimated parameters.

Table 2: Information costs and uncertainty

Quantity	Estimate	Actual data	Calibrated data
Both specific	0.0033	$[0, 8]$	$[0, 0.0555]$
Task only	0.0164	$[0, 0.5953]$	$[0.0123, 0.3017]$
Workload only	0.0079	$[0, 2.1085]$	$[0.0031, 0.1107]$
Neither specific	0.0134	$[0, 8]$	$[0.0043, 0.2540]$
Wage coefficient δ	0.0017	$[10^{-6}, 1.2694]$	$[0.0015, 0.0421]$

Notes: Ranges are numerical approximations to projections of the accepted parameter set.

5.3 Wage responses under different information costs

The estimated costs allow me to examine how cheaper assessment changes workers' responses to rewards. I double the posted reward for all 137 postings and, holding the payoff coefficients and taste distribution fixed, compare estimated information costs, half those costs, and full information. For each of these three cost specifications, I compute choice probabilities at both the original and doubled reward, allowing workers to adjust their information acquisition when the reward changes.

Let $\bar{b}_1^\kappa$ and $\bar{b}_2^\kappa$ be the equally weighted average task-taking probabilities at the original and doubled reward, with information costs $\kappa\widehat{\lambda}$ for $\kappa \in \{1, 1/2, 0\}$. The reported elasticity is the log-arc elasticity

$$\mathcal{E}^\kappa = \frac{\log(\bar{b}_2^\kappa / \bar{b}_1^\kappa)}{\log 2}. \tag{5.2}$$

This elasticity compares the change in the log of the average task-taking probability with the change in log rewards over the wage doubling. The percentage-point effect, $100(\bar{b}_2^\kappa - \bar{b}_1^\kappa)$, instead reports the absolute change in task-taking probability.

Table 3: Wage doubling for all 137 postings

Information regime	Before (%)	After (%)	Change (pp)	Elasticity
Estimated costs	2.62	3.20	0.577	0.287
Half costs	8.90	9.32	0.426	0.068
Full information	12.87	13.21	0.335	0.037

Notes: The same postings and estimated payoffs enter every regime. Rewards double for all postings; the original centering constants remain fixed. Probabilities are averaged with equal posting weights. Elasticity is defined by (5.2).

Table 3 shows that cheaper assessment changes both participation and its wage responsiveness. Doubling rewards raises the average task-taking probability by about 22.0% at estimated costs, 4.8% at half costs, and 2.6% under full information, with elasticities of 0.287, 0.068, and 0.037. At the original reward, the average task-taking probability rises from 2.62% to 8.90% and 12.87% as costs fall. Lower information costs therefore raise this probability at the original reward but reduce its increase when rewards double.

These results are compatible. Cheaper assessment can help workers recognize that a task suits them and take it at the original reward, so fewer of these workers change their decision when the wage increases. Under full information and the estimated payoff distribution, relatively few additional workers find the task worthwhile when the reward doubles. The absolute increase in the average task-taking probability falls from 0.577 to 0.335 percentage points, and the higher starting probability further reduces the percentage response. Workers can consequently become more willing to take jobs while becoming less responsive to higher wages.

Cheaper assessment can also reveal that an assignment is unsuitable and lead a worker to decline a task she would otherwise take. The net participation and elasticity effects therefore depend on the payoff distribution and rewards. Table 3 reports those effects for this application.

To assess uncertainty in these comparisons, I report nominal 95% numerical confidence ranges in Table 4. The full-information effect compares task-taking probabilities at zero and estimated costs at the original reward; the elasticity difference is the estimated-cost elasticity

minus the full-information elasticity. The actual sample leaves substantial uncertainty about the effects, while the calibrated column reports ranges for the larger simulated sample.

Table 4: Counterfactual estimates and uncertainty

Quantity	Estimate	Actual data	Calibrated data
Wage effect (pp)	0.577	[0.000, 18.675]	[0.527, 1.299]
Full-information effect (pp)	10.253	[−4.390, 14.945]	[3.970, 14.532]
Estimated-cost elasticity	0.287	[0.000, 2.353]	[0.072, 0.305]
Half-cost elasticity	0.068	[0.000, 2.196]	[0.025, 0.196]
Full-information elasticity	0.037	[0.000, 1.931]	[0.010, 0.131]
Elasticity difference	0.250	[−0.504, 1.627]	[0.000, 0.277]

5.4 Wage responses under a Cardell taste-shock distribution

The counterfactuals also depend on the distribution of uncertain payoffs. To examine this dependence, I estimate the model using a Cardell distribution, a family used by Brown and Jeon (2024). I normalize it to mean zero and variance one and estimate its shape jointly with the other parameters. I use the same postings, utility controls, and cost categories as in the flexible specification.

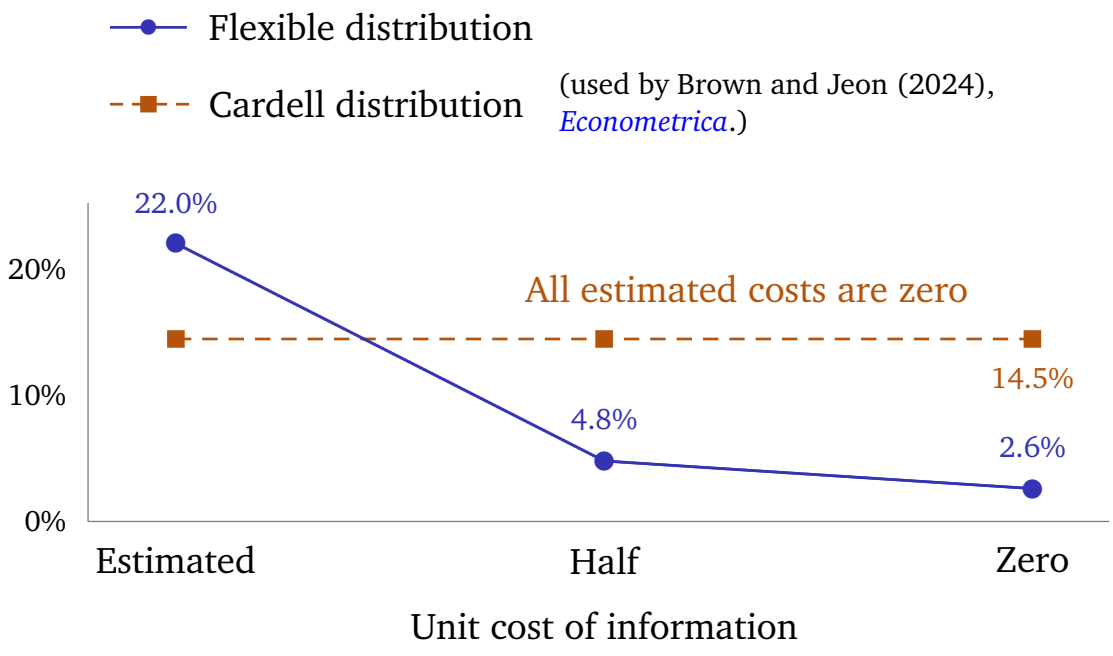


Figure 3: Percentage response to wage doubling under flexible and Cardell taste distributions. Each model is estimated on the same sample. The Cardell estimates assign zero cost to all four categories, so its three information regimes coincide.

The Cardell estimates assign zero information cost to every category. They predict a 14.5% increase in the average task-taking probability when rewards double, with log-arc elasticity approximately 0.195. Since the estimated costs are already zero, halving or removing them leaves that prediction unchanged. The flexible specification produces distinct responses under estimated, halved, and zero information costs in Figure 3.

The difference in counterfactuals is accompanied by a difference in the estimation criterion. The flexible specification attains 7.711×10^{-5}, compared with 2.550×10^{-4} under Cardell, a reduction of approximately 70%. Under the common criterion, the more flexible distribution accommodates positive information costs and distinct counterfactual responses while reducing the residual discrepancy. This comparison shows how distributional assumptions affect the application; it is not a formal test of the Cardell family.

6 Conclusion

I establish semiparametric identification of the taste-shock distribution and information-cost function in an RI discrete choice model. The construction links observed choice probabilities to the distribution of a switch point. Its common mean, full-information variance, and covariance with the curve's log odds identify the payoff coefficients and information costs. Deconvolution then recovers the taste-shock distribution. The argument uses optimal Shannon information acquisition without imposing parametric forms on either unknown function.

The same RI optimality equation provides an estimator based on market shares. Its consistency proof bounds the effect of share-estimation error and uses a continuous criterion based on interior population shares to address the discontinuity of the original loss. Test inversion provides confidence regions that allow zero information costs and zero probability weights in the taste-distribution sieve. The resulting counterfactual intervals have coverage under the sampling, trimming, and approximation conditions stated in the paper.

In the Mechanical Turk application, the estimated cost of assessing a task varies across description categories. With a flexible taste distribution, cheaper assessment raises the average task-taking probability at the original reward while reducing wage responsiveness. The Cardell specification assigns zero costs and gives different counterfactual predictions. These estimates illustrate why the distribution of uncertain payoffs matters for empirical RI analysis, while the broad actual-data uncertainty ranges show how much the available observations leave unresolved.

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A Identification for choices among multiple alternatives

This appendix states the model, assumptions, and identification theorem for the extension in Section 2.4.1. The proof is in Section C.5.

The alternatives are $\{0, 1, \ldots, J\}$, with $J \geq 1$, and

$$v_j(\Omega; z) = a_j(z) + \Omega_j, \qquad a_j(z) = \alpha_j + \delta_j z_j \quad (j = 1, \ldots, J), \qquad a_0(z) = \Omega_0 = 0.$$

Write $\alpha = (\alpha_1, \ldots, \alpha_J)' \in \mathbb{R}^J$, $\delta = (\delta_1, \ldots, \delta_J)' \in \mathbb{R}^J$, and $a(z) = (a_1(z), \ldots, a_J(z))' \in \mathbb{R}^J$. The joint taste distribution G has finite second moments, $\mathbb{E}_G \Omega = 0$, and $\mathrm{Var}_G(\Omega_1) = 1$. Slope relevance means $\delta_j \neq 0$ for every j. I orient each shifter so that $\delta_j > 0$. The unit cost of information is a continuous function $\lambda : \mathcal{X} \to [0, \infty)$ of the cost shifter x. Write $\mathcal{Z}_x = \mathrm{supp}(Z \mid X = x)$ and $\mathcal{X}_+ = \{x \in \mathcal{X} : \lambda(x) > 0\}$.

At positive cost, $b(z; x) = (b^0(z; x), \ldots, b^J(z; x))$ solves

$$\max_{q \in \Delta_J} \Phi(q; a(z), x), \qquad \Phi(q; a, x) = \lambda(x) \mathbb{E}_G \log \left\{ q_0 + \sum_{j=1}^{J} q_j e^{(a_j + \Omega_j)/\lambda(x)} \right\}, \tag{A.1}$$

where $\Delta_J = \{q \in [0,1]^{J+1} : \sum_{j=0}^{J} q_j = 1\}$. At zero cost, individuals select the alternative with the highest payoff. The consideration set is

$$\mathcal{A}(z; x) = \{j \in \{0, \ldots, J\} : b^j(z; x) > 0\}.$$

The observed population object is

$$\mathcal{B}_{\text{obs}} = \left\{ \left\{ (b^0(z; x), \ldots, b^J(z; x)) \right\}_{z \in \mathcal{Z}_x} \right\}_{x \in \mathcal{X}}.$$

As in the binary model, changes in observed payoffs and information costs must leave the joint taste distribution unchanged, and individuals must use that distribution as their prior.

Assumption A1: Independent taste shocks / common prior. $\Omega \perp (Z, X)$, $\Omega \sim G$, and individuals use G as their common prior.

To ensure a unique optimal share vector, I rule out exact linear dependence among the exponentiated taste terms.

Assumption A2: Exponential non-collinearity. For each $\ell \in \{\lambda(x) : x \in \mathcal{X}_+\}$, if a vector $u = (u_0, \ldots, u_J) \in \mathbb{R}^{J+1}$ satisfies

$$u_0 + \sum_{j=1}^{J} u_j e^{\Omega_j/\ell} = 0 \quad G\text{-almost surely},$$

then $u = 0$.

For example, a joint taste density that is positive on a nonempty open set satisfies A2, including when the taste shocks are correlated.

To recover the relative payoffs, I use observed choice situations in which each alternative can be chosen. The following condition guarantees such a point at every x with positive cost.

Assumption A3: Interior full-consideration point.

$$\text{For every } x \in \mathcal{X}_+, \quad \exists z_x^\circ \in \operatorname{int}(\mathcal{Z}_x): \quad b^j(z_x^\circ; x) > 0 \quad (j = 0, \ldots, J).$$

Continuity turns the point in A3 into a neighborhood where all choice probabilities can be compared. Alternatives may still be excluded at other covariate values.

The same low-cost variation as in ID2 supplies a comparison with choices made when information is nearly free.

Assumption A4: Nearly free attention.

$$\inf_{x \in \mathcal{X}} \lambda(x) = 0.$$

The next assumption controls the upper tails of tastes and their pairwise differences. Any bounded taste distribution satisfies it.

Assumption A5: Exponential moments. For every $t > 0$,

$$\mathbb{E}_G e^{t\Omega_k} < \infty, \qquad \mathbb{E}_G e^{t(\Omega_k - \Omega_j)} < \infty \quad (j \neq k,\ j, k \in \{1, \ldots, J\}).$$

At zero cost, the unrestricted taste distribution need not generate analytic choice curves, so I require full conditional support of Z. I also rule out pairwise payoff ties, so the full-information probabilities do not depend on a tie-breaking rule. The model permits $\lambda(x) = 0$ at every $x \in \mathcal{X}$; the proof treats this full-information case separately.

Assumption A6: Choice at zero information cost.

(a) For every $x \in \mathcal{X}$ with $\lambda(x) = 0$, $\mathcal{Z}_x = \mathbb{R}^J$.

(b) If the unit cost of information satisfies $\lambda(x) = 0$ at some cost-shifter value $x \in \mathcal{X}$, every pairwise taste difference has a continuous distribution:

$$\mathbb{P}_G(\Omega_j - \Omega_k = t) = 0 \quad \text{for every } t \in \mathbb{R} \text{ and } 0 \le j \ne k \le J.$$

These conditions support two identification arguments. If some x has $\lambda(x) > 0$, I recover the payoff coefficients from binary comparisons and low-cost variation, then recover G from the default probability. If $\lambda(x) = 0$ for every x, full conditional support permits direct full-information comparisons. The following theorem covers both cases.

Theorem 4 (Identification with multiple alternatives)**.** *Suppose A1–A6 hold,* $\delta_j \ne 0$ *for* $j = 1, \ldots, J$, $\mathbb{E}_G\|\Omega\|^2 < \infty$, $\mathbb{E}_G\Omega = 0$, *and* $\mathrm{Var}_G(\Omega_1) = 1$. *The observed family* $\mathcal{B}_{\text{obs}}$ *identifies*

$$\theta = (\alpha, \delta, \lambda, G).$$

The complete proof of Theorem 4 is in Section C.5.

B Asymptotic validity of the confidence intervals

This appendix states the formal conditions for the binary test-inversion procedure in Section 4. The market and within-market sampling conditions are those stated in Sections 2 and 3, with a common sample size N_M in each market. It uses the statistic (4.2) and projected endpoints (4.4). The trimming thresholds $\eta \in (0, 1/2)$ and $a_* > 0$ are fixed, as defined in the main text.

The test relies on the sampling behavior of the residual under the null. Since the population residual vanishes, its first-order fluctuation comes from replacing the true share with its estimate. At a true interior share, a Taylor expansion gives

$$r_m(\theta_0, \widehat{s}_m) = A_m(\theta_0, s_m)(\widehat{s}_m - s_m) + O_p((\widehat{s}_m - s_m)^2). \tag{B.1}$$

The leading sampling variance is

$$\frac{A_m(\theta_0, s_m)^2 s_m(1-s_m)}{N_M}.$$

Standardizing the residual by this variance avoids differentiation with respect to the constrained structural coordinates.

At an interior true share and positive cost,

$$A_m(\theta_0, s_m) = -\frac{\mathrm{Var}_{G_0}\{p_{\theta_0}(\Omega; W_m)\}}{s_m(1-s_m)}. \tag{B.2}$$

At zero cost, the residual subtracts its share input from a full-information probability, so the derivative is -1. These expressions explain why informative interior markets can have a nonzero derivative. The trimming rule requires sufficiently informative markets, without requiring every market to be retained.

For inference, errors must remain small after residuals are standardized and summed across markets. I therefore use a joint growth rate stronger than the one needed for consistency.

Assumption F1: Joint growth. Each market has N_M choices, with

$$M, N_M \to \infty, \qquad M/\sqrt{N_M} \to 0.$$

F1 controls the accumulated share-estimation errors. It ensures agreement between trimming based on estimated shares and trimming based on population shares across the markets. It also implies the weaker rate needed to control the Taylor-expansion and variance-estimation errors. Sieve approximation introduces a separate source of error, addressed in F4.

Write $s(w) = \pi(z, x)$ and, at an interior share, let

$$A_0(w) = \partial_q r(\theta_0; w, s(w)).$$

The test also needs markets whose residual variation can be standardized reliably. F2 requires a positive fraction of market types to lie strictly inside both trimming cutoffs at the truth.

Assumption F2: A positive fraction of informative markets.

$$\mathbb{P}\{\eta < s(W) < 1 - \eta,\ |A_0(W)| > a_*\} > 0.$$

Take a deterministic sieve approximant $\theta_M^* = (\alpha_M^*, \delta_M^*, \lambda_M^*, G_M^*) \in \Theta_M$ with $d(\theta_M^*, \theta_0) \to 0$, as in E3. For $0 < s(w) < 1$, define

$$A_M(w) = \partial_q r(\theta_M^*; w, s(w))$$

and its population-retained set

$$\mathcal{R}_M = \{w : \eta \le s(w) \le 1 - \eta,\ |A_M(w)| \ge a_*\}. \tag{B.3}$$

The set $\mathcal{R}_M$ describes which market types would be retained when testing the sieve approximant if population shares were available. It provides the population comparison for (4.1), whose share cutoff and derivative evaluation both use estimated shares.

E3, the nonzero slope in ID1(c), and conditional continuity in ID3(a) give $A_M(W) \to A_0(W)$ almost surely on interior shares. At a zero-cost limit, exact payoff ties occur with probability zero across markets. The strict inequalities in F2 therefore imply

$$\liminf_{M \to \infty} \mathbb{P}\{W \in \mathcal{R}_M\} > 0. \tag{B.4}$$

The strict inequalities in F2 are useful here: a market lying strictly inside the cutoffs at the truth remains retained once the approximant is sufficiently close. This supplies a positive limiting fraction of population-retained markets.

The remaining trimming problem concerns markets close to a cutoff, where a small share error may change whether a market is retained. To measure that proximity, I define the unscaled distance for interior $s(w)$:

$$\Delta_M(w) = \min\left\{|s(w) - \eta|,\ |s(w) - (1 - \eta)|, \left||A_M(w)| - a_*\right|\right\}. \tag{B.5}$$

Set $\Delta_M(w) = \eta$ when $s(w) \in \{0, 1\}$. At these true endpoints the sample share is exact, so no sampling error can change whether a market is retained.

Assumption F3: Little probability near the trimming cutoffs. There exist $C < \infty$, $t_0 > 0$ and an integer M_0 such that

$$\mathbb{P}\{\Delta_M(W) \le t\} \le Ct \quad \text{for every } M \ge M_0, \quad 0 < t \le t_0.$$

F3 bounds the probability of such markets uniformly along the approximating sequence. Convergence of A_M to A_0 alone does not provide this margin bound, so it is imposed separately at the approximant. On the trimmed share interval, the RI residual derivative has a uniform Lipschitz bound in the share input. That bound is absorbed into constants when bounding the probability that share error changes whether a market is retained.

Even with trimming controlled, a sieve approximant can leave a nonzero residual at population shares. This systematic discrepancy must be small relative to the sampling fluctuation used to calibrate the test.

Assumption F4: Negligible residual bias. Define

$$\rho_M = \sup_{w \in \mathcal{R}_M} |r(\theta_M^*; w, s(w))|. \tag{B.6}$$

Require $M^{1/4}\sqrt{N_M}\,\rho_M \to 0$.

F4 requires the approximant to nearly satisfy the population optimality equations on the same set $\mathcal{R}_M$ used for the population trimming comparison. It controls residual bias after standardization; F3 instead controls which markets enter the statistic. If $\theta_0 \in \Theta_M$ and $\theta_M^* = \theta_0$, then $\rho_M = 0$. For an infinite-dimensional truth, the sieve must be rich enough that its residual bias is negligible at the required rate.

Together, F1–F4 make the statistic at the approximant have the same centered and scaled limit as a chi-square variable whose degrees of freedom equal the retained-market count. Its acceptance probability therefore approaches $1 - \gamma$. This establishes acceptance of an approximant in the sieve; one further step is needed to cover a function of the infinite-dimensional truth.

Assumption F5: Negligible counterfactual approximation error. Let $\phi_M^* = \phi(\theta_M^*)$ and $E_M = \{\theta_M^* \in \mathcal{C}^s_{M,\gamma}\}$. For some deterministic $\varepsilon_M \downarrow 0$,

$$|\phi_M^* - \phi(\theta_0)| = o(\varepsilon_M),$$

$$\mathbb{P}\big\{[\phi_M^* - \varepsilon_M, \phi_M^* + \varepsilon_M] \subseteq [L_M, U_M] \mid E_M\big\} \to 1.$$

On acceptance, ϕ_M^* lies in the projected interval. The true value $\phi(\theta_0)$ may differ slightly because of sieve approximation. F5 requires that difference to be small relative to the space available around ϕ_M^* inside the interval.

F1–F4 establish the acceptance probability of the sieve approximant. F5 ensures that, conditional on this acceptance, the reported interval covers the true counterfactual value with probability approaching one. Continuity of ϕ and convergence of the approximant alone do not guarantee that the interval extends far enough on both sides of the approximant's counterfactual value.

Theorem 3 (Counterfactual coverage)**.** *Under the stated sampling assumptions, E3, ID1(c), ID3(a), and F1–F5, with compact* Θ *and continuous* $\phi : \Theta \to \mathbb{R}$*,*

$$\liminf_{M\to\infty} \mathbb{P}\{\phi(\theta_0) \in [L_M, U_M]\} \geq 1 - \gamma.$$

Section C.4 gives the proof. Theorem 3 obtains coverage from sampling variation in market shares and derivatives with respect to those shares. It requires no asymptotic normality of the sieve estimator, so parameter boundaries can remain in the tested space. How informative the resulting interval is depends on how many parameter values the observed shares can reject.

C Proofs

C.1 Auxiliary results

I first state and prove lemmas for the binary RI model used in the proofs of Theorems 1–3. The first two establish properties of choice curves and switch-point moments for identification. The next four concern residual continuity, replacing population shares with sample shares, binomial sampling moments, and a conditional central limit theorem. I restate the binary choice problem and residual so that these lemmas can be read independently of the main text.

I write $w = (z, x)$, $\theta = (\alpha, \delta, \lambda, G)$, and

$$v(\omega; z) = \alpha + \delta z + \omega, \qquad \sigma(u) = (1 + e^{-u})^{-1}, \qquad \ell(q) = \log\{q/(1-q)\} \quad (0 < q < 1).$$

For a fixed value x of the cost shifter with $\lambda(x) > 0$, the population choice probability $b(z, x; \theta)$ maximizes

$$q \longmapsto \lambda(x) \int \log\{1 - q + q e^{v(\omega;z)/\lambda(x)}\}\, dG(\omega), \qquad q \in [0, 1]. \tag{C.1.1}$$

At zero cost, I use the full-information comparison and its right-continuous CDF version when defining switch points. When discussing market shares, $s(w) = b(z, x; \theta_0)$ denotes the true population share. The residual used below is $r(\theta; w, q) = \Psi(\theta; w, q) - q$, where

$$\Psi(\theta; w, q) = \begin{cases} \displaystyle\int \sigma\left(\ell(q) + \frac{v(\omega; z)}{\lambda(x)}\right) dG(\omega), & \lambda(x) > 0,\ 0 < q < 1, \\ q, & \lambda(x) > 0,\ q \in \{0, 1\}, \\ \mathbb{P}_G\{v(\Omega; z) > 0\}, & \lambda(x) = 0. \end{cases}$$

The uses of dominated convergence and interchange of integrals below follow Billingsley (1995, Theorems 16.4 and 18.3); in each application, I give the relevant bound.

Lemma C.1.1 (Complete binary choice curves)**.** *Fix x, write $\lambda_x = \lambda(x) > 0$, and suppose G is nondegenerate with $\mathbb{E}_G|\Omega| < \infty$ and $\delta > 0$. The maximizer $b(z, x; \theta)$ in (C.1.1) is unique and continuous in z. Define its set of utility-shifter values giving interior shares by*

$$\mathcal{I}_{\theta,x} := \{z \in \mathbb{R} : 0 < b(z, x; \theta) < 1\}.$$

The set $\mathcal{I}_{\theta,x}$ is a nonempty open interval. The function $z \mapsto b(z, x; \theta)$ is real analytic and strictly increasing on $\mathcal{I}_{\theta,x}$, with range $(0, 1)$. It equals zero to the left of this interval and one to the right. Moreover,

$$\begin{aligned} h_{\theta,x} : \mathcal{I}_{\theta,x} &\longrightarrow \mathbb{R}, \\ z &\longmapsto \alpha + \delta z + \lambda_x \ell(b(z, x; \theta)) \end{aligned}$$

is an increasing bijection. If two positive-cost choice curves $z \mapsto b(z, x; \theta)$ and $z \mapsto b(z, x; \widetilde{\theta})$ agree on a nonempty open interval where both take values in $(0, 1)$, then $b(z, x; \theta) = b(z, x; \widetilde{\theta})$ for every $z \in \mathbb{R}$.

Proof. For this proof only, I suppress the fixed x and write $\lambda = \lambda_x$, $b(z) = b(z, x; \theta)$, and $h(z) = h_{\theta,x}(z)$.

I first describe the solution with $0 < b(z) < 1$ as the inverse of an increasing function. This will also determine where $b(z)$ equals zero or one, and why observations on an interval with $0 < b(z) < 1$ determine $b(z)$ for every $z \in \mathbb{R}$.

Write $\bar{a} = \alpha + \delta z$ and let $L = -\Omega + \lambda\varepsilon$, where ε is standard logistic and independent of Ω. With $\sigma(u) = (1 + e^{-u})^{-1}$,

$$F_L(h) = \mathbb{P}(L \le h) = \int \sigma\left(\frac{h + \omega}{\lambda}\right) dG(\omega).$$

The integrand is strictly increasing in h, lies in $(0, 1)$, and tends to zero and one at the two ends of $\mathbb{R}$. Bounded convergence gives the same limits for F_L. Differentiation under the integral gives

$$F_L'(h) = \frac{1}{\lambda} \int \sigma\left(\frac{h + \omega}{\lambda}\right)\left\{1 - \sigma\left(\frac{h + \omega}{\lambda}\right)\right\} dG(\omega) > 0.$$

The derivative is bounded by $1/(4\lambda)$. Thus F_L is continuous and strictly increasing from zero to one, and its inverse $F_L^{-1}(q)$ is defined for every $q \in (0, 1)$.

For real analyticity, apply Lemma C.1.7 with $J = 1$ and mixing variable $-\Omega$. The difference of two independent standard Gumbel variables is standard logistic (Train, 2002, Section 3.1), so that lemma applies directly to $L = -\Omega + \lambda\varepsilon$. In particular, F_L is real analytic. This smoothing property does not require G to have a density.

I next relate each interior share to the known payoff index that makes it optimal.

For $q \in (0,1)$ and $\omega \in \mathbb{R}$, define

$$p_q(\omega) = \sigma\left(\frac{F_L^{-1}(q) + \omega}{\lambda}\right),$$

and define the payoff-index map

$$a : (0,1) \longrightarrow \mathbb{R}, \qquad q \longmapsto a(q) = F_L^{-1}(q) - \lambda \ell(q).$$

By construction $\mathbb{E}_G p_q(\Omega) = q$. Hence

$$\begin{aligned}\mathbb{E}_G[p_q(\Omega)(1-p_q(\Omega))] &= \mathbb{E}_G p_q(\Omega) - \mathbb{E}_G[p_q(\Omega)^2] \\ &= q - q^2 - \mathrm{Var}_G(p_q(\Omega)) = q(1-q) - \mathrm{Var}_G(p_q(\Omega)).\end{aligned}$$

Since $\omega \mapsto p_q(\omega)$ is strictly increasing and G is nondegenerate, $\mathrm{Var}_G(p_q(\Omega)) > 0$. Consequently

$$0 < F_L'(F_L^{-1}(q)) = \frac{q(1-q) - \mathrm{Var}_G(p_q(\Omega))}{\lambda} < \frac{q(1-q)}{\lambda}.$$

Differentiating $a(q)$ now yields

$$a'(q) = \frac{1}{F_L'(F_L^{-1}(q))} - \frac{\lambda}{q(1-q)} > 0. \tag{C.1.2}$$

The real-analytic inverse-function theorem (Krantz and Parks, 2002, Theorem 1.5.3) applies because $F_L'(h) > 0$ for every $h \in \mathbb{R}$. Together with closure under composition, it makes $a(q)$ analytic. Equation (C.1.2) makes it strictly increasing. Its image $I_a = a((0,1))$ is a nonempty open interval, which may be bounded or unbounded on either side.

The function $a(q)$ also determines when the maximizing share equals zero or one.

Let the share objective, in payoff units, be

$$J_{\bar{a}}(q) = \lambda \int \log\{1 - q + q e^{(\bar{a}+\omega)/\lambda}\}\, dG(\omega), \qquad 0 \le q \le 1.$$

The bound

$$\left|\lambda \log\{1 - q + q e^{(\bar{a}+\omega)/\lambda}\}\right| \le |\bar{a}| + |\omega|$$

is integrable because $\mathbb{E}_G|\Omega| < \infty$. Lemma C.1.8, with $J = 1$, therefore gives finiteness and continuity of $q \mapsto J_{\bar{a}}(q)$ on $[0,1]$. Nondegeneracy of G also gives A2 in this binary case: a relation $u_0 + u_1 e^{\Omega/\lambda} = 0$ with $u_1 \neq 0$ would force Ω to be constant, while $u_1 = 0$ requires $u_0 = 0$. Lemma C.1.9 then gives strict concavity and a unique optimal share.

For $q \in (0,1)$, choose $\eta > 0$ so that $q \in [\eta, 1-\eta]$. With $R(\omega) = e^{(\bar{a}+\omega)/\lambda} > 0$,

$$\left| \lambda \frac{R(\omega)-1}{1-q+qR(\omega)} \right| \leq \frac{\lambda}{\eta}.$$

This bound justifies differentiation under the integral on any such share interval. The logit identity then gives

$$\begin{aligned} J'_{\bar{a}}(q) &= \lambda \int \frac{e^{(\bar{a}+\omega)/\lambda} - 1}{1-q+qe^{(\bar{a}+\omega)/\lambda}} \, dG(\omega) \\ &= \frac{\lambda}{q(1-q)} \{F_L(\bar{a} + \lambda \ell(q)) - q\}. \end{aligned}$$

Because $F_L(a(q) + \lambda \ell(q)) = q$ and F_L is strictly increasing,

$$\operatorname{sign} J'_{\bar{a}}(q) = \operatorname{sign}\{\bar{a} - a(q)\}.$$

If $\bar{a} \in I_a$, there is exactly one q with $\bar{a} = a(q)$. The derivative is positive below that share and negative above it, so the unique optimum is $q = a^{-1}(\bar{a})$. If $\bar{a}$ is at or below a finite left endpoint of I_a, every interior derivative is negative and the optimum is zero. If $\bar{a}$ is at or above a finite right endpoint, the optimum is one.

The inverse shares approach the appropriate endpoints as well. To see this at a finite left endpoint $L = \inf I_a$, take any sequence $\bar{a}_n \in I_a$ with $\bar{a}_n \to L$ and put $q_n = a^{-1}(\bar{a}_n)$. If q_n did not tend to zero, some subsequence would satisfy $q_n \geq \varepsilon$ for a fixed $\varepsilon \in (0,1)$. Monotonicity of a would give $\bar{a}_n = a(q_n) \geq a(\varepsilon) > L$ along that subsequence, contradicting $\bar{a}_n \to L$. Thus $q_n \to 0$. At a finite right endpoint $R = \sup I_a$, the same reasoning uses $q_n \leq 1-\varepsilon$ and $\bar{a}_n \leq a(1-\varepsilon) < R$ to show $q_n \to 1$.

Thus the interior utility-shifter interval is

$$I = \{z : \alpha + \delta z \in I_a\}, \qquad b(z) = a^{-1}(\alpha + \delta z) \quad (z \in I).$$

The inverse-function theorem and (C.1.2) make b analytic and strictly increasing on I. As q ranges over $(0,1)$, the corresponding z ranges over I, so $b(I) = (0,1)$. The limits just proved show that b remains continuous where it equals zero or one. Finally, on I,

$$h(z) = F_L^{-1}(b(z)).$$

Since F_L^{-1} maps $(0,1)$ onto $\mathbb{R}$, h maps I bijectively and increasingly onto $\mathbb{R}$.

It remains to show why equality on an observed open interval where both probabilities lie in $(0,1)$ implies equality for every real z.

Consider two such curves $z \mapsto b_1(z)$ and $z \mapsto b_2(z)$ with interior intervals I_1 and I_2, agreeing on a nonempty observed open interval. The identity theorem says that two real-analytic functions agreeing on an open interval agree throughout their connected common domain. By Krantz and Parks (2002, Corollary 1.2.6), they therefore agree on $I_1 \cap I_2$.

Write $I_i = (L_i, R_i)$, allowing an endpoint to be infinite. If $L_1 < L_2$, their nonempty intersection implies $L_2 < R_1$. Thus L_2 is finite and lies in I_1. Choose $z_n \in I_1 \cap I_2$ with $z_n \downarrow L_2$. The second curve satisfies $b_2(z_n) \to 0$, whereas continuity of the first gives $b_1(z_n) \to b_1(L_2) > 0$. This contradicts their equality on the intersection. Interchanging the curves rules out $L_2 < L_1$, so $L_1 = L_2$. The corresponding argument at the right endpoints uses the limiting share one and gives $R_1 = R_2$. Hence $I_1 = I_2$, and their zero and one extensions also agree. □

Lemma C.1.2 (Switch-point moments)**.** *Suppose $\mathbb{E}_G\Omega = 0$, $\mathrm{Var}_G(\Omega) = \tau^2 \in (0, \infty)$, and $\delta > 0$. Fix x with $\lambda_x = \lambda(x) \geq 0$, and write $\pi(z, x) = b(z, x; \theta)$ for every $z \in \mathbb{R}$. If $\lambda_x = 0$, suppose this curve is continuous. If $\lambda_x > 0$, set*

$$I_x = \{z : 0 < \pi(z, x) < 1\}, \qquad h_x : I_x \to \mathbb{R}, \quad h_x(z) = \alpha + \delta z + \lambda_x \ell(\pi(z, x)).$$

For $\Omega \sim G$ and an independent standard logistic ε, define the switch point

$$S_x = \begin{cases} h_x^{-1}(-\Omega + \lambda_x \varepsilon), & \lambda_x > 0, \\ -(\alpha + \Omega)/\delta, & \lambda_x = 0. \end{cases}$$

Its CDF is $\pi(\cdot, x)$. The moments

$$\mu(x) = \mathbb{E}S_x, \qquad V(x) = \mathrm{Var}(S_x), \qquad C(x) = \mathrm{Cov}(S_x, \ell(\pi(S_x, x)))$$

are finite, $V(x) > 0$, $C(x) > 0$, and

$$\mu(x) = -\frac{\alpha}{\delta}, \qquad \delta^2 V(x) + 2\delta\lambda_x C(x) = \tau^2. \tag{C.1.3}$$

Moreover,

$$0 \leq \tau^2 - \delta^2 V(x) \leq \frac{2\pi\tau}{\sqrt{3}}\lambda_x, \tag{C.1.4}$$

and $V(x) = \tau^2/\delta^2$ if and only if $\lambda_x = 0$.

Proof. I first verify that the switch point has the stated CDF. At positive cost and $z \in I_x$, the

Matějka–McKay conditional choice formula and its average give

$$\begin{aligned}\pi(z,x) &= \int \frac{\pi(z,x)e^{(\alpha+\delta z+\omega)/\lambda_x}}{1-\pi(z,x)+\pi(z,x)e^{(\alpha+\delta z+\omega)/\lambda_x}}\,dG(\omega)\\ &= \int \sigma\left(\frac{h_x(z)+\omega}{\lambda_x}\right)dG(\omega)\\ &= \mathbb{P}\{h_x(z) \geq -\Omega+\lambda_x\varepsilon\}.\end{aligned}$$

Lemma C.1.1 makes h_x an increasing bijection from I_x onto $\mathbb{R}$. Thus

$$\mathbb{P}(S_x \leq z) = \pi(z,x) \qquad (z \in I_x).$$

Since S_x lies in I_x almost surely, the equality also holds to its left and right, where both probabilities are zero and one, respectively. Continuity gives equality at finite endpoints. At zero cost, $S_x = -(\alpha+\Omega)/\delta$ gives

$$\mathbb{P}(S_x \leq z) = \mathbb{P}_G(\alpha+\delta z+\Omega \geq 0) = \pi(z,x),$$

where continuity rules out a positive probability of a payoff tie at any z. This also shows that S_x is nondegenerate: at positive cost its CDF increases from zero to one on I_x, and at zero cost its variance is $\tau^2/\delta^2 > 0$.

The continuous CDF implies that $\pi(S_x,x)$ is uniform on $(0,1)$. To verify this directly, fix $u \in (0,1)$ and choose z_u with $\pi(z_u,x) = u$. Any interval on which the CDF equals u has probability zero. Consequently

$$\mathbb{P}\{\pi(S_x,x) \leq u\} = \mathbb{P}(S_x \leq z_u) = u.$$

In particular, the variable

$$\eta_x = \ell(\pi(S_x,x))$$

is well defined almost surely and is standard logistic. Both ε and η_x have mean zero and variance $\pi^2/3$. These moments follow, for example, from the representation of a standard logistic variable as the difference of two independent standard Gumbel variables, whose variance is $\pi^2/6$ (Train, 2002, Section 3.1). The variables ε and η_x need not be independent.

Let $L_x = -\Omega+\lambda_x\varepsilon$. Applying h_x to the definition of S_x at positive cost gives

$$L_x = \alpha+\delta S_x+\lambda_x\eta_x. \tag{C.1.5}$$

At zero cost, the same identity follows directly from $S_x = -(\alpha+\Omega)/\delta$. Independence of Ω and

ε gives

$$\mathbb{E}L_x = 0, \qquad \mathbb{E}L_x^2 = \mathrm{Var}(L_x) = \tau^2 + \lambda_x^2 \frac{\pi^2}{3} < \infty.$$

Rearranging (C.1.5) bounds the switch-point second moment:

$$\mathbb{E}S_x^2 \le \frac{3}{\delta^2}\left\{\mathbb{E}L_x^2 + \alpha^2 + \lambda_x^2 \mathbb{E}\eta_x^2\right\} < \infty.$$

Thus $\mu(x)$, $V(x)$, and $C(x)$ are finite. Taking means in (C.1.5) gives $0 = \alpha + \delta\mu(x)$. Taking variances gives

$$\tau^2 + \lambda_x^2 \frac{\pi^2}{3} = \delta^2 V(x) + 2\delta\lambda_x C(x) + \lambda_x^2 \frac{\pi^2}{3}.$$

The common logistic variance terms cancel, proving (C.1.3). The covariance term remains because S_x and η_x generally depend on each other.

I next establish $C(x) > 0$. Let S_x' be an independent copy of S_x. Expansion and independence give

$$2C(x) = \mathbb{E}\big[(S_x - S_x')\{\ell(\pi(S_x, x)) - \ell(\pi(S_x', x))\}\big].$$

The product is nonnegative because the CDF and log odds are increasing. The two CDF values are independent continuous uniform variables, so they differ almost surely. Monotonicity then makes the product strictly positive almost surely. Its expectation is finite by the second-moment bound, and hence $C(x) > 0$.

It follows from (C.1.3) that

$$\tau^2 - \delta^2 V(x) = 2\delta\lambda_x C(x) \ge 0,$$

with equality exactly when $\lambda_x = 0$. Cauchy–Schwarz and this inequality give

$$C(x) \le \frac{\pi}{\sqrt{3}}\sqrt{V(x)}, \qquad \delta\sqrt{V(x)} \le \tau.$$

Therefore

$$0 \le \tau^2 - \delta^2 V(x) = 2\delta\lambda_x C(x) \le \frac{2\pi}{\sqrt{3}}\lambda_x \delta\sqrt{V(x)} \le \frac{2\pi\tau}{\sqrt{3}}\lambda_x,$$

which proves (C.1.4). □

Lemma C.1.3 (Residual derivatives and continuity)**.** *For each $\eta \in (0, 1/2)$ there is a finite constant K_η, independent of θ and w, such that*

$$|r(\theta; w, q)| \le 1, \qquad |\partial_q r(\theta; w, q)| + |\partial_q^2 r(\theta; w, q)| \le K_\eta \tag{C.1.6}$$

for $q \in [\eta, 1-\eta]$. In particular, $q \mapsto \partial_q r(\theta; w, q)$ is uniformly Lipschitz on this interval. Fix $w = (z, x)$ and $q \in (0, 1)$, and suppose $d(\theta_n, \theta) \to 0$, where $\theta = (\alpha, \delta, \lambda, G)$. If $\lambda(x) > 0$, or if $\lambda(x) = 0$ and $G\{-\alpha - \delta z\} = 0$, then

$$r(\theta_n; w, q) \longrightarrow r(\theta; w, q),$$
$$\partial_q r(\theta_n; w, q) \longrightarrow \partial_q r(\theta; w, q).$$

For each fixed $\theta = (\alpha, \delta, \lambda, G)$ with $\delta \neq 0$, ID3(a) implies

$$\mathbb{P}\{\lambda(X) = 0,\ G\{-\alpha - \delta Z\} > 0\} = 0.$$

Proof. Recall $r(\theta; w, q) = \Psi(\theta; w, q) - q$. Because $\Psi(\theta; w, q)$ and q lie in $[0, 1]$, $|r(\theta; w, q)| \le 1$. I next bound derivatives with respect to the input share and then examine continuity in the parameter, including a zero-cost limit.

When $\lambda(x) > 0$, define

$$p_q(\omega) = \sigma\left(\ell(q) + \frac{\alpha + \delta z + \omega}{\lambda(x)}\right).$$

Using $\sigma'(u) = \sigma(u)(1 - \sigma(u))$ and $\ell'(q) = 1/\{q(1-q)\}$ gives

$$\partial_q p_q(\omega) = \frac{p_q(\omega)(1 - p_q(\omega))}{q(1-q)}.$$

Differentiating once more and collecting terms gives

$$\begin{aligned}\partial_q^2 p_q(\omega) &= \frac{p_q(\omega)(1 - p_q(\omega))\{(1 - 2p_q(\omega)) - (1 - 2q)\}}{q^2(1-q)^2} \\ &= \frac{2p_q(\omega)(1 - p_q(\omega))(q - p_q(\omega))}{q^2(1-q)^2}.\end{aligned}$$

Let $d_\eta = \eta(1-\eta) > 0$. For $q \in [\eta, 1-\eta]$ and $\omega \in \mathbb{R}$,

$$|\partial_q p_q(\omega)| \le \frac{1}{4d_\eta}, \qquad |\partial_q^2 p_q(\omega)| \le \frac{1}{2d_\eta^2}.$$

These bounds do not involve the payoff, the cost, or G. They justify differentiation under the

integral. Since

$$\partial_q r(\theta; w, q) = \int \partial_q p_q(\omega)\, dG(\omega) - 1,$$
$$\partial_q^2 r(\theta; w, q) = \int \partial_q^2 p_q(\omega)\, dG(\omega).$$

the constant $K_\eta = 1 + 1/(4d_\eta) + 1/(2d_\eta^2)$ suffices in (C.1.6). When $\lambda(x) = 0$, $\Psi(\theta; w, q)$ is independent of q, so $\partial_q r(\theta; w, q) = -1$ and $\partial_q^2 r(\theta; w, q) = 0$. The mean-value theorem also gives

$$|\partial_q r(\theta; w, t) - \partial_q r(\theta; w, q)| \le K_\eta |t - q|$$

for two shares in the trimmed interval.

When $\lambda_0(x) > 0$, set $\theta = \theta_0$ and $q = s(w) \in (0, 1)$. The true share satisfies $\mathbb{E}_{G_0} p_q(\Omega) = q$. Thus

$$\begin{aligned}\partial_q r(\theta_0; w, q) &= \frac{\mathbb{E}_{G_0}[p_q(\Omega)(1 - p_q(\Omega))]}{q(1-q)} - 1 \\ &= \frac{q(1-q) - \mathrm{Var}_{G_0}(p_q(\Omega))}{q(1-q)} - 1 = -\frac{\mathrm{Var}_{G_0}(p_q(\Omega))}{q(1-q)}.\end{aligned}$$

This verifies (B.2).

I now turn from share derivatives to continuity in the candidate parameter.

For the sequence in the statement, the representation theorem in Billingsley (1995, Theorem 25.6) allows weak convergence $G_n \Rightarrow G$ to be represented by random variables $\Omega_n \sim G_n$ and $\Omega \sim G$ on a common probability space with $\Omega_n \to \Omega$ almost surely. This representation lets me compare the integrands directly. Metric convergence also gives

$$\alpha_n \to \alpha, \qquad \delta_n \to \delta, \qquad \lambda_n(x) \to \lambda(x).$$

Write $v_n = \alpha_n + \delta_n z + \Omega_n$ and $v = \alpha + \delta z + \Omega$. The integrand defining $\Psi(\theta_n; w, q)$ is

$$p_n(q) = \begin{cases} \sigma(\ell(q) + v_n/\lambda_n(x)), & \lambda_n(x) > 0, \\ \mathbf{1}\{v_n > 0\}, & \lambda_n(x) = 0. \end{cases}$$

If $\lambda(x) > 0$, then $\lambda_n(x) > 0$ eventually. Ordinary continuity gives, almost surely,

$$p_n(q) \longrightarrow p(q) := \sigma(\ell(q) + v/\lambda(x)),$$
$$\partial_q p_n(q) = \frac{p_n(q)(1 - p_n(q))}{q(1-q)} \longrightarrow \frac{p(q)(1 - p(q))}{q(1-q)}.$$

The integrands satisfy $0 \le p_n(q) \le 1$ and $|\partial_q p_n(q)| \le 1/\{4q(1-q)\}$. Because q is fixed in $(0,1)$, bounded convergence applies to both expectations. Subtracting q from the first expectation and 1 from the second proves the two limits in the statement.

If $\lambda(x) = 0$, then, outside the payoff-tie event $v = 0$, $v_n \to v$ with the same sign eventually. Along terms with $\lambda_n(x) > 0$,

$$\begin{aligned} v > 0 &\implies \ell(q) + v_n/\lambda_n(x) \to +\infty, \quad p_n(q) \to 1, \\ v < 0 &\implies \ell(q) + v_n/\lambda_n(x) \to -\infty, \quad p_n(q) \to 0. \end{aligned}$$

In either case

$$\partial_q p_n(q) = \frac{p_n(q)(1-p_n(q))}{q(1-q)} \to 0.$$

Along terms with $\lambda_n(x) = 0$, $p_n(q) = \mathbf{1}\{v_n > 0\}$ and $\partial_q p_n(q) = 0$. Therefore $p_n(q) \to \mathbf{1}\{v > 0\}$ and $\partial_q p_n(q) \to 0$ along the full sequence. The condition $G\{-\alpha - \delta z\} = 0$ makes these limits almost sure. Applying bounded convergence with the same bounds gives

$$\begin{aligned} r(\theta_n; w, q) &\longrightarrow \mathbb{P}_G(v > 0) - q = r(\theta; w, q), \\ \partial_q r(\theta_n; w, q) &\longrightarrow -1 = \partial_q r(\theta; w, q). \end{aligned}$$

To apply this continuity argument across markets for a fixed $\theta = (\alpha, \delta, \lambda, G)$, I must show that $G\{-\alpha - \delta Z\} > 0$ has probability zero.

Let $\mathcal{D}_G = \{\omega : G\{\omega\} > 0\}$ be the set of point masses of a fixed G. This set is countable. With $\delta \neq 0$, a payoff tie requires

$$Z \in \mathcal{T}_\theta := \{(-\alpha - \omega)/\delta : \omega \in \mathcal{D}_G\}.$$

The set $\mathcal{T}_\theta$ is also countable. ID3(a) gives $\mathbb{P}(Z \in \mathcal{T}_\theta \mid X = x) = 0$ at each x, and integration over X gives probability zero across markets. This assertion is for each fixed candidate; the argument does not require a single tie-free event for every candidate simultaneously. □

For the share-sampling results, markets have characteristics W_m, population shares $s_m = s(W_m)$, and deterministic sizes $N_{m,M}$. Conditional on the characteristics, the count $N_{m,M}\widehat{s}_m$ is binomial with success probability s_m. The two criteria are

$$\widehat{Q}_M(\theta) = M^{-1} \sum_m r(\theta; W_m, \widehat{s}_m)^2, \qquad Q_M(\theta) = M^{-1} \sum_m r(\theta; W_m, s_m)^2.$$

The parameter-free share bound is

$$H(t,s)=\begin{cases}\min\{1,\frac{1}{2}|\ell(t)-\ell(s)|+2|t-s|\}, & t,s\in(0,1),\\ 0, & t=s\in\{0,1\},\\ 1, & \text{otherwise.}\end{cases}$$

Lemma C.1.4 (Share substitution)**.** *Under the sampling conditions and E1, the function* $H:[0,1]^2\to[0,1]$ *defined above satisfies*

$$\sup_{\theta\in\Theta}|\widehat{Q}_M(\theta)-Q_M(\theta)|\le\frac{1}{M}\sum_{m=1}^{M}H(\widehat{s}_m,s_m)\to_p 0.$$

Proof. I first bound $|r(\theta;w,t)^2-r(\theta;w,s)^2|$ by a function of t and s that does not depend on θ. I then show that the average bound $M^{-1}\sum_m H(\widehat{s}_m,s_m)$ tends to zero.

For two share inputs t,s, write $r_t=r(\theta;w,t)$ and $r_s=r(\theta;w,s)$. Since $|r_t|,|r_s|\le 1$,

$$\begin{aligned}|r_t^2-r_s^2|&=|r_t-r_s||r_t+r_s|\\&\le 2|r_t-r_s|\le 2|\Psi(\theta;w,t)-\Psi(\theta;w,s)|+2|t-s|.\end{aligned}$$

When $\lambda(x)>0$ and $t,s\in(0,1)$, $\sup_u|\sigma'(u)|=1/4$ implies

$$|\Psi(\theta;w,t)-\Psi(\theta;w,s)|\le\tfrac{1}{4}|\ell(t)-\ell(s)|.$$

Thus $|r_t^2-r_s^2|\le\frac{1}{2}|\ell(t)-\ell(s)|+2|t-s|$. When $\lambda(x)=0$, $\Psi(\theta;w,q)$ does not depend on q, so $|r_t^2-r_s^2|\le 2|t-s|$. Since $r_t^2,r_s^2\in[0,1]$, $|r_t^2-r_s^2|\le 1$ in all cases. If $t=s\in\{0,1\}$, then $r_t^2=r_s^2$. If either input equals zero or one and $t\neq s$, I use the bound one. These inequalities give exactly $|r_t^2-r_s^2|\le H(t,s)$.

Applying this market by market,

$$|\widehat{Q}_M(\theta)-Q_M(\theta)|\le\frac{1}{M}\sum_m H(\widehat{s}_m,s_m).$$

The right-hand side contains no candidate parameter, so taking the supremum over θ leaves the same bound.

It remains to show $\mathbb{E}[M^{-1}\sum_m H(\widehat{s}_m, s_m)] \to 0$.

For a fixed market type w, let n denote its within-market sample size and define

$$h_n(w) = \mathbb{E}[H(\widehat{s}, s(w)) \mid W = w], \qquad n\widehat{s} \mid W = w \sim \text{Binomial}(n, s(w)).$$

When $0 < s(w) < 1$, binomial share consistency gives $\widehat{s} \to_p s(w)$ as $n \to \infty$. Since $H(t, s(w)) \to 0$ as $t \to s(w)$ and $0 \le H \le 1$, for every $\varepsilon > 0$,

$$h_n(w) \le \varepsilon + \mathbb{P}\{H(\widehat{s}, s(w)) > \varepsilon \mid W = w\}, \qquad \limsup_{n\to\infty} h_n(w) \le \varepsilon.$$

Letting $\varepsilon \downarrow 0$ gives $h_n(w) \to 0$. When $s(w) \in \{0, 1\}$, the sampled share equals $s(w)$ exactly, so $h_n(w) = 0$.

The minimum sample size $N_M = \min_m N_{m,M}$ tends to infinity by E1. Even though the sizes can differ across markets,

$$\begin{aligned}\mathbb{E}\left[\frac{1}{M}\sum_m H(\widehat{s}_m, s_m)\right] &= \int \frac{1}{M}\sum_m h_{N_{m,M}}(w)\, dP_W(w) \\ &\le \int \sup_{n \ge N_M} h_n(w)\, dP_W(w) \longrightarrow 0.\end{aligned}$$

The inequality uses $N_{m,M} \ge N_M$ for every market. For each fixed w, $h_n(w) \to 0$ means that all sufficiently late terms are small, so $\sup_{n \ge N_M} h_n(w) \to 0$. This supremum is bounded by one; bounded convergence justifies the last integral limit. Finally, Markov's inequality gives

$$\mathbb{P}\left\{\frac{1}{M}\sum_m H(\widehat{s}_m, s_m) > \varepsilon\right\} \le \frac{1}{\varepsilon}\mathbb{E}\left[\frac{1}{M}\sum_m H(\widehat{s}_m, s_m)\right] \longrightarrow 0.$$

□

Lemma C.1.5 (Binomial sampling moments). *Fix $\eta \in (0, 1/2)$ and $N \in \mathbb{N}$. Let $N\widehat{s} \sim$ Binomial(N, s) with $s \in [\eta, 1-\eta]$, and set $\xi = \sqrt{N}(\widehat{s}-s)/\sqrt{s(1-s)}$. Then*

$$\mathbb{E}\xi = 0, \qquad \mathbb{E}\xi^2 = 1,$$
$$\mathbb{E}\xi^3 = \frac{1-2s}{\sqrt{Ns(1-s)}}, \qquad \mathbb{E}\xi^4 = 3 + \frac{1-6s(1-s)}{Ns(1-s)}.$$

For each fixed $p \geq 2$, $\mathbb{E}|\xi|^p$ is uniformly bounded and $\mathbb{E}|\widehat{s}-s|^p \leq C_p N^{-p/2}$, uniformly over this share interval. The same statements hold conditionally on market characteristics.

Proof. Write $N\widehat{s} = \sum_{i=1}^{N} Y_i$, with independent Bernoulli variables of success probability s, and let $B_i = Y_i - s$. Put $D = s(1-s)$. The two possible values of B_i give

$$\mathbb{E}B_i = 0, \quad \mathbb{E}B_i^2 = D, \quad \mathbb{E}B_i^3 = D(1-2s), \quad \mathbb{E}B_i^4 = D(1-3D).$$

Also $\xi = (ND)^{-1/2} \sum_i B_i$. Independence and centering immediately give $\mathbb{E}\xi = 0$ and $\mathbb{E}\xi^2 = 1$.

In the expansion of the third power, any term with an index appearing only once has expectation zero. Only terms with all three indices equal remain:

$$\mathbb{E}\left(\sum_i B_i\right)^3 = N\mathbb{E}B_1^3.$$

In the fourth power, the nonzero terms have either all four indices equal or two pairs of equal indices. There are $3N(N-1)$ terms of the latter type, so

$$\mathbb{E}\left(\sum_i B_i\right)^4 = N\mathbb{E}B_1^4 + 3N(N-1)(\mathbb{E}B_1^2)^2.$$

Dividing by $(ND)^{3/2}$ and $(ND)^2$, respectively, gives

$$\mathbb{E}\xi^3 = \frac{1-2s}{\sqrt{ND}}, \qquad \mathbb{E}\xi^4 = \frac{1-3D}{ND} + 3\frac{N-1}{N} = 3 + \frac{1-6D}{ND}.$$

For higher absolute moments, Hoeffding (1963, Theorem 2), applied to both tails of the Bernoulli average, and $D \geq \eta(1-\eta)$ give

$$\mathbb{P}(|\xi| > t) = \mathbb{P}\left(|\widehat{s}-s| > t\sqrt{D/N}\right) \leq 2e^{-2Dt^2} \leq 2e^{-2\eta(1-\eta)t^2}.$$

Using the tail-integral formula,

$$\mathbb{E}|\xi|^p = \int_0^\infty p t^{p-1} \mathbb{P}(|\xi| > t)\, dt \le 2p \int_0^\infty t^{p-1} e^{-2\eta(1-\eta)t^2}\, dt < \infty.$$

The bound depends only on p and η. Since $\widehat{s} - s = \sqrt{D/N}\,\xi$, it also gives the stated share-error bound. Conditional on the market characteristics, s is fixed and the observations remain independent Bernoulli variables, so every calculation applies conditionally as well. □

Lemma C.1.6 (Conditional central limit theorem)**.** *Let $\mathcal{F}_M = \sigma(W_1, \dots, W_M)$. For each M, suppose there are k_M conditionally independent variables $X_{M,1}, \dots, X_{M,k_M}$, centered given $\mathcal{F}_M$. The number of summands k_M is finite and $\mathcal{F}_M$-measurable. Set*

$$v_M^2 = \sum_{j=1}^{k_M} \mathbb{E}[X_{M,j}^2 \mid \mathcal{F}_M].$$

All conditional moments used below are finite. Suppose $\mathbb{P}(v_M > 0) \to 1$ and, for some $p > 2$,

$$\Lambda_M := \frac{\sum_{j=1}^{k_M} \mathbb{E}[|X_{M,j}|^p \mid \mathcal{F}_M]}{v_M^p} \to_p 0,$$

with the ratio defined as zero when $v_M = 0$. Then

$$S_M := \begin{cases} v_M^{-1} \sum_{j=1}^{k_M} X_{M,j}, & v_M > 0, \\ 0, & v_M = 0, \end{cases} \qquad S_M \Rightarrow N(0,1).$$

Proof. The conditioning separates two sources of randomness. Once the market characteristics are fixed, the number of summands and their conditional moments are fixed; the sampled choices, and hence the summands themselves, remain random. The ordinary Lyapunov theorem (Billingsley, 1995, Theorem 27.3) can then be applied to these conditional distributions. I spell out why convergence of the random Lyapunov ratio in probability is sufficient.

For a real number t, define the conditional CDF

$$F_M(t) = \mathbb{P}(S_M \le t \mid \mathcal{F}_M).$$

I will first show $F_M(t) \to_p \Phi(t)$, where Φ is the standard normal CDF. The ordinary Lyapunov theorem concerns deterministic arrays of distributions. Here Λ_M, k_M, and v_M depend on market characteristics, so I first obtain almost-sure convergence along a subsequence.

Take any deterministic subsequence of sample sizes (M_ℓ). Since

$$\Lambda_M + \mathbf{1}\{v_M = 0\} \to_p 0,$$

there is a further deterministic subsequence (M_{ℓ_r}) along which

$$\Lambda_{M_{\ell_r}} + \mathbf{1}\{v_{M_{\ell_r}} = 0\} \longrightarrow 0 \quad \text{almost surely.}$$

This follows from the subsequence characterization of convergence in probability (Billingsley, 1995, Theorem 20.5). The indicator implies $v_{M_{\ell_r}} > 0$ eventually, almost surely.

Choose conditional distributions given $\mathcal{F}_M$. In this application they are supplied directly by the finite product of binomial sampling distributions after $W_1, \ldots, W_M$ have been fixed. On a probability-one set, conditional independence, centering, and the moment identities hold for every sample size; a single such set can be used because there are countably many sample sizes.

Now fix a realization ω in this set and in the almost-sure convergence event above. For sample size M_{ℓ_r}, the conditional distribution has exactly $k_{M_{\ell_r}}(\omega)$ independent centered summands and total variance $v_{M_{\ell_r}}(\omega)^2$. Its ordinary Lyapunov ratio is the number $\Lambda_{M_{\ell_r}}(\omega)$, which tends to zero. Theorem 27.3 of Billingsley (1995) therefore gives

$$F_{M_{\ell_r}}(t)(\omega) \longrightarrow \Phi(t) \qquad \text{for every real } t.$$

Thus every subsequence of $F_M(t)$ has a further subsequence converging almost surely to $\Phi(t)$. This proves

$$F_M(t) \to_p \Phi(t).$$

No independence between samples of different sizes is needed: the Lyapunov theorem uses independence among the summands within each sample.

The conditional convergence just obtained also implies unconditional convergence.

Because $0 \le F_M(t) \le 1$, for every $\varepsilon > 0$,

$$\mathbb{E}|F_M(t) - \Phi(t)| \le \varepsilon + \mathbb{P}\{|F_M(t) - \Phi(t)| > \varepsilon\}.$$

The probability tends to zero. Letting $\varepsilon \downarrow 0$ gives $\mathbb{E}|F_M(t) - \Phi(t)| \to 0$. Finally, the tower property of conditional expectation (Billingsley, 1995, Theorem 34.4) yields

$$\mathbb{P}(S_M \le t) = \mathbb{E}F_M(t), \qquad |\mathbb{P}(S_M \le t) - \Phi(t)| \le \mathbb{E}|F_M(t) - \Phi(t)| \longrightarrow 0.$$

The standard normal CDF is continuous at every real t, so this is precisely $S_M \Rightarrow N(0,1)$. □

I now state and prove lemmas for the RI model with multiple alternatives, used in the identification proof of Theorem 4. Throughout these lemmas, I use the setup and notation from Appendix A. The first gives the joint distribution of the Gumbel differences used in that proof. Its special case $J = 1$ also gives the logistic distribution used in the binary model. The subsequent lemmas establish continuity and uniqueness of the optimal share vector and show how an interior share vector determines the known payoff indices.

Lemma C.1.7 (The Gumbel-difference distribution and its convolution)**.** *Let $\eta_0, \ldots, \eta_J$ be independent standard Gumbel variables, with CDF $\mathbb{P}(\eta_j \leq y) = \exp(-e^{-y})$, and put $\zeta = (\zeta_1, \ldots, \zeta_J)'$, where $\zeta_j = \eta_j - \eta_0$. Its CDF F_ζ, density f_ζ, and characteristic function φ_ζ satisfy, for $y, t \in \mathbb{R}^J$,*

$$F_\zeta(y) = \frac{1}{1 + \sum_{j=1}^J e^{-y_j}}, \tag{C.1.7}$$

$$f_\zeta(y) = \frac{J!\, e^{-\sum_j y_j}}{(1 + \sum_j e^{-y_j})^{J+1}}, \tag{C.1.8}$$

$$\varphi_\zeta(t) = \Gamma\left(1 + i\sum_{j=1}^J t_j\right)\prod_{j=1}^J \Gamma(1 - it_j) \neq 0 \quad (t \in \mathbb{R}^J). \tag{C.1.9}$$

If $\Omega \in \mathbb{R}^J$ is any random vector independent of $\eta_0, \ldots, \eta_J$ and $\lambda > 0$, then the CDF F_H and density f_H of $H = \Omega + \lambda\zeta$ are real analytic on $\mathbb{R}^J$.

Proof. The inequality $\zeta \leq y$ means $\eta_j - \eta_0 \leq y_j$ for every $j = 1, \ldots, J$. Thus $\{\zeta \leq y\}$ is the event that alternative zero is chosen when the known payoffs are $0, -y_1, \ldots, -y_J$ and the payoff shocks are $\eta_0, \ldots, \eta_J$. The multinomial logit formula (Train, 2002, Eq. (3.6)) therefore gives (C.1.7). Differentiating once in each coordinate gives (C.1.8).

A standard Gumbel variable has characteristic function $\Gamma(1 - it)$ (Janson, 2010, Section 3). Independence gives

$$\varphi_\zeta(t) = \mathbb{E}e^{-i(\sum_j t_j)\eta_0}\prod_j \mathbb{E}e^{it_j\eta_j} = \Gamma\left(1 + i\sum_j t_j\right)\prod_j \Gamma(1 - it_j).$$

The gamma function Γ has no zeros, and all its arguments here have real part one, away from its poles (NIST, 2026, Section 5.2(i)). This proves (C.1.9), including nonvanishing.

For the analyticity claim, let G denote the distribution of Ω. Conditioning on Ω gives

$$F_H(h) = \int \frac{1}{1 + \sum_{j=1}^J \exp\{(\omega_j - h_j)/\lambda\}}\, dG(\omega).$$

This is a mixed logit probability for the reference alternative, with known payoff indices $-h_j/\lambda$

and random intercepts Ω_j/λ. By Iaria and Wang (2025, Theorem 1(a)), the non-reference choice probabilities are real analytic for any mixing distribution. Since F_H is one minus their sum, it is real analytic on $\mathbb{R}^J$. Its density $f_H = \partial_{h_1} \cdots \partial_{h_J} F_H$ is real analytic as well, because derivatives preserve real analyticity. No density or moment restriction on G is needed. □

Lemma C.1.8 (Continuity of the share objective and its unique maximizer)**.** *Fix $J \geq 1$ and x with $\lambda(x) > 0$. Let $\Omega \in \mathbb{R}^J$ have distribution G satisfying*

$$\mathbb{E}_G \|\Omega\|_1 = \sum_{j=1}^{J} \mathbb{E}_G |\Omega_j| < \infty.$$

For $a \in \mathbb{R}^J$ and $\Delta_J = \{q \in [0,1]^{J+1} : \sum_{j=0}^{J} q_j = 1\}$, define

$$\Phi(q; a, x) = \lambda(x) \mathbb{E}_G \log \left\{ q_0 + \sum_{j=1}^{J} q_j e^{(a_j + \Omega_j)/\lambda(x)} \right\}.$$

Then $(q, a) \mapsto \Phi(q; a, x)$ is finite and continuous on $\Delta_J \times \mathbb{R}^J$, including the boundary of Δ_J. For every a, a maximizer over Δ_J exists. If that maximizer is unique for every a, the map

$$q^*(\cdot; x) : \mathbb{R}^J \longrightarrow \Delta_J, \qquad a \longmapsto q^*(a; x) := \arg\max_{q \in \Delta_J} \Phi(q; a, x)$$

is continuous.

Proof. Write $\lambda = \lambda(x)$ and define the integrand

$$g(q, a, \omega) = \lambda \log \left\{ q_0 + \sum_{j=1}^{J} q_j e^{(a_j + \omega_j)/\lambda} \right\}.$$

Because the nonnegative weights $q_0, \ldots, q_J$ sum to one,

$$\min\{0, a_1 + \omega_1, \ldots, a_J + \omega_J\} \leq g(q, a, \omega) \leq \max\{0, a_1 + \omega_1, \ldots, a_J + \omega_J\}.$$

In particular,

$$|g(q, a, \omega)| \leq \|a\|_\infty + \|\omega\|_1.$$

Since $\mathbb{E}_G \|\Omega\|_1 < \infty$, the bound is integrable for fixed a, so $\Phi(q; a, x)$ is finite for every (q, a).

To prove joint continuity, take arbitrary sequences $q_n \to q$ in Δ_J and $a_n \to a$ in $\mathbb{R}^J$. The sequence a_n is bounded, so there is $R < \infty$ with $\|a_n\|_\infty \leq R$ for all n. For each fixed ω, the

sum inside the logarithm is strictly positive, even if some coordinates of q are zero. Thus

$$g(q_n, a_n, \omega) \longrightarrow g(q, a, \omega), \qquad |g(q_n, a_n, \omega)| \le R + \|\omega\|_1.$$

The bound is independent of n and integrable under G. Dominated convergence gives $\Phi(q_n; a_n, x) \to \Phi(q; a, x)$, proving joint continuity. For fixed a, continuity on the compact simplex gives existence of a maximizer by the extreme value theorem (Munkres, 2000, Theorem 27.4).

Now suppose the maximizer is unique for every a. Fix any $a_n \to a$ and write $q_n^* = q^*(a_n; x)$. I show $q_n^* \to q^*(a; x)$. Take any subsequence $\{q_{n_k}^*\}$. Compactness of Δ_J gives a further subsequence $q_{n_{k_\ell}}^* \to \bar{q} \in \Delta_J$. For every fixed $q \in \Delta_J$, optimality at $a_{n_{k_\ell}}$ gives

$$\Phi(q_{n_{k_\ell}}^*; a_{n_{k_\ell}}, x) \ge \Phi(q; a_{n_{k_\ell}}, x).$$

Joint continuity permits passage to the limit on both sides:

$$\Phi(\bar{q}; a, x) \ge \Phi(q; a, x) \qquad \text{for every } q \in \Delta_J.$$

Hence $\bar{q}$ maximizes the objective at a. Uniqueness gives $\bar{q} = q^*(a; x)$. If the original sequence q_n^* did not converge to $q^*(a; x)$, some subsequence would stay at least a fixed positive distance from $q^*(a; x)$. The compactness argument just given would produce a further subsequence converging to $q^*(a; x)$, a contradiction. Therefore $q^*(a_n; x) \to q^*(a; x)$. □

Lemma C.1.9 (Uniqueness under exponential non-collinearity)**.** *For the distribution G, cost $\lambda(x) > 0$, simplex Δ_J, and objective $\Phi(q; a, x)$ in Lemma C.1.8, suppose A2 holds at $\lambda(x)$: for $u \in \mathbb{R}^{J+1}$,*

$$u_0 + \sum_{j=1}^{J} u_j e^{\Omega_j/\lambda(x)} = 0 \quad G\text{-almost surely} \quad \Longrightarrow \quad u = 0.$$

For every $a \in \mathbb{R}^J$, the function $q \mapsto \Phi(q; a, x)$ is strictly concave on Δ_J. Its maximizer $q^*(a; x)$ is unique, and the map $a \mapsto q^*(a; x)$ is continuous.

Proof. Fix $a \in \mathbb{R}^J$ and write $\lambda = \lambda(x)$. For $q \in \Delta_J$, put

$$D_q(\Omega) = q_0 + \sum_{j=1}^{J} q_j e^{a_j/\lambda} e^{\Omega_j/\lambda} > 0.$$

Take $q, \widetilde{q} \in \Delta_J$ with $q \neq \widetilde{q}$. If $D_q(\Omega) = D_{\widetilde{q}}(\Omega)$ almost surely, then

$$(q_0 - \widetilde{q}_0) + \sum_{j=1}^{J} (q_j - \widetilde{q}_j) e^{a_j/\lambda} e^{\Omega_j/\lambda} = 0 \quad G\text{-almost surely.}$$

Apply A2 with $u_0 = q_0 - \widetilde{q}_0$ and $u_j = (q_j - \widetilde{q}_j) e^{a_j/\lambda}$ for $j = 1, \ldots, J$. It gives $u_j = 0$ for every j. Since $e^{a_j/\lambda} > 0$, this would imply $q = \widetilde{q}$, a contradiction. Therefore

$$\mathbb{P}_G\{D_q(\Omega) \neq D_{\widetilde{q}}(\Omega)\} > 0.$$

For any $t \in (0,1)$, strict concavity of the logarithm gives

$$\log\{t D_q(\Omega) + (1-t) D_{\widetilde{q}}(\Omega)\} \geq t \log D_q(\Omega) + (1-t) \log D_{\widetilde{q}}(\Omega),$$

with a strict inequality on the event just displayed. All three logarithms are integrable by Lemma C.1.8. Taking expectations and multiplying by $\lambda > 0$ therefore gives

$$\Phi(tq + (1-t)\widetilde{q}; a, x) > t\Phi(q; a, x) + (1-t)\Phi(\widetilde{q}; a, x).$$

This is strict concavity. Since a maximizer exists, it is unique. The continuity conclusion now follows from Lemma C.1.8. □

Lemma C.1.10 (Interior shares determine a unique payoff vector)**.** *Use the cost $\lambda(x) > 0$, distribution G, and objective $\Phi(q; a, x)$ in Lemma C.1.8, and suppose A2 holds at $\lambda(x)$. Write*

$$\begin{aligned}
&\mathcal{D}_J = \{p \in [0,1]^J : \textstyle\sum_{j=1}^J p_j \le 1\},\\
&\mathcal{D}_J^\circ = \{p \in \mathcal{D}_J : p_j > 0 \ (j = 1,\dots,J),\ \textstyle\sum_{j=1}^J p_j < 1\},\\
&p_0 = 1 - \textstyle\sum_{j=1}^J p_j, \qquad q(p) = (p_0, p_1, \dots, p_J)',
\end{aligned}$$

so that $\mathcal{D}_J$ uses the J free coordinates of Δ_J. Let $\bar{b}(a; x)$ be the unique maximizer of $\Phi(\cdot; a, x)$, and set

$$\mathcal{W}_x = \{a \in \mathbb{R}^J : \bar{b}^j(a; x) > 0 \ (j = 0, \dots, J)\}.$$

For every $p \in \mathcal{D}_J^\circ$, the function

$$H(a; p, x) = \Phi(q(p); a, x) - p'a \qquad (a \in \mathbb{R}^J)$$

is strictly convex and tends to $+\infty$ as $\|a\|_\infty \to \infty$. It has a unique minimizer $a^*(p; x)$. The set $\mathcal{W}_x$ is nonempty and open, and the maps

$$\begin{aligned}
B_x : \mathcal{W}_x \longrightarrow \mathcal{D}_J^\circ, \quad a \longmapsto B_x(a) = (\bar{b}^1(a; x), \dots, \bar{b}^J(a; x))',\\
a^*(\cdot; x) : \mathcal{D}_J^\circ \longrightarrow \mathcal{W}_x, \quad p \longmapsto a^*(p; x)
\end{aligned}$$

are mutually inverse bijections.

Proof. Lemmas C.1.8 and C.1.9 give a unique share vector $\bar{b}(a; x)$ and continuity of $a \mapsto \bar{b}(a; x)$ on $\mathbb{R}^J$. Consequently $\mathcal{W}_x$ is open: if every $\bar{b}^j(a; x) > 0$, continuity keeps every $\bar{b}^j(\widetilde{a}; x)$ positive for $\widetilde{a}$ sufficiently close to a. I show that each $p \in \mathcal{D}_J^\circ$ is attained at exactly one payoff vector a by minimizing $a \mapsto H(a; p, x)$.

Fix $p \in \mathcal{D}_J^\circ$, write $\lambda = \lambda(x)$, and define

$$\begin{aligned}
D(a, \omega; p) &= p_0 + \sum_{j=1}^J p_j e^{(a_j + \omega_j)/\lambda},\\
\pi_0(a, \omega; p) &= p_0 / D(a, \omega; p),\\
\pi_j(a, \omega; p) &= p_j e^{(a_j + \omega_j)/\lambda} / D(a, \omega; p) \quad (j = 1, \dots, J).
\end{aligned}$$

All $\pi_j(a, \omega; p)$ are positive and sum to one. Differentiating with respect to a gives

$$\begin{aligned}
\nabla_a H(a; p, x) &= \mathbb{E}_G \pi_{1:J}(a, \Omega; p) - p,\\
\nabla_a^2 H(a; p, x) &= \frac{1}{\lambda} \mathbb{E}_G \{\operatorname{diag}(\pi_{1:J}) - \pi_{1:J} \pi_{1:J}'\},
\end{aligned}$$

where $\pi_{1:J} = (\pi_1, \dots, \pi_J)'$, evaluated at $(a, \Omega; p)$ in the second line. These differentiations under the expectation are valid because $0 < \pi_j < 1$ and

$$\partial_{a_k} \pi_j = \lambda^{-1} \pi_j (\mathbf{1}\{j = k\} - \pi_k) \quad (j, k = 1, \dots, J)$$

is bounded independently of Ω.

For $u \in \mathbb{R}^J$, put $u_0 = 0$. The Hessian's quadratic form is

$$u' \nabla_a^2 H(a; p, x) u = \frac{1}{\lambda} \mathbb{E}_G \left\{ \sum_{j=0}^{J} \pi_j u_j^2 - \left(\sum_{j=0}^{J} \pi_j u_j \right)^2 \right\}.$$

For each Ω, the expression in braces is the variance of the numbers $u_0, \dots, u_J$ under the positive probabilities $\pi_0, \dots, \pi_J$. It can be zero only if all these numbers are equal. Since $u_0 = 0$, it is strictly positive for every $u \neq 0$. Thus $\nabla_a^2 H(a; p, x)$ is positive definite and $H(\cdot; p, x)$ is strictly convex.

To establish existence of its minimizer, put $p_{\min} = \min_{0 \le j \le J} p_j > 0$ and $a_0 = 0$. Choosing the largest of $a_0, \dots, a_J$ in the denominator gives

$$\lambda \log D(a, \Omega; p) \ge \lambda \log p_{\min} + \max_{0 \le j \le J} a_j - \|\Omega\|_\infty.$$

The first-moment assumption implies $\mathbb{E}_G \|\Omega\|_\infty < \infty$. Writing $a_{\max} = \max_{0 \le j \le J} a_j$ and $a_{\min} = \min_{0 \le j \le J} a_j$, I obtain

$$H(a; p, x) \ge a_{\max} - \sum_{j=0}^{J} p_j a_j + \lambda \log p_{\min} - \mathbb{E}_G \|\Omega\|_\infty,$$

$$\begin{aligned} a_{\max} - \sum_{j=0}^{J} p_j a_j &= \sum_{j=0}^{J} p_j (a_{\max} - a_j) \\ &\ge p_{\min} (a_{\max} - a_{\min}) \ge p_{\min} \|a\|_\infty. \end{aligned}$$

The last inequality uses $a_0 = 0$. Therefore $H(a; p, x) \to +\infty$ as $\|a\|_\infty \to \infty$. In particular, outside a sufficiently large closed ball its value exceeds $H(0; p, x)$. Continuity and the extreme value theorem on that ball give a minimizer; strict convexity makes it unique. Denote it by $a^*(p; x)$.

I next show that the maximizing shares at $a^*(p; x)$ are exactly $q(p)$. The first-order condition for minimizing H gives $\mathbb{E}_G \pi_j(a^*(p; x), \Omega; p) = p_j$ for $j = 1, \dots, J$. Summing these equations and taking complements gives the same equality for $j = 0$. Since all p_j are positive, division

gives

$$\mathbb{E}_G \frac{e^{(a_j^*(p;x)+\Omega_j)/\lambda}}{D(a^*(p;x),\Omega;p)} = 1 \quad (j = 1,\dots,J), \qquad \mathbb{E}_G \frac{1}{D(a^*(p;x),\Omega;p)} = 1.$$

The derivative of the share objective in its free coordinates is

$$\partial_{p_j} \Phi(q(p);a,x) = \lambda \mathbb{E}_G \frac{e^{(a_j+\Omega_j)/\lambda} - 1}{D(a,\Omega;p)}.$$

This derivative is valid for $p \in \mathcal{D}_J^\circ$: each ratio $e^{(a_j+\Omega_j)/\lambda}/D$ is bounded by $1/p_j$, and $1/D \le 1/p_0$, with locally uniform bounds. The preceding equalities make this derivative zero for every j at $(a^*(p;x),p)$. Strict concavity in the shares then gives

$$\bar{b}(a^*(p;x);x) = q(p), \qquad B_x(a^*(p;x)) = p.$$

Thus $a^*(p;x) \in \mathcal{W}_x$, and $\mathcal{W}_x$ is nonempty.

Conversely, take $a \in \mathcal{W}_x$ and put $p = B_x(a)$. Since every component of $q(p)$ is positive, the equations $\partial_{p_j}\Phi(q(p);a,x) = 0$ give a common number k:

$$\mathbb{E}_G \frac{e^{(a_j+\Omega_j)/\lambda}}{D(a,\Omega;p)} = \mathbb{E}_G \frac{1}{D(a,\Omega;p)} =: k \quad (j = 1,\dots,J).$$

Multiplying by p_j, including p_0 for the default equation, and summing shows $k = \mathbb{E}_G[D/D] = 1$. Hence $\mathbb{E}_G \pi_j(a,\Omega;p) = p_j$ for every j and $\nabla_a H(a;p,x) = 0$. The strictly convex function $H(\cdot;p,x)$ has only one stationary point, so $a = a^*(p;x)$. This proves both inverse identities and the claimed bijection. □

Lemma C.1.11 (Smooth inverse shares and the value function)**.** *Under the conditions and notation of Lemma C.1.10, the mutually inverse maps B_x and $a^*(\cdot;x)$ are real analytic. Define the optimized value and its convex conjugate by*

$$V_x(a) = \max_{q\in\Delta_J} \Phi(q;a,x), \qquad V_x^*(p) = \sup_{a\in\mathbb{R}^J}\{p'a - V_x(a)\}.$$

The function V_x is finite and convex on $\mathbb{R}^J$. For $p \in \mathcal{D}_J^\circ$ and $a \in \mathcal{W}_x$,

$$V_x^*(p) = -H(a^*(p;x);p,x),$$
$$\nabla_a V_x(a) = B_x(a), \qquad \nabla_p V_x^*(p) = a^*(p;x).$$

The matrix $D_a B_x(a)$ is symmetric and positive definite for every $a \in \mathcal{W}_x$.

Proof. I first establish the differentiability needed for the gradient identities. Fix x and write $\lambda = \lambda(x)$. For $(a,p) \in \mathbb{R}^J \times \mathcal{D}_J^\circ$, define

$$c_j(a,p) = -a_j + \lambda \log(p_0/p_j), \qquad Q_j(c) = \mathbb{E}_G \frac{e^{(\Omega_j - c_j)/\lambda}}{1 + \sum_{k=1}^J e^{(\Omega_k - c_k)/\lambda}} \quad (j = 1,\dots,J),$$

and $Q_0(c) = 1 - \sum_{j=1}^J Q_j(c)$. These mixed logit probabilities are real analytic in c by Iaria and Wang (2025, Theorem 1(a)). Since $(a,p) \mapsto c(a,p)$ is real analytic on the stated domain, the following two first-order-condition maps are jointly real analytic:

$$F_j(a,p) := \partial_{p_j}\Phi(q(p);a,x) = \lambda\{Q_j(c(a,p))/p_j - Q_0(c(a,p))/p_0\},$$
$$L_j(a,p) := \partial_{a_j} H(a;p,x) = Q_j(c(a,p)) - p_j, \quad j = 1,\dots,J.$$

Write $F = (F_1,\dots,F_J)'$ and $L = (L_1,\dots,L_J)'$.

To apply the implicit-function theorem to $F(a,p) = 0$, I verify that $D_p F(a,p)$ is nonsingular. Put

$$t_j(a,\Omega) = e^{(a_j+\Omega_j)/\lambda} - 1, \qquad t(a,\Omega) = (t_1(a,\Omega),\dots,t_J(a,\Omega))',$$

and recall the denominator

$$D(a,\Omega;p) = p_0 + \sum_{j=1}^J p_j e^{(a_j+\Omega_j)/\lambda}.$$

Differentiating the share objective gives

$$D_p F(a,p) = -\lambda \mathbb{E}_G \frac{t(a,\Omega)t(a,\Omega)'}{D(a,\Omega;p)^2}.$$

The ratios t_j/D are locally uniformly bounded by $1/p_j + 1/p_0$, which justifies differentiation under the expectation. For a nonzero $u \in \mathbb{R}^J$, the equality $u't(a,\Omega) = 0$ almost surely would imply

$$-\sum_{j=1}^{J} u_j + \sum_{j=1}^{J} u_j e^{a_j/\lambda} e^{\Omega_j/\lambda} = 0 \quad G\text{-almost surely.}$$

A2 would make every $u_j e^{a_j/\lambda}$ zero, contradicting $u \neq 0$. Thus $u't$ is nonzero with positive probability, and

$$u' D_p F(a,p) u = -\lambda \mathbb{E}_G\{(u't)^2/D^2\} < 0.$$

The matrix $D_p F(a,p)$ is therefore negative definite. The real-analytic implicit-function theorem (Krantz and Parks, 2002, Theorem 2.5.1), applied at every $(a, B_x(a))$ with $a \in \mathcal{W}_x$, makes $B_x(a)$ real analytic on $\mathcal{W}_x$.

For the inverse, $D_a L(a,p) = \nabla_a^2 H(a;p,x)$ is positive definite by the preceding lemma. The same theorem applied to $L(a,p) = 0$ makes $a^*(p;x)$ real analytic on $\mathcal{D}_J^\circ$. The local solutions agree on overlaps because $B_x(a)$ and $a^*(p;x)$ are unique at each a and p, respectively.

I now relate these inverse maps to V_x and V_x^*. For fixed q, the function $a \mapsto \Phi(q;a,x)$ is convex, because the logarithm of a sum of exponentials with nonnegative weights is convex. Its maximum over q is convex as well. The bound in Lemma C.1.8 gives

$$|V_x(a)| \leq \|a\|_\infty + \mathbb{E}_G\|\Omega\|_1 < \infty.$$

On $\mathcal{W}_x$, differentiate $V_x(a) = \Phi(q(B_x(a)); a, x)$. The derivative through $B_x(a)$ vanishes because the share first-order conditions are $F(a, B_x(a)) = 0$. The remaining derivative is

$$\nabla_a V_x(a) = \mathbb{E}_G \pi_{1:J}(a, \Omega; B_x(a)) = B_x(a),$$

where the final equality was proved in Lemma C.1.10.

Fix $p \in \mathcal{D}_J^\circ$. For every $a \in \mathbb{R}^J$,

$$p'a - V_x(a) \leq p'a - \Phi(q(p); a, x) = -H(a;p,x) \leq -H(a^*(p;x);p,x).$$

Both inequalities are equalities at $a = a^*(p;x)$ because $\bar{b}(a^*(p;x);x) = q(p)$. Taking the supremum proves the expression for $V_x^*(p)$. Equivalently,

$$V_x^*(p) = p'a^*(p;x) - V_x(a^*(p;x)).$$

Differentiating this identity is now legitimate. The chain rule gives

$$\nabla_p V_x^*(p) = a^*(p;x) + D_p a^*(p;x)'\{p - \nabla_a V_x(a^*(p;x))\} = a^*(p;x),$$

since $\nabla_a V_x(a^*(p;x)) = B_x(a^*(p;x)) = p$.

Finally, $D_a B_x(a) = \nabla_a^2 V_x(a)$ is symmetric and positive semidefinite by convexity of V_x. Differentiating the inverse identity gives

$$D_p a^*(B_x(a);x)\, D_a B_x(a) = I_J.$$

Hence $D_a B_x(a)$ is nonsingular. A symmetric positive semidefinite matrix with no zero eigenvalue is positive definite, proving the last claim. □

I return to the binary RI model to justify the joint continuity of $(z,x) \mapsto b(z,x;\theta_0)$ used in the discussion preceding Theorem 1. Assumption ID1(e) imposes continuity only in z. The next result derives joint continuity from the RI problem, continuity of λ_0, and the existing zero-cost support condition in ID3(b).

Lemma C.1.12 (Joint continuity of the true choice probability)**.** *Suppose ID1 holds, with ID1(e) stated as continuity of $z \mapsto b(z,x;\theta_0)$ on $\mathcal{Z}_x$ whenever $\lambda_0(x) = 0$. Suppose also that $\mathcal{Z}_x = \mathbb{R}$ at every such x, as required by ID3(b). Then the map*

$$\operatorname{supp}(Z,X) \longrightarrow [0,1], \qquad (z,x) \longmapsto b(z,x;\theta_0)$$

is jointly continuous. In particular, for every sequence (z_n,x_n) in this support converging to $(z_\star,x_\star)$ in the support,

$$b(z_n,x_n;\theta_0) \longrightarrow b(z_\star,x_\star;\theta_0). \tag{C.1.10}$$

Proof. The taste distribution is the same G_0 along the sequence by ID1(a). Write

$$\begin{aligned} a_n &= \alpha_0 + \delta_0 z_n, \quad \lambda_n = \lambda_0(x_n), \quad b_n = b(z_n,x_n;\theta_0), \\ a_\star &= \alpha_0 + \delta_0 z_\star, \quad \lambda_\star = \lambda_0(x_\star), \quad b_\star = b(z_\star,x_\star;\theta_0). \end{aligned}$$

Then $a_n \to a_\star$ and $\lambda_n \to \lambda_\star$ by ID1(d). The normalization in ID1(b) gives $\mathbb{E}_{G_0}|\Omega| \le (\mathbb{E}_{G_0}\Omega^2)^{1/2} = 1$.

First suppose $\lambda_\star > 0$. Then $\lambda_n > 0$ eventually. For $q \in [0,1]$ and $\lambda_0(x) > 0$, define the share objective

$$J(q; z, x) = \lambda_0(x) \int \log\left\{1 - q + q e^{(\alpha_0 + \delta_0 z + \omega)/\lambda_0(x)}\right\} dG_0(\omega).$$

If $q_n \to q$, the integrand at (q_n, z_n, x_n) converges pointwise to the integrand at $(q, z_\star, x_\star)$. It also satisfies

$$\left|\lambda_n \log\{1 - q_n + q_n e^{(a_n+\omega)/\lambda_n}\}\right| \le |a_n| + |\omega|.$$

The sequence (a_n) is bounded, so the right-hand side has an integrable bound independent of n. Dominated convergence gives $J(q_n; z_n, x_n) \to J(q; z_\star, x_\star)$.

The variance normalization makes G_0 nondegenerate, so the binary maximizer is unique by Lemma C.1.9. The same continuity-of-maximizers argument proved in Lemma C.1.8, using the compact set $[0,1]$ and the objective convergence just established, gives $b_n \to b_\star$.

Now suppose $\lambda_\star = 0$. By ID3(b), $\mathcal{Z}_{x_\star} = \mathbb{R}$. The full-information rule in Matějka and McKay (2015, Theorem 1) gives

$$b(z, x_\star; \theta_0) = \mathbb{P}_{G_0}\{\alpha_0 + \delta_0 z + \Omega > 0\}.$$

Since $\delta_0 \neq 0$, a positive mass at $-\alpha_0 - \delta_0 z_\star$ would give this function a jump at $z_\star$. Continuity in z in ID1(e) therefore implies

$$\mathbb{P}_{G_0}\{a_\star + \Omega = 0\} = 0.$$

I first consider terms with $\lambda_n = 0$. At these terms, $b_n = \mathbb{P}_{G_0}\{a_n + \Omega > 0\}$. Since $a_n \to a_\star$ and the limiting payoff tie has probability zero, bounded convergence gives $b_n \to \mathbb{P}_{G_0}\{a_\star + \Omega > 0\} = b_\star$.

For terms with $\lambda_n > 0$, fix any trial share $q \in (0,1)$. Outside the probability-zero limiting payoff tie, the logistic integrand converges to $\mathbf{1}\{a_\star + \omega > 0\}$. Since it lies in $[0,1]$, bounded convergence gives

$$r(\theta_0; (z_n, x_n), q) = \int \sigma\left(\ell(q) + \frac{a_n + \omega}{\lambda_n}\right) dG_0(\omega) - q \longrightarrow b_\star - q. \tag{C.1.11}$$

The derivative identity proved in Lemma C.1.1 gives

$$\partial_q J(q; z_n, x_n) = \frac{\lambda_n}{q(1-q)} r(\theta_0; (z_n, x_n), q).$$

For $q < b_\star$, the derivative is positive eventually and concavity gives $b_n > q$; for $q > b_\star$, it gives $b_n < q$. Letting fixed trial shares approach $b_\star$ proves $b_n \to b_\star$. If $b_\star = 0$, only the upper bounds $b_n < q$ are needed; if $b_\star = 1$, only the lower bounds $b_n > q$ are needed. Combining the terms with zero and positive costs proves (C.1.10) and completes the proof. □

C.2 Proof of Theorem 1

Proof of Theorem 1. Without loss of generality, assume $\delta_0 > 0$. Fix $x \in \mathcal{X}$. If $\mathcal{Z}_x = \mathbb{R}$, the function $z \mapsto \pi(z,x)$ is identified for every $z \in \mathbb{R}$. If $\mathcal{Z}_x \neq \mathbb{R}$, ID3(b) implies $\lambda_0(x) > 0$ and supplies a nonempty open interval $J_x \subseteq \operatorname{int}(\mathcal{Z}_x)$ such that

$$0 < \pi(z,x) < 1 \qquad (z \in J_x).$$

Lemma C.1.1 then identifies $z \mapsto b(z,x;\theta_0)$ on $\mathbb{R}$ from its values on J_x. Thus, for every $x \in \mathcal{X}$, the function

$$\pi(\cdot,x) : \mathbb{R} \longrightarrow [0,1], \qquad z \longmapsto b(z,x;\theta_0)$$

is identified. I use this notation also for $z \notin \mathcal{Z}_x$. The function is continuous on $\mathbb{R}$: at positive cost this follows from Lemma C.1.1, and at zero cost from ID1(e) and $\mathcal{Z}_x = \mathbb{R}$ in ID3(b).

For x with $\lambda_0(x) > 0$, define

$$I_x = \{z \in \mathbb{R} : 0 < \pi(z,x) < 1\}, \qquad \begin{aligned} h_x : I_x &\longrightarrow \mathbb{R}, \\ z &\longmapsto \alpha_0 + \delta_0 z + \lambda_0(x)\ell(\pi(z,x)). \end{aligned}$$

For $z \in I_x$, the conditional choice probability in (2.6) can be written as

$$\begin{aligned} p_{\theta_0}(\omega;z,x) &= \frac{\pi(z,x)e^{(\alpha_0+\delta_0 z+\omega)/\lambda_0(x)}}{1-\pi(z,x)+\pi(z,x)e^{(\alpha_0+\delta_0 z+\omega)/\lambda_0(x)}} \\ &= \sigma\left(\frac{h_x(z)+\omega}{\lambda_0(x)}\right). \end{aligned}$$

Let ε be standard logistic and independent of $\Omega \sim G_0$. Since σ is its CDF, averaging this equation over G_0 gives

$$\begin{aligned} \pi(z,x) &= \int \mathbb{P}\left\{\varepsilon \le \frac{h_x(z)+\omega}{\lambda_0(x)}\right\} dG_0(\omega) \\ &= \mathbb{P}\{h_x(z) \ge -\Omega + \lambda_0(x)\varepsilon\}. \end{aligned} \tag{C.2.1}$$

Lemma C.1.1 makes $h_x : I_x \to \mathbb{R}$ an increasing bijection, so the switch point in Definition 2 is well defined:

$$S_x = \begin{cases} h_x^{-1}(-\Omega + \lambda_0(x)\varepsilon), & \lambda_0(x) > 0, \\ -(\alpha_0+\Omega)/\delta_0, & \lambda_0(x) = 0. \end{cases}$$

For positive cost and $z \in I_x$, (C.2.1) gives

$$\mathbb{P}(S_x \le z) = \mathbb{P}\{-\Omega + \lambda_0(x)\varepsilon \le h_x(z)\} = \pi(z,x).$$

Since $S_x \in I_x$ almost surely, this equality extends to $z \notin I_x$, with equality at any finite endpoints by continuity. At zero cost, continuity of $\pi(\cdot, x)$ and $\delta_0 > 0$ imply

$$\mathbb{P}_{G_0}(\Omega = -\alpha_0 - \delta_0 z) = 0 \qquad (z \in \mathbb{R}).$$

Consequently, the definition of S_x gives

$$\mathbb{P}(S_x \le z) = \mathbb{P}_{G_0}(\alpha_0 + \delta_0 z + \Omega \ge 0) = \pi(z, x) \qquad (z \in \mathbb{R})$$

at zero cost as well. Thus the identified function $\pi(\cdot, x)$ is the CDF of S_x for every x.

Lemma C.1.2 gives the finite, identified moments

$$\mu(x) = \mathbb{E}S_x, \qquad V(x) = \mathrm{Var}(S_x), \qquad C(x) = \mathrm{Cov}(S_x, \ell(\pi(S_x, x))) > 0.$$

They are integrals with respect to the identified CDF $\pi(\cdot, x)$. The lemma also gives

$$\mu(x) = -\frac{\alpha_0}{\delta_0}, \qquad \delta_0^2 V(x) + 2\delta_0 \lambda_0(x) C(x) = 1. \tag{C.2.2}$$

In particular, $V(x) \le 1/\delta_0^2$ at every x. For each positive integer n, ID2 supplies $x_n \in \mathcal{X}$ with $\lambda_0(x_n) < 1/n$. The variance bound in Lemma C.1.2 gives

$$0 \le 1 - \delta_0^2 V(x_n) \le \frac{2\pi}{\sqrt{3}} \lambda_0(x_n) < \frac{2\pi}{\sqrt{3}\, n} \longrightarrow 0.$$

Hence

$$\sup_{x \in \mathcal{X}} V(x) = \frac{1}{\delta_0^2}.$$

The supremum need not be attained. This equality and the common mean identify

$$\delta_0 = \left(\sup_{x \in \mathcal{X}} V(x) \right)^{-1/2}, \qquad \alpha_0 = -\delta_0 \mu(x). \tag{C.2.3}$$

The sequence x_n proves the equality but is not an input to these formulas. With δ_0 identified, solving (C.2.2) gives

$$\lambda_0(x) = \frac{1 - \delta_0^2 V(x)}{2\delta_0 C(x)}, \qquad x \in \mathcal{X}. \tag{C.2.4}$$

The denominator is positive, so this formula also identifies which values of x have zero cost.

I next define a map from the identified functions to G_0. For every x with $\lambda_0(x) > 0$, both h_x and the distribution of S_x are now identified. The equality $h_x(S_x) = -\Omega + \lambda_0(x)\varepsilon$ and

independence of Ω and ε give

$$\begin{aligned}\int_{I_x} e^{-ith_x(s)}\,d\pi(s,x) &= \mathbb{E}e^{-ith_x(S_x)}\\ &= \mathbb{E}\big[e^{it\Omega}e^{-it\lambda_0(x)\varepsilon}\big]\\ &= \varphi_{G_0}(t)\varphi_\varepsilon(-\lambda_0(x)t),\end{aligned}$$

where $\varphi_{G_0}(t)=\int e^{it\omega}\,dG_0(\omega)$. The integral in the first line is with respect to the switch-point distribution with CDF $s\mapsto\pi(s,x)$. The standard logistic characteristic function φ_ε is nonzero at every real argument. Specifically, Janson (2013, Example B.13, Eq. (B.47)) gives

$$\varphi_\varepsilon(t)=\begin{cases}\pi t/\sinh(\pi t), & t\neq 0,\\ 1, & t=0.\end{cases}$$

For every x with $\lambda_0(x)=0$, the definition of S_x instead gives

$$\Omega=-\alpha_0-\delta_0 S_x,\qquad \varphi_{G_0}(t)=\int_{\mathbb{R}} e^{-it(\alpha_0+\delta_0 s)}\,d\pi(s,x).$$

In this case continuity also gives the CDF formula

$$G_0(w)=1-\pi\left(\frac{-\alpha_0-w}{\delta_0},x\right)\qquad (w\in\mathbb{R}).$$

These equalities apply at every x and do not require selecting one value of the cost shifter. To state the resulting map precisely, define from the identified functions

$$\psi_x(t)=\begin{cases}\dfrac{\int_{I_x} e^{-ith_x(s)}\,d\pi(s,x)}{\varphi_\varepsilon(-\lambda_0(x)t)}, & \lambda_0(x)>0,\\ \displaystyle\int_{\mathbb{R}} e^{-it(\alpha_0+\delta_0 s)}\,d\pi(s,x), & \lambda_0(x)=0.\end{cases}\tag{C.2.5}$$

All terms in (C.2.5) are determined by $\pi(\cdot,x)$ and the parameter formulas (C.2.3)–(C.2.4). The preceding equalities give $\psi_x=\varphi_{G_0}$ for every $x\in\mathcal{X}$.

Define[8]

$$\mathcal{K}\Big(\big\{\{\pi(z,x)\}_{z\in\mathcal{Z}_x}\big\}_{x\in\mathcal{X}}\Big)=\psi. \tag{C.2.6}$$

Since $\psi=\varphi_{G_0}$, characteristic-function uniqueness identifies G_0 (Billingsley, 1995, Theorem 26.2).

Finally, let m denote the common value of $\mu(x)$ and set $v_*=\sup_{x\in\mathcal{X}}V(x)$. Both are identified, and $0<v_*<\infty$. The calculations above give the following explicit map in Definition 1:

$$\begin{aligned}\mathcal{T}\Big(\big\{\{\pi(z,x)\}_{z\in\mathcal{Z}_x}\big\}_{x\in\mathcal{X}}\Big)&=\left(-\frac{m}{\sqrt{v_*}},\quad \frac{1}{\sqrt{v_*}},\quad \left[x\mapsto\frac{v_*-V(x)}{2\sqrt{v_*}\,C(x)}\right],\quad G_0\right)\\&=(\alpha_0,\delta_0,\lambda_0,G_0).\end{aligned} \tag{C.2.7}$$

Every function on the right is obtained from the input probabilities by the preceding constructions. This proves the theorem. □

[8] Let Π denote the input probability family in Definition 1. Applying (C.2.5) at each $x\in\mathcal{X}$ defines the map $\Pi\mapsto\bigcup_x\{\psi_x\}$, where ψ_x denotes the function $t\mapsto\psi_x(t)$. The preceding calculation shows that this union is $\{\varphi_{G_0}\}$. Composing this map with $\{f\}\mapsto f$ defines $\mathcal{K}$, with output denoted by ψ.

C.3 Proof of Theorem 2

The estimator $\widehat{\theta}_M$ minimizes $\theta \mapsto \widehat{Q}_M(\theta)$, which uses estimated shares and can be discontinuous when $\lambda(x) = 0$. I compare it with the two criteria formed using population shares $s_m = s(W_m)$:

$$Q_M(\theta) = \frac{1}{M}\sum_m r(\theta; W_m, s_m)^2, \qquad Q_M^\circ(\theta) = \frac{1}{M}\sum_m f_\theta(W_m),$$

where

$$f_\theta(w) = \mathbf{1}\{0 < s(w) < 1\} r(\theta; w, s(w))^2, \qquad Q^\circ(\theta) = \mathbb{E} f_\theta(W).$$

The criterion $Q_M^\circ(\theta)$ excludes markets with $s_m \in \{0, 1\}$ and satisfies $0 \le Q_M^\circ(\theta) \le Q_M(\theta)$ for every $\theta \in \Theta$. Following Chen (2007, Section 3.1, proof of Theorem 3.1 and Remark 3.1(4)), I show that $Q_M^\circ(\widehat{\theta}_M) \to_p 0$ and that $Q^\circ(\theta)$ is bounded away from zero when $d(\theta, \theta_0)$ is bounded away from zero. The following lemmas establish continuity and uniform convergence for Q_M°, the unique zero of Q°, and $Q_M(\theta_M^*) \to_p 0$ for the sieve approximant.

Lemma C.3.1 (Continuity and uniform convergence of the auxiliary criterion)**.** *Under the sampling conditions, ID3(a), and E2, Q° is continuous on Θ and*

$$\sup_{\theta \in \Theta} |Q_M^\circ(\theta) - Q^\circ(\theta)| \to_p 0.$$

Proof. Fix $\theta = (\alpha, \delta, \lambda, G) \in \Theta$ and a sequence $\vartheta_n \to \theta$ in the metric d. On $\{0 < s(W) < 1\}$, Lemma C.1.3 gives continuity of $\vartheta \mapsto r(\vartheta; W, s(W))$ at θ, except possibly on

$$\{\lambda(X) = 0,\ G\{-\alpha - \delta Z\} > 0\}.$$

This event means that, at zero information cost, the candidate distribution puts positive probability on equal payoffs for the two alternatives. Since $\delta \neq 0$, ID3(a) and Lemma C.1.3 give it probability zero. On $\{s(W) \in \{0, 1\}\}$, $f_\vartheta(W) = 0$ for every ϑ, so continuity holds automatically. Therefore

$$f_{\vartheta_n}(W) \to f_\theta(W) \quad \text{almost surely}, \qquad 0 \le f_\vartheta(W) \le 1.$$

Bounded convergence yields $Q^\circ(\vartheta_n) \to Q^\circ(\theta)$. This proves continuity of Q° on the compact parameter space.

For each fixed θ, the preceding argument gives almost-sure continuity of $\vartheta \mapsto f_\vartheta(W)$ at θ. The bound $0 \le f_\vartheta(W) \le 1$ gives an integrable envelope, the market characteristics are i.i.d., and E2 makes Θ compact under d. The uniform-convergence argument of Newey and McFadden (1994, Lemma 2.4) therefore gives the result. Although that chapter writes its parameters as vectors, its proof covers the parameter space by finitely many neighborhoods and bounds the variation of the loss within each neighborhood. These steps use only compactness

and continuity in the parameter metric, so they apply here to d, giving

$$\sup_{\theta\in\Theta} |Q_M^\circ(\theta) - Q^\circ(\theta)| \to_p 0.$$

The result uses continuity at each fixed parameter with probability one; it does not require $\vartheta \mapsto r(\vartheta; w, s(w))^2$ to be continuous at every $\theta \in \Theta$ for every w. As usual, the supremum may be interpreted in outer probability if needed, as in the qualification following that lemma.

□

The next lemma shows that $Q^\circ(\theta) = 0$ identifies θ_0. Its proof must account for the exclusion of markets with $s(W) \in \{0, 1\}$ and for candidates satisfying $\lambda(x) = 0$.

Lemma C.3.2 (The auxiliary criterion has a unique zero)**.** *Under ID1–ID3 and E2, $Q^\circ(\theta) = 0$ for $\theta \in \Theta$ implies $\theta = \theta_0$.*

Proof. Suppose $Q^\circ(\theta) = 0$. Since its integrand is nonnegative,

$$r(\theta; W, s(W)) = 0 \quad \text{almost surely on } \{0 < s(W) < 1\}.$$

To make the conditional statement precise, define

$$g_\theta(x) = \mathbb{E}\big[\mathbf{1}\{0 < s(W) < 1\} r(\theta; W, s(W))^2 \mid X = x\big].$$

The tower property gives $\mathbb{E} g_\theta(X) = Q^\circ(\theta) = 0$. Because $g_\theta(X) \geq 0$, it follows that $g_\theta(X) = 0$ almost surely. Hence the set $\mathcal{B} = \{x \in \mathcal{X} : g_\theta(x) = 0\}$ has probability one. For each $x \in \mathcal{B}$, the conditional nonnegative integrand is zero almost surely, giving the stated residual equality for almost every z with $0 < s(z, x) < 1$ under $Z \mid X = x$.

At a point with $0 < s(z, x) < 1$ and $\lambda(x) > 0$, $r(\theta; (z, x), s(z, x)) = 0$ is the first-order condition for the strictly concave share objective. It implies $b(z, x; \theta) = s(z, x)$. When $\lambda(x) = 0$, the same conclusion follows from

$$r(\theta; (z, x), s(z, x)) = \mathbb{P}_G\{\alpha + \delta z + \Omega > 0\} - s(z, x).$$

For $x \in \mathcal{B}$, the equality $b(z, x; \theta) = s(z, x)$ therefore holds at a dense set of points in every observed open interval with $0 < s(z, x) < 1$: every nonempty open subinterval has positive conditional probability under $Z \mid X = x$.

Replace Z by $-Z$ if necessary so that $\delta_0 > 0$. Fix $x \in \mathcal{B}$. There are observed agreement points $z_1 < z_2$ with $0 < s(z_1, x) < s(z_2, x) < 1$. When $\lambda_0(x) > 0$, Lemma C.1.1 and ID3(b) supply an observed interval where $s(\cdot, x)$ is strictly increasing. When $\lambda_0(x) = 0$, ID3(b) gives

$\mathcal{Z}_x = \mathbb{R}$, and the continuous, nonconstant full-information CDF also supplies such agreement points. If $\delta < 0$, $b(\cdot, x; \theta)$ would be nonincreasing and could not equal s at both z_1 and z_2. Thus $\delta > 0$ as well.

Now take any open interval contained in $\text{int}(\mathcal{Z}_x)$ on which $0 < s(z, x) < 1$. For any point z in the interval, choose agreement points $z_n^- \uparrow z$ and $z_n^+ \downarrow z$. Since $\delta > 0$ and $\delta_0 > 0$, monotonicity gives

$$s(z_n^-, x) = b(z_n^-, x; \theta) \leq b(z, x; \theta) \leq b(z_n^+, x; \theta) = s(z_n^+, x).$$

The true choice function $s(\cdot, x)$ is continuous, so both outer terms tend to $s(z, x)$. Therefore $b(z, x; \theta) = s(z, x)$ at every point of the interval. This argument also rules out a jump in a zero-cost candidate curve at such a point.

I next show $b(z, x; \theta) = s(z, x)$ for every $z \in \mathbb{R}$, at each $x \in \mathcal{B}$. Define the interval

$$I_x = \{z \in \mathbb{R} : 0 < s(z, x) < 1\}.$$

The argument depends on the support $\mathcal{Z}_x$ and the candidate cost $\lambda(x)$.

If $\mathcal{Z}_x = \mathbb{R}$, the preceding argument gives equality on all of I_x. At a finite left boundary of I_x, $s(z, x)$ tends to zero; at a finite right boundary it tends to one. Monotonicity and $0 \leq b(z, x; \theta) \leq 1$ then force equality also where $s(z, x) \in \{0, 1\}$. This case includes every x with $\lambda_0(x) = 0$, by ID3(b).

If $\mathcal{Z}_x \neq \mathbb{R}$, ID3(b) implies $\lambda_0(x) > 0$ and provides a nonempty open interval in $\mathcal{Z}_x$ on which $0 < s(z, x) < 1$. When $\lambda(x) > 0$ also, Lemma C.1.1 gives $b(z, x; \theta) = b(z, x; \theta_0)$ for every $z \in \mathbb{R}$.

It remains to consider $\mathcal{Z}_x \neq \mathbb{R}$ and $\lambda(x) = 0$. The positive-cost and full-support alternatives in E2(c) do not apply in this case. Its third alternative therefore makes $z \mapsto b(z, x; \theta)$ real analytic on $\mathbb{R}$ and gives $0 < b(z, x; \theta) < 1$ at every finite z. Both curves are analytic on I_x, and they agree on an observed open subinterval. The identity theorem (Krantz and Parks, 2002, Corollary 1.2.6) gives agreement throughout I_x. If its left endpoint a were finite, continuity would imply

$$b(a, x; \theta) = \lim_{z \downarrow a} b(z, x; \theta_0) = 0,$$

contradicting $b(a, x; \theta) > 0$. A finite right endpoint similarly contradicts $b(z, x; \theta) < 1$ at every finite z. Thus $I_x = \mathbb{R}$, and the complete curves agree in this case too.

I have now shown

$$F_x(z) := b(z, x; \theta) = b(z, x; \theta_0) \quad (z \in \mathbb{R},\ x \in \mathcal{B}).$$

The common curve F_x is continuous: at positive true cost this follows from Lemma C.1.1, and

at zero true cost it follows from ID1(e) and ID3(b). Hence the switch points constructed in Lemma C.1.2 for the candidate and the truth have the same continuous CDF F_x. In particular, their means, variances, and covariances

$$\mu(x) = \mathbb{E}S_x, \qquad V(x) = \mathrm{Var}(S_x), \qquad C(x) = \mathrm{Cov}(S_x, \ell(F_x(S_x)))$$

are common to both parameter values on $\mathcal{B}$. The lemma's zero-cost continuity condition therefore holds for the candidate as well as the truth wherever it is needed.

To apply the moment identities on $\mathcal{B}$, I need costs to approach zero on this set. Let $\lambda^\dagger \in \{\lambda, \lambda_0\}$. For every $\varepsilon > 0$, ID2 gives $x_0 \in \mathcal{X}$ with $\lambda^\dagger(x_0) < \varepsilon/2$. Continuity of $\lambda^\dagger$ gives a relative open neighborhood U of x_0 on which $\lambda^\dagger(x) < \varepsilon$. Because $x_0 \in \mathrm{supp}(X)$,

$$\mathbb{P}(X \in U) > 0, \qquad \mathbb{P}(X \in U \cap \mathcal{B}) = \mathbb{P}(X \in U) > 0.$$

Thus $\mathcal{B}$ contains a point x with $\lambda^\dagger(x) < \varepsilon$. Applying this argument to λ and λ_0 proves

$$\inf_{x \in \mathcal{B}} \lambda(x) = \inf_{x \in \mathcal{B}} \lambda_0(x) = 0.$$

Lemma C.1.2, applied separately to the candidate and the truth, now gives

$$\sup_{x \in \mathcal{B}} V(x) = \frac{1}{\delta^2} = \frac{1}{\delta_0^2}.$$

Since $\delta > 0$ and $\delta_0 > 0$, this equality yields $\delta = \delta_0$. The common mean gives $\alpha = \alpha_0$, and the covariance identity gives

$$\lambda(x) = \frac{1 - \delta_0^2 V(x)}{2\delta_0 C(x)} = \lambda_0(x) \qquad (x \in \mathcal{B}).$$

If there is $x \in \mathcal{B}$ with $\lambda(x) = \lambda_0(x) > 0$, the deconvolution argument in Section C.2 gives $G = G_0$. If instead $\lambda(x) = \lambda_0(x) = 0$ for every $x \in \mathcal{B}$, the common switch-point distribution gives $G = G_0$ directly through $\Omega \overset{d}{=} -\alpha_0 - \delta_0 S_x$.

Finally, λ and λ_0 agree on the full-measure set $\mathcal{B} \subseteq \mathrm{supp}(X)$, hence everywhere on that support. If they differed at some point, continuity would give a neighborhood with a nonzero difference and positive support probability, contradicting their almost-everywhere equality. Consequently $\theta = \theta_0$, and Q° has a unique zero. □

To compare $\widehat{\theta}_M$ with the feasible sieve approximant $\theta_M^* \in \Theta_M$, I also need $Q_M(\theta_M^*) \to_p 0$. This criterion includes markets with $s_m \in \{0, 1\}$.

Lemma C.3.3 (Vanishing loss at the sieve approximant). *Under the sampling conditions, ID1–ID3, and E3,*

$$\mathbb{E}[r(\theta_M^*; W, s(W))^2] \to 0, \qquad Q_M(\theta_M^*) \to_p 0.$$

Proof. Write

$$\theta_M^* = (\alpha_M^*, \delta_M^*, \lambda_M^*, G_M^*).$$

On $\{0 < s(W) < 1\}$, E3 and Lemma C.1.3 give

$$r(\theta_M^*; W, s(W)) \longrightarrow 0.$$

For $w = (z, x)$ with $s(w) \in \{0, 1\}$, consider $\lambda_0(x) > 0$ and $\lambda_0(x) = 0$ separately. If $\lambda_0(x) > 0$, $\lambda_M^*(x) > 0$ eventually, and $r(\theta_M^*; w, s(w)) = 0$ by (3.1). If $\lambda_0(x) = 0$, terms with $\lambda_M^*(x) > 0$ also have $r(\theta_M^*; w, s(w)) = 0$. Along terms with $\lambda_M^*(x) = 0$, use the indicator-convergence argument in the proof of Lemma C.1.3. It uses convergence of $(\alpha_M^*, \delta_M^*, G_M^*)$ and $G_0\{-\alpha_0 - \delta_0 z\} = 0$. This part of the argument compares the indicators of $\alpha_M^* + \delta_M^* z + \Omega > 0$ and $\alpha_0 + \delta_0 z + \Omega > 0$ and therefore also applies when $s(w) \in \{0, 1\}$. It gives

$$\Psi(\theta_M^*; w, s(w)) = \mathbb{P}_{G_M^*}\{\alpha_M^* + \delta_M^* z + \Omega > 0\} \longrightarrow \mathbb{P}_{G_0}\{\alpha_0 + \delta_0 z + \Omega > 0\} = s(w).$$

Subtracting $s(w)$ gives $r(\theta_M^*; w, s(w)) \to 0$ along these terms as well. By ID1(c), ID3(a), and Lemma C.1.3,

$$\mathbb{P}\{\lambda_0(X) = 0,\ G_0\{-\alpha_0 - \delta_0 Z\} > 0\} = 0.$$

Thus the limits just proved hold almost surely over W.

Every population share satisfies either $0 < s(W) < 1$ or $s(W) \in \{0, 1\}$. The preceding arguments therefore establish

$$r(\theta_M^*; W, s(W)) \longrightarrow 0 \quad \text{almost surely.}$$

Lemma C.1.3 also gives

$$|r(\theta_M^*; W, s(W))| \le 1.$$

Since θ_M^* is deterministic and every W_m has the same distribution as W, bounded convergence yields

$$\mathbb{E}Q_M(\theta_M^*) = \mathbb{E}[r(\theta_M^*; W, s(W))^2] \longrightarrow 0.$$

Markov's inequality therefore gives $Q_M(\theta_M^*) \to_p 0$. This argument evaluates Q_M only at θ_M^*; it uses no uniform convergence of $\theta \mapsto Q_M(\theta)$. □

Proof of Theorem 2. The preceding lemmas allow me to show that $Q_M^\circ(\widehat{\theta}_M) \to_p 0$, then use the

unique minimum of Q° to obtain $\widehat{\theta}_M \to_p \theta_0$ in the metric d. The inequalities below connect the criterion $\widehat{Q}_M$ minimized by the estimator to Q°_M. Let

$$\zeta_M = \sup_{\theta\in\Theta} |\widehat{Q}_M(\theta) - Q_M(\theta)|.$$

Lemma C.1.4 gives $\zeta_M = o_p(1)$. Since $\theta^*_M \in \Theta_M$, minimization up to the allowed error $\epsilon_M = o_p(1)$ gives

$$\begin{aligned} 0 \le Q^\circ_M(\widehat{\theta}_M) &\le Q_M(\widehat{\theta}_M) \\ &\le \widehat{Q}_M(\widehat{\theta}_M) + \zeta_M \\ &\le \widehat{Q}_M(\theta^*_M) + \zeta_M + \epsilon_M \\ &\le Q_M(\theta^*_M) + 2\zeta_M + \epsilon_M = o_p(1). \end{aligned}$$

The two copies of ζ_M arise from replacing $\widehat{s}_m$ by s_m at $\widehat{\theta}_M$ and at θ^*_M. Uniform convergence of Q°_M to Q° then gives

$$0 \le Q^\circ(\widehat{\theta}_M) \le Q^\circ_M(\widehat{\theta}_M) + \sup_{\theta\in\Theta} |Q^\circ_M(\theta) - Q^\circ(\theta)| \to_p 0.$$

For $\varepsilon > 0$, let $K_\varepsilon = \{\theta \in \Theta : d(\theta, \theta_0) \ge \varepsilon\}$. If this set is empty, the desired probability is zero. Otherwise, the extreme value theorem (Munkres, 2000, Theorem 27.4), applied to the compact set K_ε, and the unique zero imply

$$\kappa_\varepsilon := \min_{\theta\in K_\varepsilon} Q^\circ(\theta) > 0.$$

Therefore

$$\mathbb{P}\{d(\widehat{\theta}_M, \theta_0) \ge \varepsilon\} \le \mathbb{P}\{Q^\circ(\widehat{\theta}_M) \ge \kappa_\varepsilon\} \longrightarrow 0.$$

This is $d(\widehat{\theta}_M, \theta_0) \to_p 0$. □

The same comparison proves consistency with bounded positive weights. Let $\widehat{\theta}_M^w$ approximately minimize (3.7), with error $\epsilon_M^w = o_p(1)$, and suppose $0 < c \le w(\theta; w, q) \le C < \infty$ uniformly over (θ, w, q). The weight bounds give

$$c\widehat{Q}_M(\widehat{\theta}_M^w) \le \widehat{Q}_M^w(\widehat{\theta}_M^w) \le \widehat{Q}_M^w(\theta_M^*) + \epsilon_M^w \le C\widehat{Q}_M(\theta_M^*) + \epsilon_M^w.$$

Share substitution and Lemma C.3.3 imply

$$\widehat{Q}_M(\widehat{\theta}_M^w) \le \frac{C}{c}\{Q_M(\theta_M^*) + \zeta_M\} + \frac{\epsilon_M^w}{c} \to_p 0.$$

Hence

$$Q_M^\circ(\widehat{\theta}_M^w) \le \widehat{Q}_M(\widehat{\theta}_M^w) + \zeta_M \to_p 0.$$

Lemma C.3.1 then gives $Q^\circ(\widehat{\theta}_M^w) \to_p 0$. For $\varepsilon > 0$ with $K_\varepsilon \neq \varnothing$,

$$\mathbb{P}\{d(\widehat{\theta}_M^w, \theta_0) \ge \varepsilon\} \le \mathbb{P}\{Q^\circ(\widehat{\theta}_M^w) \ge \kappa_\varepsilon\} \longrightarrow 0.$$

If $K_\varepsilon = \varnothing$, the probability on the left is zero.

C.4 Proof of Theorem 3

I first show that the test accepts the deterministic sieve approximant θ_M^* with probability approaching $1-\gamma$. I then use F5 to show that the reported interval covers $\phi(\theta_0)$ with at least this limiting probability. The first part requires three comparisons: population and sample trimming must retain the same markets; a statistic formed using population shares must have a normal limit; and replacing its population quantities by estimates must have a negligible effect.

In the sample with M markets, N_M is the common within-market sample size in F1. Write $w = (z, x)$, $s(w) = \pi(z, x)$, and $s_m = s(W_m)$ for population shares, and $\widehat{s}_m$ for their sample estimates. Fix the trimming thresholds $\eta \in (0, 1/2)$ and $a_* > 0$. The sigma-field

$$\mathcal{F}_M = \sigma(W_1, \ldots, W_M)$$

contains the observed market characteristics. Conditional on $\mathcal{F}_M$, the population shares are fixed and the sampled choices are independent across markets. In particular, $N_M \widehat{s}_m \mid \mathcal{F}_M$ is binomial with success probability s_m.

For $q \in (0, 1)$, write $A_m(\theta, q) = \partial_q r(\theta; W_m, q)$. For $0 < s(w) < 1$, define

$$A_M(w) = \partial_q r(\theta_M^*; w, s(w)), \qquad \mathcal{R}_M = \{w : \eta \le s(w) \le 1-\eta,\ |A_M(w)| \ge a_*\}.$$

These are the market types retained using population shares. Their observed indices and number are

$$\mathcal{A}_M = \{m \le M : W_m \in \mathcal{R}_M\}, \qquad k_M = |\mathcal{A}_M|.$$

I also write

$$\widehat{\mathcal{A}}_M = \widehat{\mathcal{A}}_M(\theta_M^*), \qquad \widehat{k}_M = |\widehat{\mathcal{A}}_M|,$$

for the retained indices and number obtained from estimated shares when testing θ_M^*. Thus $\mathcal{A}_M$ depends on market characteristics and population shares, whereas $\widehat{\mathcal{A}}_M$ also depends on share-estimation error. For the proof only, put $d_M = \max(k_M, 1)$ so that the denominators $\sqrt{d_M}$ and $\sqrt{2d_M}$ remain positive. The implemented test continues to use $\widehat{k}_M$, and accepts when that count is zero.

For each $m \in \mathcal{A}_M$, define

$$\xi_{m,M} = \frac{\sqrt{N_M}(\widehat{s}_m - s_m)}{\sqrt{s_m(1-s_m)}}, \qquad \beta_{m,M} = \frac{\sqrt{N_M}\, r(\theta_M^*; W_m, s_m)}{A_M(W_m)\sqrt{s_m(1-s_m)}}. \tag{C.4.1}$$

The variable $\xi_{m,M}$ rescales the share error $\widehat{s}_m - s_m$. Conditional on $\mathcal{F}_M$, it has mean zero and

variance one. The quantity $\beta_{m,M}$ rescales $r(\theta_M^*; W_m, s_m)$. It is fixed conditional on $\mathcal{F}_M$ and would equal zero if $r(\theta_M^*; W_m, s_m) = 0$.

I will compare the implemented statistic with

$$T_M^\circ = \sum_{m \in \mathcal{A}_M} (\xi_{m,M} + \beta_{m,M})^2.$$

In T_M°, the population shares s_m determine both $\mathcal{A}_M$ and the denominators in $\xi_{m,M}$ and $\beta_{m,M}$. The sampled shares $\widehat{s}_m$ still enter through $\xi_{m,M}$.

Lemma C.4.1 (A positive fraction of markets is retained)**.** *Under E3, ID1(c), ID3(a), and F2, there is $p_* > 0$ such that*

$$\mathbb{P}\{k_M \geq p_* M/2\} \to 1, \qquad d_M \to_p \infty, \qquad M/d_M = O_p(1).$$

Proof. Recall that

$$A_M(w) = \partial_q r(\theta_M^*; w, s(w)), \qquad A_0(w) = \partial_q r(\theta_0; w, s(w)).$$

E3 and Lemma C.1.3 imply $A_M(W) \to A_0(W)$ almost surely whenever $0 < s(W) < 1$, except possibly on

$$\{\lambda_0(X) = 0,\ G_0\{-\alpha_0 - \delta_0 Z\} > 0\}.$$

Lemma C.1.3 gives this event probability zero: ID1(c) makes $\delta_0 \neq 0$, and the continuous conditional distribution of Z in ID3(a) excludes the countably many values at which G_0 puts positive probability on a payoff tie. Thus $A_M(W) \to A_0(W)$ almost surely on $\{0 < s(W) < 1\}$.

Define the event that both trimming inequalities hold strictly:

$$\mathcal{B} = \{\eta < s(W) < 1 - \eta,\ |A_0(W)| > a_*\}.$$

Its probability $p_0 = \mathbb{P}(\mathcal{B})$ is positive by F2. For a market type in $\mathcal{B}$, the share already satisfies the population trimming inequalities, and derivative convergence gives $|A_M(W)| \geq a_*$ for all sufficiently large M. Consequently,

$$\liminf_{M \to \infty} \mathbf{1}\{W \in \mathcal{R}_M\} \geq \mathbf{1}\{\mathcal{B}\} \quad \text{almost surely.}$$

Fatou's inequality shows that the probability of satisfying the population trimming conditions stays bounded away from zero:

$$\liminf_{M \to \infty} p_M = \liminf_{M \to \infty} \mathbb{E}\mathbf{1}\{W \in \mathcal{R}_M\} \geq \mathbb{E} \liminf_{M \to \infty} \mathbf{1}\{W \in \mathcal{R}_M\} \geq p_0 > 0,$$

where $p_M = \mathbb{P}(W \in \mathcal{R}_M)$. Choose $p_* > 0$ with $p_M \geq p_*$ for all sufficiently large M.

The indicators $\mathbf{1}\{W_m \in \mathcal{R}_M\}$ are i.i.d. within each sample because θ_M^* and $\mathcal{R}_M$ are deterministic and $W_1, \ldots, W_M$ are i.i.d. Therefore

$$\mathbb{E}(k_M/M) = p_M, \qquad \mathrm{Var}(k_M/M) = \frac{p_M(1-p_M)}{M} \leq \frac{1}{4M}.$$

For large M, Chebyshev's inequality implies

$$\mathbb{P}\{k_M < p_* M/2\} \leq \mathbb{P}\{|k_M/M - p_M| > p_*/2\} \leq \frac{1}{p_*^2 M} \longrightarrow 0.$$

Hence the number of retained markets diverges with probability approaching one. In particular,

$$\mathbb{P}(k_M = d_M) \to 1, \qquad d_M \to_p \infty, \qquad M/d_M = O_p(1).$$

□

For $m \in \mathcal{A}_M$, the population shares s_m and derivatives $A_M(W_m)$ satisfy $s_m \in [\eta, 1-\eta]$ and $|A_M(W_m)| \geq a_*$. F4 bounds $r(\theta_M^*; W_m, s_m)$ and hence its standardized value $\beta_{m,M}$.

Lemma C.4.2 (The standardized sieve bias is negligible)**.** *Under F4, with empty sums and maxima set to zero,*

$$\max_{m \in \mathcal{A}_M} |\beta_{m,M}| \to 0, \qquad \nu_M := \sum_{m \in \mathcal{A}_M} \beta_{m,M}^2 = o(\sqrt{M}).$$

Together with Lemma C.4.1, $\nu_M / \sqrt{d_M} \to_p 0$.

Proof. On $\mathcal{A}_M$, the trimming rules give

$$s_m(1-s_m) \geq \eta(1-\eta), \qquad |A_M(W_m)| \geq a_*.$$

With $\rho_M = \sup_{w \in \mathcal{R}_M} |r(\theta_M^*; w, s(w))|$ from F4, these bounds give

$$|\beta_{m,M}| \leq \frac{\sqrt{N_M}\rho_M}{a_* \sqrt{\eta(1-\eta)}}.$$

Write $\nu_M = \sum_{m \in \mathcal{A}_M} \beta_{m,M}^2$ for the total squared standardized bias. With constants depending only on the fixed trimming bounds,

$$\max_{m \in \mathcal{A}_M} |\beta_{m,M}| \leq C\sqrt{N_M}\rho_M \longrightarrow 0, \qquad \nu_M := \sum_{m \in \mathcal{A}_M} \beta_{m,M}^2 \leq CMN_M\rho_M^2 = o(\sqrt{M}). \tag{C.4.2}$$

An empty maximum or sum is defined as zero. To verify both rates, F4 states $M^{1/4}\sqrt{N_M}\rho_M \to 0$, so

$$\sqrt{N_M}\rho_M = M^{-1/4}\{M^{1/4}\sqrt{N_M}\rho_M\} \to 0,$$

and

$$\frac{\nu_M}{\sqrt{M}} \leq C\sqrt{M}N_M\rho_M^2 = C\{M^{1/4}\sqrt{N_M}\rho_M\}^2 \to 0.$$

Combining the last bound with Lemma C.4.1 also yields $\nu_M/\sqrt{d_M} \to_p 0$. □

Before studying the statistic's distribution, I show that estimating shares does not change which markets enter it asymptotically.

Lemma C.4.3 (Agreement of population and sample trimming)**.** *Under F1 and F3,*

$$\mathbb{P}\{\widehat{\mathcal{A}}_M \neq \mathcal{A}_M\} \to 0.$$

Proof. Fix a market m and write $e_{m,M} = \widehat{s}_m - s_m$ for its share error. The sample and population trimming rules can disagree for two reasons. First, $\widehat{s}_m$ and s_m can lie on different sides of either η or $1-\eta$. Such a crossing requires

$$\min\{|s_m - \eta|,\ |s_m - (1-\eta)|\} \leq |e_{m,M}|.$$

Second, if both $\widehat{s}_m$ and s_m lie in $[\eta, 1-\eta]$, the absolute residual derivative can cross a_*. The uniform Lipschitz bound in Lemma C.1.3 gives

$$\left||A_m(\theta_M^*, \widehat{s}_m)| - |A_M(W_m)|\right| \leq K_\eta |e_{m,M}|.$$

A derivative crossing therefore requires

$$\left||A_M(W_m)| - a_*\right| \leq K_\eta |e_{m,M}|.$$

For w with $0 < s(w) < 1$, define its distance to the cutoffs by

$$\Delta_M(w) = \min\{|s(w) - \eta|,\ |s(w) - (1-\eta)|, \left||A_M(w)| - a_*\right|\}.$$

When $s(w) \in \{0, 1\}$, set $\Delta_M(w) = \eta$. This is the quantity in F3. It is the smallest distance to any trimming cutoff. For a fixed $K \geq \max(1, K_\eta)$, disagreement between the sample and population trimming rules implies

$$\Delta_M(W_m) \leq K|e_{m,M}|.$$

At true shares zero and one, the sample share is exact and both rules discard the market, so no difference is possible.

Applying Hoeffding (1963, Theorem 2) to the independent Bernoulli choices conditional on $\mathcal{F}_M$, and using a union bound for the two tails, gives

$$\mathbb{P}\{|e_{m,M}| \geq u \mid \mathcal{F}_M\} \leq 2e^{-2N_M u^2}.$$

Substituting $u = \Delta_M(W_m)/K$ and then taking expectations shows

$$\mathbb{P}(\text{the trimming rules disagree for market } m) \leq 2\mathbb{E}e^{-cN_M \Delta_M(W)^2}, \qquad c = 2/K^2.$$

I next bound this expectation using F3. For every nonnegative number u,

$$e^{-cN_M u^2} = \int_u^\infty 2cN_M t e^{-cN_M t^2}\, dt.$$

Taking expectations and interchanging the nonnegative integrals yields

$$\begin{aligned}
\mathbb{E}e^{-cN_M \Delta_M(W)^2} &= \int_0^\infty 2cN_M t e^{-cN_M t^2} \mathbb{P}\{\Delta_M(W) \leq t\}\, dt \\
&\leq 2cCN_M \int_0^{t_0} t^2 e^{-cN_M t^2}\, dt + e^{-cN_M t_0^2} \\
&\leq \frac{2cC}{\sqrt{N_M}} \int_0^\infty u^2 e^{-cu^2}\, du + e^{-cN_M t_0^2} = O(N_M^{-1/2}).
\end{aligned}$$

The second line uses F3 for $t \leq t_0$ and the bound by one for larger t; the third line uses $u = \sqrt{N_M}\, t$. The constants absorb the derivative's Lipschitz bound, while Δ_M remains the unscaled cutoff distance.

A union bound over the M markets and F1 now give

$$\mathbb{P}\{\widehat{\mathcal{A}}_M \neq \mathcal{A}_M\} \leq CM/\sqrt{N_M} \longrightarrow 0. \tag{C.4.3}$$

Thus, with probability approaching one, replacing the population shares changes none of the trimming decisions. □

The next result studies T_M°, which uses s_m in the standardization and $\mathcal{A}_M$ to choose the markets. Conditioning on $\mathcal{F}_M$ fixes $\mathcal{A}_M$ and leaves independent share errors as its source of randomness.

Lemma C.4.4 (Normal limit with population standardization)**.** *Under the sampling conditions, E3, ID1(c), ID3(a), and F1–F4,*

$$\frac{T_M^\circ - d_M}{\sqrt{2d_M}} \Rightarrow N(0,1).$$

Proof. Because $\mathbb{E}(\xi_{m,M} \mid \mathcal{F}_M) = 0$ and $\mathbb{E}(\xi_{m,M}^2 \mid \mathcal{F}_M) = 1$,

$$\mathbb{E}\{(\xi_{m,M} + \beta_{m,M})^2 \mid \mathcal{F}_M\} = 1 + \beta_{m,M}^2.$$

Define the centered contribution

$$X_{m,M} = (\xi_{m,M} + \beta_{m,M})^2 - (1 + \beta_{m,M}^2) = (\xi_{m,M}^2 - 1) + 2\beta_{m,M}\xi_{m,M}.$$

Conditional on $\mathcal{F}_M$, these contributions are independent and have mean zero. In particular,

$$T_M^\circ - k_M - \nu_M = \sum_{m \in \mathcal{A}_M} X_{m,M}.$$

To calculate their conditional variances, expand the last expression for $X_{m,M}$:

$$\begin{aligned}\operatorname{Var}(X_{m,M} \mid \mathcal{F}_M) &= \mathbb{E}(\xi_{m,M}^4 \mid \mathcal{F}_M) - 1 + 4\beta_{m,M}^2 \mathbb{E}(\xi_{m,M}^2 \mid \mathcal{F}_M) \\ &\quad + 4\beta_{m,M}\mathbb{E}(\xi_{m,M}^3 - \xi_{m,M} \mid \mathcal{F}_M) \\ &= 2 + 4\beta_{m,M}^2 + 4\beta_{m,M}\mathbb{E}(\xi_{m,M}^3 \mid \mathcal{F}_M) + O(N_M^{-1}).\end{aligned}$$

The final equality uses the binomial formulas in Lemma C.1.5; their bounds are uniform because $s_m \in [\eta, 1-\eta]$ for every $m \in \mathcal{A}_M$. The same lemma gives a uniformly bounded sixth moment of $\xi_{m,M}$. Using $|a+b|^3 \le 4(|a|^3 + |b|^3)$, it follows that

$$\begin{aligned}\mathbb{E}(|X_{m,M}|^3 \mid \mathcal{F}_M) &\le 4\mathbb{E}(|\xi_{m,M}^2 - 1|^3 \mid \mathcal{F}_M) + 32|\beta_{m,M}|^3 \mathbb{E}(|\xi_{m,M}|^3 \mid \mathcal{F}_M) \\ &\le C(1 + |\beta_{m,M}|^3).\end{aligned}$$

Let the total conditional variance be

$$\nu_M^2 = \sum_{m \in \mathcal{A}_M} \operatorname{Var}(X_{m,M} \mid \mathcal{F}_M).$$

The bound in Lemma C.4.2 supplies the deterministic sequence

$$\bar{\beta}_M := \frac{\sqrt{N_M}\rho_M}{a_* \sqrt{\eta(1-\eta)}} \to 0, \qquad |\beta_{m,M}| \le \bar{\beta}_M \quad (m \in \mathcal{A}_M).$$

Lemma C.1.5 also gives $|\mathbb{E}(\xi_{m,M}^3 \mid \mathcal{F}_M)| \le C/\sqrt{N_M}$ uniformly over these markets. Substituting both bounds into the variance expansion yields

$$\left|\mathrm{Var}(X_{m,M} \mid \mathcal{F}_M) - 2\right| \le C\left(\bar{\beta}_M^2 + \frac{\bar{\beta}_M}{\sqrt{N_M}} + \frac{1}{N_M}\right).$$

The right-hand side is deterministic, tends to zero, and is the same for every retained market. Summing over $\mathcal{A}_M$ gives

$$|v_M^2 - 2k_M| \le Ck_M\left(\bar{\beta}_M^2 + \frac{\bar{\beta}_M}{\sqrt{N_M}} + \frac{1}{N_M}\right).$$

On the event $k_M > 0$, divide by $2k_M$. Lemma C.4.1 gives $\mathbb{P}(k_M = d_M > 0) \to 1$, so

$$\frac{v_M^2}{2d_M} \to_p 1.$$

For large M, on $\{k_M \ge p_* M/2\}$, $v_M^2 \ge k_M$ and $\sum_{m\in\mathcal{A}_M} \mathbb{E}(|X_{m,M}|^3 \mid \mathcal{F}_M) \le Ck_M$. Therefore

$$\frac{\sum_{m\in\mathcal{A}_M} \mathbb{E}(|X_{m,M}|^3 \mid \mathcal{F}_M)}{v_M^3} \le \frac{Ck_M}{k_M^{3/2}} = \frac{C}{\sqrt{k_M}} \longrightarrow 0$$

on that event. Its probability tends to one, so the ratio converges to zero in probability, and $\mathbb{P}(v_M > 0) \to 1$. As in Lemma C.1.6, the ratio is defined as zero when $v_M = 0$.

To apply that lemma, order the indices in $\mathcal{A}_M$ as

$$\mathcal{A}_M = \{m_{M,1}, \dots, m_{M,k_M}\}, \qquad m_{M,1} < \cdots < m_{M,k_M}.$$

There are k_M summands in this sample. Once the market characteristics are fixed, the index list, its length, and the conditional distributions of the contributions $X_{m_{M,1},M}, \dots, X_{m_{M,k_M},M}$ are fixed. The contributions themselves remain random because the choices are sampled. Thus the conditional independence, centering, and third-moment condition just verified are precisely the hypotheses of Lemma C.1.6.

Applying Lemma C.1.6 to $X_{m_{M,1},M}, \dots, X_{m_{M,k_M},M}$ gives

$$\frac{T_M^\circ - k_M - v_M}{v_M} \Rightarrow N(0,1).$$

Here the ratio on the left is defined as zero when $v_M = 0$. Since $v_M/\sqrt{2d_M} \to_p 1$, Slutsky's

theorem (Billingsley, 1995, Section 25, in particular Theorem 25.4 and Example 25.8) yields

$$\frac{T_M^\circ - k_M - \nu_M}{\sqrt{2d_M}} \Rightarrow N(0,1).$$

By Lemma C.4.1, $k_M = d_M$ with probability approaching one. Lemma C.4.2 gives $\nu_M/\sqrt{d_M} \to_p 0$. Hence

$$\frac{T_M^\circ - d_M}{\sqrt{2d_M}} \Rightarrow N(0,1). \tag{C.4.4}$$

□

It remains to check the estimated denominators $A_m(\theta_M^*, \widehat{s}_m)\sqrt{\widehat{s}_m(1-\widehat{s}_m)}$ and residuals $r(\theta_M^*; W_m, \widehat{s}_m)$ in $T_M(\theta_M^*)$. Their replacement error must be small relative to $\sqrt{d_M}$, the scale in the preceding normal limit.

Lemma C.4.5 (Estimated standardization has a negligible effect)**.** *Under the conditions of Lemma C.4.4,*

$$\frac{T_M(\theta_M^*) - T_M^\circ}{\sqrt{d_M}} \to_p 0.$$

Proof. For $m \in \mathcal{A}_M \cap \widehat{\mathcal{A}}_M$, define the share error $e_{m,M} = \widehat{s}_m - s_m$ and the standardized residual

$$Z_{m,M} = \frac{\sqrt{N_M}\, r(\theta_M^*; W_m, \widehat{s}_m)}{A_m(\theta_M^*, \widehat{s}_m)\sqrt{\widehat{s}_m(1-\widehat{s}_m)}}.$$

Its square is the market's contribution to the implemented statistic (4.2). For this calculation I write the population and estimated denominators as

$$D_{m,M} = A_M(W_m)\sqrt{s_m(1-s_m)}, \qquad \widehat{D}_{m,M} = A_m(\theta_M^*, \widehat{s}_m)\sqrt{\widehat{s}_m(1-\widehat{s}_m)}.$$

Both have absolute value at least $a_*\sqrt{\eta(1-\eta)} > 0$. Their signs need not be positive; only this lower bound on their absolute values is used.

Both shares, and the interval between them, lie in $[\eta, 1-\eta]$. Taylor's theorem and the uniform second-derivative bound in Lemma C.1.3 give

$$r(\theta_M^*; W_m, \widehat{s}_m) = r(\theta_M^*; W_m, s_m) + A_M(W_m)e_{m,M} + u_{m,M}/\sqrt{N_M}, \qquad |u_{m,M}| \le C\sqrt{N_M}\, e_{m,M}^2.$$

By the definitions of $\beta_{m,M}$ and $\xi_{m,M}$, this is

$$\sqrt{N_M}\, r(\theta_M^*; W_m, \widehat{s}_m) = D_{m,M}(\beta_{m,M} + \xi_{m,M}) + u_{m,M}.$$

To bound the change in the denominator, define

$$F_{m,M} : [\eta, 1-\eta] \longrightarrow \mathbb{R}, \qquad q \longmapsto F_{m,M}(q) = A_m(\theta_M^*, q)\sqrt{q(1-q)}.$$

The product rule gives

$$F'_{m,M}(q) = \partial_q^2 r(\theta_M^*; W_m, q)\sqrt{q(1-q)} + A_m(\theta_M^*, q)\frac{1-2q}{2\sqrt{q(1-q)}}.$$

For $q \in [\eta, 1-\eta]$, $q(1-q) \geq \eta(1-\eta) > 0$. Lemma C.1.3 bounds $|\partial_q r(\theta_M^*; W_m, q)|$ and $|\partial_q^2 r(\theta_M^*; W_m, q)|$ by K_η, so $|F'_{m,M}(q)| \leq C_\eta$, uniformly over m, M, and q in that interval. The mean-value theorem therefore gives

$$|\widehat{D}_{m,M} - D_{m,M}| = |F_{m,M}(\widehat{s}_m) - F_{m,M}(s_m)| \leq C_\eta |e_{m,M}|.$$

Dividing the Taylor expression by $\widehat{D}_{m,M}$ and subtracting $\beta_{m,M} + \xi_{m,M}$ gives the explicit remainder:

$$R_{m,M} = \frac{(D_{m,M} - \widehat{D}_{m,M})(\beta_{m,M} + \xi_{m,M}) + u_{m,M}}{\widehat{D}_{m,M}}.$$

Put $d_* = a_*\sqrt{\eta(1-\eta)} > 0$. Since $|\widehat{D}_{m,M}| \geq d_*$, the preceding bounds imply

$$|R_{m,M}| \leq d_*^{-1}\left\{C_\eta |e_{m,M}|(|\beta_{m,M}| + |\xi_{m,M}|) + C\sqrt{N_M}e_{m,M}^2\right\},$$
$$|\xi_{m,M}| = \frac{\sqrt{N_M}|e_{m,M}|}{\sqrt{s_m(1-s_m)}} \leq \frac{\sqrt{N_M}|e_{m,M}|}{\sqrt{\eta(1-\eta)}}.$$

Substituting the second inequality into the first shows that the term $|e_{m,M}||\xi_{m,M}|$ is bounded by a constant times $\sqrt{N_M}e_{m,M}^2$. Hence

$$Z_{m,M} = \beta_{m,M} + \xi_{m,M} + R_{m,M}, \qquad |R_{m,M}| \leq C\{\sqrt{N_M}e_{m,M}^2 + |\beta_{m,M}|\,|e_{m,M}|\}. \tag{C.4.5}$$

For $m \in \mathcal{A}_M \setminus \widehat{\mathcal{A}}_M$, define $R_{m,M} = 0$. The bound on $|R_{m,M}|$ then holds for every $m \in \mathcal{A}_M$. Using the second and fourth share-error moments from Lemma C.1.5,

$$\begin{aligned}\mathbb{E}(R_{m,M}^2 \mid \mathcal{F}_M) &\leq C\{N_M\mathbb{E}(e_{m,M}^4 \mid \mathcal{F}_M) + \beta_{m,M}^2\mathbb{E}(e_{m,M}^2 \mid \mathcal{F}_M)\} \\ &\leq C(1+\beta_{m,M}^2)/N_M.\end{aligned}$$

Let

$$B_M = \sum_{m\in\mathcal{A}_M} R_{m,M}^2, \qquad V_M = 2\sqrt{T_M^\circ B_M} + B_M.$$

On the event $\widehat{\mathcal{A}}_M = \mathcal{A}_M$, the implemented statistic sums $(\beta_{m,M} + \xi_{m,M} + R_{m,M})^2$ over exactly the same indices as T_M°. Expanding those squares and applying Cauchy–Schwarz gives

$$|T_M(\theta_M^*) - T_M^\circ| \le 2\sqrt{T_M^\circ B_M} + B_M = V_M.$$

The conditional expectations satisfy

$$\mathbb{E}(T_M^\circ \mid \mathcal{F}_M) = k_M + \nu_M, \qquad \mathbb{E}(B_M \mid \mathcal{F}_M) \le C(k_M + \nu_M)/N_M.$$

Another application of Cauchy–Schwarz, this time to the conditional expectation, gives

$$\begin{aligned}\mathbb{E}(V_M \mid \mathcal{F}_M) &\le 2\sqrt{\mathbb{E}(T_M^\circ \mid \mathcal{F}_M)\mathbb{E}(B_M \mid \mathcal{F}_M)} + \mathbb{E}(B_M \mid \mathcal{F}_M) \\ &\le C(k_M + \nu_M)/\sqrt{N_M}.\end{aligned}$$

Here $k_M \le M$, and the deterministic bound on ν_M in Lemma C.4.2 is $o(\sqrt{M})$. Taking expectations therefore yields

$$\mathbb{E}V_M \le C\{M + o(\sqrt{M})\}/\sqrt{N_M}.$$

This error must be small compared with the statistic's scale $\sqrt{d_M}$, rather than small in absolute value. For any $\varepsilon > 0$, bound separately the probability that the trimming rules disagree and the probability that too few markets satisfy the population trimming conditions:

$$\begin{aligned}\mathbb{P}\left\{\frac{|T_M(\theta_M^*) - T_M^\circ|}{\sqrt{d_M}} > \varepsilon\right\} &\le \mathbb{P}(\widehat{\mathcal{A}}_M \ne \mathcal{A}_M) + \mathbb{P}(k_M < p_* M/2) \\ &\quad + \frac{\mathbb{E}V_M}{\varepsilon\sqrt{p_* M/2}}.\end{aligned}$$

The first two terms vanish by Lemmas C.4.1 and C.4.3. The third is $O(\sqrt{M/N_M}) + o(N_M^{-1/2})$ and vanishes because F1 implies $M/N_M \to 0$. Consequently,

$$\frac{T_M(\theta_M^*) - T_M^\circ}{\sqrt{d_M}} \to_p 0. \qquad \square$$

Proof of Theorem 3. Lemmas C.4.4 and C.4.5 give

$$\frac{T_M(\theta_M^*) - d_M}{\sqrt{2d_M}} \Rightarrow N(0,1).$$

I first use this limit to obtain $\mathbb{P}\{\theta_M^* \in \mathcal{C}_{M,\gamma}^s\}$. F5 then shows that, conditional on this acceptance, $[L_M, U_M]$ contains $\phi(\theta_0)$ with probability approaching one. For deterministic integers $k \to \infty$, let $Z_1, \ldots, Z_k$ be i.i.d. $N(0,1)$ and set $C_k = \sum_{j=1}^k Z_j^2 \sim \chi_k^2$. The squared variables Z_j^2 have

mean one and variance two, so the ordinary central limit theorem (Billingsley, 1995, Theorem 27.1) gives $(C_k - k)/\sqrt{2k} \Rightarrow N(0,1)$. Since the standard normal CDF is continuous and strictly increasing, the corresponding quantiles satisfy

$$\frac{\chi^2_{1-\gamma,k} - k}{\sqrt{2k}} \longrightarrow z_{1-\gamma},$$

where $z_{1-\gamma}$ is the standard normal $(1-\gamma)$ quantile. To justify using the random integer d_M, write $c_k = (\chi^2_{1-\gamma,k} - k)/\sqrt{2k}$. For every $\varepsilon > 0$, choose K such that $|c_k - z_{1-\gamma}| < \varepsilon$ for all $k \geq K$. Then

$$\mathbb{P}\{|c_{d_M} - z_{1-\gamma}| \geq \varepsilon\} \leq \mathbb{P}(d_M < K) \longrightarrow 0,$$

because $d_M \to_p \infty$. This proves

$$\frac{\chi^2_{1-\gamma,d_M} - d_M}{\sqrt{2d_M}} \to_p z_{1-\gamma}.$$

Define the acceptance event $E_M = \{\theta^*_M \in \mathcal{C}^s_{M,\gamma}\}$. On $\{\widehat{\mathcal{A}}_M = \mathcal{A}_M,\ k_M = d_M\}$, the test's actual count $\widehat{k}_M$ equals d_M, so acceptance is equivalently

$$\frac{T_M(\theta^*_M) - d_M}{\sqrt{2d_M}} \leq \frac{\chi^2_{1-\gamma,d_M} - d_M}{\sqrt{2d_M}}.$$

The left side converges in distribution to a standard normal and the right side converges in probability to $z_{1-\gamma}$. Slutsky's theorem therefore gives

$$\frac{T_M(\theta^*_M) - \chi^2_{1-\gamma,d_M}}{\sqrt{2d_M}} \Rightarrow N(-z_{1-\gamma}, 1).$$

Since this limiting distribution assigns probability zero to zero, the probability that the displayed difference is nonpositive tends to $\Phi(z_{1-\gamma}) = 1-\gamma$. The event excluded in making the comparison has probability tending to zero. Hence

$$\mathbb{P}(E_M) = \mathbb{P}\{\theta^*_M \in \mathcal{C}^s_{M,\gamma}\} \longrightarrow 1-\gamma. \tag{C.4.6}$$

The test still accepts θ^*_M when $\widehat{\mathcal{A}}_M = \varnothing$. Lemmas C.4.1 and C.4.3 show that the probability of this event tends to zero.

Write $\phi^*_M = \phi(\theta^*_M)$ and define the interval-inclusion event

$$\mathcal{H}_M = \{[\phi^*_M - \varepsilon_M, \phi^*_M + \varepsilon_M] \subseteq [L_M, U_M]\}.$$

F5 gives

$$|\phi_M^* - \phi(\theta_0)| = o(\varepsilon_M), \qquad \mathbb{P}(\mathcal{H}_M \mid E_M) \to 1.$$

The bound $|\phi_M^* - \phi(\theta_0)| = o(\varepsilon_M)$ means that, for all sufficiently large M, $\phi(\theta_0)$ lies in $[\phi_M^* - \varepsilon_M, \phi_M^* + \varepsilon_M]$. On $E_M \cap \mathcal{H}_M$, that interval is contained in the interval $[L_M, U_M]$. Thus, for all sufficiently large M,

$$\begin{aligned}\mathbb{P}\{\phi(\theta_0) \in [L_M, U_M]\} &\geq \mathbb{P}(E_M \cap \mathcal{H}_M) \\ &= \mathbb{P}(E_M)\mathbb{P}(\mathcal{H}_M \mid E_M).\end{aligned}$$

Using (C.4.6) and F5 proves

$$\liminf_{M\to\infty} \mathbb{P}\{\phi(\theta_0) \in [L_M, U_M]\} \geq 1 - \gamma.$$

If $\mathcal{C}^s_{M,\gamma} = \varnothing$, the interval $[L_M, U_M]$ is taken to be empty. On E_M, $\theta_M^* \in \mathcal{C}^s_{M,\gamma}$, so this accepted set is nonempty. Compactness of Θ and continuity of ϕ ensure that the values $\phi(\theta)$ for $\theta \in \mathcal{C}^s_{M,\gamma}$ remain bounded. This proves the stated coverage probability for $[L_M, U_M]$ without requiring asymptotic normality of $\widehat{\theta}_M$. □

C.5 Proof of Theorem 4

I use the model and notation in Appendix A. Alternative 0, whose payoff is normalized to zero, is the default alternative in the notation below. Write $\alpha = (\alpha_1, \ldots, \alpha_J)'$ and $\delta = (\delta_1, \ldots, \delta_J)'$. The two parameter values $\theta = (\alpha, \delta, \lambda, G)$ and $\widetilde{\theta} = (\widetilde{\alpha}, \widetilde{\delta}, \widetilde{\lambda}, \widetilde{G})$ compared in the proof must each satisfy A1–A6. If $\lambda(x) = 0$ or $\widetilde{\lambda}(x) = 0$, A6(a) requires $\mathcal{Z}_x = \mathbb{R}^J$. This is the counterpart of $\mathcal{Z}_x = \mathbb{R}$ at zero information cost in the binary theorem. Write $p_j(z_j; x)$ for the choice probability of alternative j in the binary RI problem on $\{0, j\}$, using the same marginal distribution of Ω_j under G and unit cost $\lambda(x)$. This probability describes choice when only alternatives 0 and j are available. I aim to recover $p_j(z_j; x)$ for every $z_j \in \mathbb{R}$, including values outside $\operatorname{supp}(Z_j \mid X = x)$. When $\lambda(x) > 0$, $p_j(z_j; x)$ maximizes

$$\phi_j(p; z_j, x) = \lambda(x)\mathbb{E}_G \log\left\{1 - p + pe^{(\alpha_j + \delta_j z_j + \Omega_j)/\lambda(x)}\right\} \quad (p \in [0,1]).$$

When $\lambda(x) = 0$, $p_j(z_j; x) = \mathbb{P}_G\{\alpha_j + \delta_j z_j + \Omega_j > 0\}$. Assumption A6(b) gives probability zero to equality of the two payoffs.

The following lemmas establish properties of the choice probabilities, recover the binary choice curves, and identify the joint taste distribution. The last lemma treats the case $\lambda(x) = 0$ for every $x \in \mathcal{X}$.

Lemma C.5.1 (Positive-cost shares and full consideration)**.** *Suppose A2 holds. Fix $x \in \mathcal{X}_+$ and write $\lambda = \lambda(x)$. For $a \in \mathbb{R}^J$, let $\bar{b}(a; x) = (\bar{b}^0(a; x), \ldots, \bar{b}^J(a; x))'$ maximize $q \mapsto \Phi(q; a, x)$ over Δ_J, as in (A.1). This maximizer is unique, and the map $a \mapsto \bar{b}(a; x)$ is continuous. In covariate coordinates, $b(z; x) = \bar{b}(a(z); x)$, where $a(z) = (\alpha_j + \delta_j z_j)_{j=1}^J$. The full-consideration region*

$$\mathcal{W}_x = \{a \in \mathbb{R}^J : \bar{b}^j(a; x) > 0 \text{ for every } j = 0, \ldots, J\}$$

is connected and open. Each function $a \mapsto \bar{b}^j(a; x)$ is real analytic on $\mathcal{W}_x$. The map

$$\begin{aligned} c : \mathcal{W}_x &\longrightarrow \mathbb{R}^J, \\ a &\longmapsto c(a) = (c_1(a), \ldots, c_J(a))', \\ c_j(a) &= -a_j + \lambda \log \frac{\bar{b}^0(a; x)}{\bar{b}^j(a; x)} \quad (j = 1, \ldots, J) \end{aligned}$$

is a real-analytic diffeomorphism from $\mathcal{W}_x$ onto $\mathbb{R}^J$. For every $j = 1, \ldots, J$ and $a \in \mathcal{W}_x$, $\partial \bar{b}^j(a; x)/\partial a_j > 0$.

Proof. The finite second moments of G stated in Appendix A imply

$$\mathbb{E}_G\|\Omega\|_1 = \sum_{j=1}^{J} \mathbb{E}_G|\Omega_j| \leq \sum_{j=1}^{J} (\mathbb{E}_G \Omega_j^2)^{1/2} < \infty.$$

Lemma C.1.8 therefore makes $(q,a) \mapsto \Phi(q;a,x)$ finite and continuous on $\Delta_J \times \mathbb{R}^J$. A2 and Lemma C.1.9 give a unique maximizer $\bar{b}(a;x)$ and continuity of $a \mapsto \bar{b}(a;x)$, including values of a at which the maximizing share vector lies on the boundary of Δ_J.

Lemmas C.1.10 and C.1.11 show that the map

$$B_x : \mathcal{W}_x \longrightarrow \mathcal{D}_J^\circ, \qquad a \longmapsto (\bar{b}^1(a;x), \ldots, \bar{b}^J(a;x))'$$

is a bijection with a real-analytic inverse $a^*(\cdot;x)$. In particular, $\bar{b}^j(a;x)$ is real analytic on $\mathcal{W}_x$. The set $\mathcal{D}_J^\circ$ of positive free shares with sum less than one is convex, hence connected. Its continuous image under $a^*(\cdot;x)$ is $\mathcal{W}_x$, so $\mathcal{W}_x$ is connected (Munkres, 2000, Theorem 23.5). It is open by continuity of $\bar{b}$.

I next verify the asserted inverse for the threshold map c. For any threshold vector $c \in \mathbb{R}^J$, define

$$p_0(\omega;c) = \frac{1}{1+\sum_j e^{(\omega_j - c_j)/\lambda}}, \qquad p_j(\omega;c) = e^{(\omega_j - c_j)/\lambda} p_0(\omega;c), \qquad q_j(c) = \mathbb{E}_G p_j(\Omega;c).$$

Write $q(c) = (q_0(c), \ldots, q_J(c))'$; every component $q_j(c)$ is positive. Define the map $a_x : \mathbb{R}^J \to \mathbb{R}^J$, $c \mapsto a_x(c)$, by

$$a_{x,j}(c) = -c_j + \lambda \log\{q_0(c)/q_j(c)\}. \tag{C.5.1}$$

For these indices the objective denominator satisfies

$$q_0(c) + \sum_{j=1}^{J} q_j(c) e^{(a_{x,j}(c)+\Omega_j)/\lambda} = q_0(c) \left\{ 1 + \sum_{j=1}^{J} e^{(\Omega_j - c_j)/\lambda} \right\}.$$

The conditional probabilities in the share first-order conditions are therefore precisely $p_j(\Omega;c)$. Their expectations equal $q_j(c)$ by definition. As shown in Lemma C.1.10, these equalities make every free share derivative zero. Strict concavity then gives

$$\bar{b}(a_x(c);x) = q(c), \qquad a_x(c) \in \mathcal{W}_x.$$

Substitution into the definition of $c(a)$ now gives

$$c_j(a_x(c)) = -a_{x,j}(c) + \lambda \log\{q_0(c)/q_j(c)\} = c_j.$$

Thus $c(a_x(c)) = c$ for every $c \in \mathbb{R}^J$.

For the other inverse identity, fix $a \in \mathcal{W}_x$ and write $\bar{q} = \bar{b}(a;x)$. The definition of $c_j(a)$ implies

$$e^{(\Omega_j - c_j(a))/\lambda} = \frac{\bar{q}_j}{\bar{q}_0} e^{(a_j+\Omega_j)/\lambda}.$$

Hence $p_j(\Omega; c(a))$ is the conditional probability associated with the interior optimal vector $\bar{q}$ at a. Lemma C.1.10 gives $\mathbb{E}_G p_j(\Omega; c(a)) = \bar{q}_j$ for every $j = 0, \ldots, J$. Consequently $q(c(a)) = \bar{q}$, and

$$a_{x,j}(c(a)) = -c_j(a) + \lambda \log(\bar{q}_0/\bar{q}_j) = a_j.$$

This proves $a_x(c(a)) = a$ and establishes the bijection between $\mathcal{W}_x$ and $\mathbb{R}^J$.

The functions $q_j(c)$ are real analytic by Iaria and Wang (2025, Theorem 1(a)). Thus $c \mapsto a_x(c)$ is real analytic by (C.5.1). Its inverse $a \mapsto c(a)$ is real analytic because the shares are analytic and strictly positive on $\mathcal{W}_x$. These explicit inverse maps prove the asserted analytic diffeomorphism. For completeness, with $\mathbf{q}(c) = (q_1(c), \ldots, q_J(c))'$, differentiating $B_x(a_x(c)) = \mathbf{q}(c)$ gives

$$D_c a_x(c) = \{D_a B_x(a_x(c))\}^{-1} D_c \mathbf{q}(c). \tag{C.5.2}$$

The first factor is nonsingular by Lemma C.1.11. To verify the second, put

$$\mathbf{p}(\Omega; c) = (p_1(\Omega; c), \ldots, p_J(\Omega; c))'.$$

Differentiating the probabilities under the expectation gives

$$D_c \mathbf{q}(c) = -\frac{1}{\lambda} \mathbb{E}_G\{\operatorname{diag}(\mathbf{p}(\Omega; c)) - \mathbf{p}(\Omega; c)\mathbf{p}(\Omega; c)'\}.$$

For any nonzero $u = (u_1, \ldots, u_J)' \in \mathbb{R}^J$, $-u' D_c \mathbf{q}(c) u$ is $1/\lambda$ times the expected variance of $0, u_1, \ldots, u_J$ under the positive probabilities $p_0(\Omega; c), \ldots, p_J(\Omega; c)$. That variance is positive by the argument in Lemma C.1.10. Thus $D_c \mathbf{q}(c)$ is negative definite and nonsingular.

Finally, Lemma C.1.11 gives a positive-definite share Jacobian $D_a B_x(a)$. For the coordinate vector $e_j \in \mathbb{R}^J$,

$$\frac{\partial \bar{b}^j(a;x)}{\partial a_j} = e_j' D_a B_x(a) e_j > 0.$$

Thus increasing a_j while holding the other payoff indices fixed locally increases $\bar{b}^j(a;x)$ at every $a \in \mathcal{W}_x$. □

The next lemma recovers the probability $p_j(z_j;x)$ for choice from $\{0,j\}$ using probabilities from the model with all alternatives available. At positive information cost, for a target share $p \in (0,1)$, I construct covariate vectors z_n such that every $b^i(z_n;x)$ is positive, while $b^j(z_n;x) \to p$, $b^0(z_n;x) \to 1-p$, and the probabilities of all other alternatives tend to zero. The comparison uses equality of $b(z_n;x)$ and $\widetilde{b}(z_n;x)$ only at points where every alternative has positive probability.

Lemma C.5.2 (The complete binary comparisons are identified)**.** *Two parameter values satisfying A1–A6 and generating the same $\mathcal{B}_{\text{obs}}$ have the same complete binary choice curves:*

$$p_j(z_j;x) = \widetilde{p}_j(z_j;x) \quad \textit{for every } z_j \in \mathbb{R},\ j = 1,\dots,J,\ x \in \mathcal{X}.$$

Proof. I first show that δ_j and $\widetilde{\delta}_j$ have the same sign for each $j = 1,\dots,J$. Suppose first that at least one parameter value has positive cost at some $x \in \mathcal{X}$. Interchanging the two parameter values if necessary, choose x° with $\lambda(x^\circ) > 0$. Assumption A3 supplies a covariate vector

$$z^\circ \in \operatorname{int}(\mathcal{Z}_{x^\circ}), \qquad b^i(z^\circ;x^\circ) > 0 \quad (i = 0,\dots,J).$$

Since z° lies inside the conditional support, the choice probabilities are observed in a neighborhood of z°. Continuity keeps every probability positive in a sufficiently small such neighborhood. Holding z_k fixed for $k \neq j$, Lemma C.5.1 and the chain rule give

$$\frac{\partial\, b^j(z;x^\circ)}{\partial z_j} = \delta_j \frac{\partial\, \bar{b}^j(a(z);x^\circ)}{\partial a_j}, \qquad \frac{\partial\, \bar{b}^j(a(z);x^\circ)}{\partial a_j} > 0.$$

Thus $b^j(z;x^\circ)$ strictly increases with z_j when $\delta_j > 0$ and strictly decreases when $\delta_j < 0$ in this neighborhood.

The other parameter value gives the same observed probabilities there. If $\widetilde{\lambda}(x^\circ) > 0$, the same derivative calculation applies to $\widetilde{b}^j$ and $\widetilde{\delta}_j$. If $\widetilde{\lambda}(x^\circ) = 0$, increasing z_j changes only alternative j's payoff, by $\widetilde{\delta}_j$ times the increase in z_j. Hence $\widetilde{b}^j$ is nondecreasing in z_j when $\widetilde{\delta}_j > 0$ and nonincreasing when $\widetilde{\delta}_j < 0$. It cannot agree with b^j on this neighborhood if the two coefficients have opposite signs. Therefore the two coefficients have the same sign whenever at least one of the two cost functions takes a positive value.

If instead $\lambda(x) = \widetilde{\lambda}(x) = 0$ for every $x \in \mathcal{X}$, fix any $x^* \in \mathcal{X}$ and hold $z_k = 0$ for $k \neq j$. A6(a) gives $\mathcal{Z}_{x^*} = \mathbb{R}^J$. Write $e_j \in \mathbb{R}^J$ for the jth coordinate vector. When $\delta_j > 0$, for every finite taste vector, alternative j eventually has the largest payoff as $t \to +\infty$ at $z = te_j$, and eventually loses to the default as $t \to -\infty$. Bounded convergence gives

$$\lim_{t\to+\infty} b^j(te_j;x^*) = 1, \qquad \lim_{t\to-\infty} b^j(te_j;x^*) = 0.$$

When $\delta_j < 0$, the two limits are reversed. The same argument applies to $\widetilde{\delta}_j$. Equality of the observed probabilities at every te_j therefore rules out opposite signs in this case as well. Thus, in either case,

$$\operatorname{sign}(\widetilde{\delta}_j) = \operatorname{sign}(\delta_j) \quad (j = 1, \ldots, J).$$

Replacing Z_j by $-Z_j$ whenever this common sign is negative makes both δ_j and $\widetilde{\delta}_j$ positive. I use these shifters throughout the rest of the proof.

Fix x with $\lambda(x) > 0$ and $\widetilde{\lambda}(x) > 0$. Define

$$W = \{z \in \mathbb{R}^J : b^j(z; x) > 0 \text{ for every } j = 0, \ldots, J\},$$
$$\widetilde{W} = \{z \in \mathbb{R}^J : \widetilde{b}^j(z; x) > 0 \text{ for every } j = 0, \ldots, J\}.$$

By A3, there is $z_x^\circ \in \operatorname{int}(\mathcal{Z}_x)$ at which every $b^j(z_x^\circ; x)$ is positive. The observed equality $b(z_x^\circ; x) = \widetilde{b}(z_x^\circ; x)$ makes every candidate probability positive as well. Continuity therefore supplies an open ball $D \subseteq \operatorname{int}(\mathcal{Z}_x) \cap W \cap \widetilde{W}$ around z_x°. Since D lies in the observed support, $b(z; x) = \widetilde{b}(z; x)$ for every $z \in D$. The regions W and $\widetilde{W}$ are open and connected, and $z \mapsto b(z; x)$ and $z \mapsto \widetilde{b}(z; x)$ are real analytic there, by Lemma C.5.1. Let V be the connected component of $W \cap \widetilde{W}$ containing D. Since an open Euclidean set is locally connected, its components are open (Munkres, 2000, Theorem 25.3); thus V is open.

The identity argument also applies to functions of several variables. For a fixed $j \in \{0, \ldots, J\}$, put $f(z) = b^j(z; x) - \widetilde{b}^j(z; x)$ for $z \in V$, and define

$$E_f = \{z \in V : D^\nu f(z) = 0 \text{ for every } \nu \in \mathbb{N}_0^J\},$$

where D^ν is the partial derivative with multi-index ν. For each multi-index ν, the set $\{z \in V : D^\nu f(z) = 0\}$ is closed relative to V because $D^\nu f$ is continuous. Their intersection E_f is therefore closed relative to V. If $z \in E_f$, every coefficient in the Taylor series of f at z is zero. Real analyticity makes f identically zero on a neighborhood of z; every derivative is then zero on that neighborhood, so it lies in E_f. Thus E_f is open relative to V as well. Since $f = 0$ on the open ball D, all its derivatives vanish there, and $D \subseteq E_f$. Connectedness of V now gives $E_f = V$. In particular, the derivative with multi-index zero is f itself, so $b^j(z; x) = \widetilde{b}^j(z; x)$ for every $z \in V$. Applying this to all j gives $b(z; x) = \widetilde{b}(z; x)$ on V. This is the same open-and-closed argument used for the one-dimensional identity result in Krantz and Parks (2002, Corollary 1.2.6); the connectedness criterion is given in Munkres (2000, Section 23).

This component is also closed relative to W. Indeed, if $z_n \in V$ and $z_n \to z \in W$, continuity gives $\widetilde{b}(z; x) = b(z; x)$ with every coordinate positive. Thus $z \in W \cap \widetilde{W}$. Because this intersection is open, choose $r > 0$ with $B(z, r) \subseteq W \cap \widetilde{W}$. For sufficiently large n, $z_n \in B(z, r) \cap V$. The

ball and V are connected and intersect, so their union is connected. Maximality of the connected component V implies $B(z,r) \subseteq V$, and in particular $z \in V$. This proves that V is closed relative to W. Because V is nonempty and both open and closed relative to the connected set W, $V = W$ (Munkres, 2000, Section 23); interchanging the two parameter values gives $V = \widetilde{W}$. Consequently

$$W = \widetilde{W}, \qquad b(z;x) = \widetilde{b}(z;x) \quad (z \in W). \tag{C.5.3}$$

The equality in (C.5.3) has been proved only for $z \in W$, where every $b^j(z;x)$ is positive.

Now fix $j \in \{1,\dots,J\}$ and $z_j \in \mathbb{R}$ for which $0 < p_j(z_j;x) < 1$. Put

$$p = p_j(z_j;x) \in (0,1), \qquad q_0 = 1-p, \qquad a_j = \alpha_j + \delta_j z_j, \qquad \lambda = \lambda(x),$$

and define a target threshold and rival thresholds by

$$c_j = \lambda \log(q_0/p) - a_j, \qquad c_{k,n} = -\lambda \log \varepsilon_n \quad (k \notin \{0,j\}), \qquad \varepsilon_n \downarrow 0.$$

Write $R_k = e^{\Omega_k/\lambda}$ and

$$L_n = 1 + e^{(\Omega_j - c_j)/\lambda} + \varepsilon_n \sum_{k \notin \{0,j\}} R_k.$$

The integrated conditional logits are

$$Q_{0,n} = \mathbb{E}(1/L_n), \qquad Q_{j,n} = \mathbb{E}\{e^{(\Omega_j - c_j)/\lambda}/L_n\}, \qquad Q_{k,n} = \varepsilon_n \mathbb{E}(R_k/L_n).$$

Write $Q_n = (Q_{0,n},\dots,Q_{J,n})'$. Every coordinate of Q_n is positive and their sum is one. Since the target binary share $p = p_j(z_j;x)$ lies in $(0,1)$, its first-order condition is the binary fixed-point equation

$$p = \mathbb{E}_G \frac{p e^{(a_j+\Omega_j)/\lambda}}{q_0 + p e^{(a_j+\Omega_j)/\lambda}}, \qquad q_0 = 1-p.$$

The definition $c_j = \lambda \log(q_0/p) - a_j$ gives $e^{(\Omega_j - c_j)/\lambda} = (p/q_0) e^{(a_j+\Omega_j)/\lambda}$. Substituting this relation into the preceding equation and taking the complement gives

$$\mathbb{E}\frac{1}{L_0} = q_0, \qquad \mathbb{E}\frac{e^{(\Omega_j - c_j)/\lambda}}{L_0} = p, \qquad L_0 = 1 + e^{(\Omega_j - c_j)/\lambda}.$$

For each fixed taste vector Ω, $L_n \to L_0$ as $\varepsilon_n \downarrow 0$. The integrands defining $Q_{0,n}$, $Q_{j,n}$, and $Q_{k,n}$ all lie in $[0,1]$ and converge respectively to $1/L_0$, $e^{(\Omega_j - c_j)/\lambda}/L_0$, and zero. Bounded convergence therefore gives

$$Q_{0,n} \to q_0, \qquad Q_{j,n} \to p, \qquad Q_{k,n} \to 0 \quad (k \notin \{0,j\}). \tag{C.5.4}$$

Use the construction in Lemma C.5.1 to set

$$a_{i,n} = -c_{i,n} + \lambda \log(Q_{0,n}/Q_{i,n}), \qquad z_{i,n} = (a_{i,n} - \alpha_i)/\delta_i \quad (i = 1, \dots, J),$$

with $c_{j,n} = c_j$. Each z_n belongs to W, and its optimal share vector is Q_n. In particular,

$$a_{j,n} \to -c_j + \lambda \log(q_0/p) = a_j, \qquad z_{j,n} \to z_j.$$

The rival indices need not tend to $-\infty$, but they are bounded above. To verify this without an exponential-moment assumption, put

$$m_k = \mathbb{E}(R_k/L_1) \in (0, 1/\varepsilon_1].$$

Since $L_n \le L_1$, the reconstruction formula gives

$$\frac{Q_{k,n}}{\varepsilon_n} = \mathbb{E}(R_k/L_n) \ge m_k, \qquad a_{k,n} = \lambda \log \frac{Q_{0,n}}{Q_{k,n}/\varepsilon_n} \le -\lambda \log m_k. \tag{C.5.5}$$

All indices and covariates in this sequence are finite.

By (C.5.3), the same Q_n maximizes the candidate's objective at z_n. Write $\widetilde{\lambda} = \widetilde{\lambda}(x)$ and $\widetilde{\Omega} \sim \widetilde{G}$. Its target index converges to $\widetilde{a}_j = \widetilde{\alpha}_j + \widetilde{\delta}_j z_j$. Its rival indices are also bounded above, because

$$\widetilde{a}_{k,n} = \frac{\widetilde{\delta}_k}{\delta_k} a_{k,n} + \widetilde{\alpha}_k - \frac{\widetilde{\delta}_k}{\delta_k} \alpha_k$$

and the slope ratios are positive. Its objective denominator and value evaluated at Q_n are

$$\widetilde{D}_n = Q_{0,n} + \sum_{i=1}^{J} Q_{i,n} e^{(\widetilde{a}_{i,n} + \widetilde{\Omega}_i)/\widetilde{\lambda}}, \qquad \widetilde{\Phi}_n(Q_n) = \widetilde{\lambda} \mathbb{E}_{\widetilde{G}} \log \widetilde{D}_n.$$

For every finite candidate taste realization, (C.5.4) and these bounds give

$$\widetilde{D}_n \longrightarrow q_0 + p e^{(\widetilde{a}_j + \widetilde{\Omega}_j)/\widetilde{\lambda}}.$$

Each term for a rival alternative converges to zero: its share tends to zero and its known index is bounded above. For all sufficiently large n,

$$\widetilde{\lambda} \log \widetilde{D}_n \ge \widetilde{\lambda} \log(q_0/2).$$

If $\bar{a}_i$ bound the candidate indices above for all sufficiently large n, then

$$\widetilde{\lambda} \log \widetilde{D}_n \leq \max\{0, \max_{i \geq 1}(\bar{a}_i + \widetilde{\Omega}_i)\}.$$

Together with the lower bound, this gives an absolute bound

$$\left|\widetilde{\lambda} \log \widetilde{D}_n\right| \leq \left|\widetilde{\lambda} \log(q_0/2)\right| + \max_i |\bar{a}_i| + \sum_{i=1}^{J} |\widetilde{\Omega}_i|.$$

The candidate's moment restriction gives $\mathbb{E}_{\widetilde{G}}|\widetilde{\Omega}_i| < \infty$ for every i. The right-hand side is therefore integrable and independent of n. Dominated convergence yields

$$\widetilde{\Phi}_n(Q_n) \longrightarrow \widetilde{\phi}_j(p; z_j, x).$$

For every $p' \in [0,1]$, the vector assigning probabilities $1-p'$ and p' to the default and target, and zero to the rivals, is feasible in every candidate problem. Optimality implies

$$\widetilde{\Phi}_n(Q_n) \geq \widetilde{\phi}_j(p'; z_{j,n}, x).$$

For fixed $p' \in [0,1]$, the candidate's target index satisfies $\widetilde{a}_{j,n} \to \widetilde{a}_j$. Lemma C.1.8, with $J = 1$ and distribution $\widetilde{G}$, gives

$$\widetilde{\phi}_j(p'; z_{j,n}, x) \longrightarrow \widetilde{\phi}_j(p'; z_j, x),$$

including when $p' = 0$ or $p' = 1$. The left-hand side of the optimality inequality has already been shown to converge to $\widetilde{\phi}_j(p; z_j, x)$. Taking limits on both sides therefore gives

$$\widetilde{\phi}_j(p; z_j, x) \geq \widetilde{\phi}_j(p'; z_j, x) \quad \text{for every } p' \in [0,1].$$

A2 for the candidate also rules out a nonzero relation involving only 1 and $e^{\widetilde{\Omega}_j/\widetilde{\lambda}}$. Lemma C.1.9, applied with $J = 1$ to this marginal distribution, therefore makes $\widetilde{p}_j(z_j; x)$ the unique maximizer of $\widetilde{\phi}_j(\cdot; z_j, x)$. The preceding inequality shows that p is a maximizer, so $p = \widetilde{p}_j(z_j; x)$. This holds at every point of the first curve's interior interval. By Lemma C.1.1, the functions $z_j \mapsto p_j(z_j; x)$ and $z_j \mapsto \widetilde{p}_j(z_j; x)$ are nondecreasing CDFs with limits zero and one. Their equality on $\{z_j : 0 < p_j(z_j; x) < 1\}$ forces equality also where $p_j(z_j; x)$ equals zero or one, by monotonicity. Thus $p_j(z_j; x) = \widetilde{p}_j(z_j; x)$ for every $z_j \in \mathbb{R}$.

If exactly one of $\lambda(x)$ and $\widetilde{\lambda}(x)$ is zero, A6(a) gives $\mathcal{Z}_x = \mathbb{R}^J$. Thus the observed equality $b(z; x) = \widetilde{b}(z; x)$ holds for every $z \in \mathbb{R}^J$. Write $p_j^+(z_j; x)$ for the binary probability under the parameter value with positive cost, and write $p_j^{\text{zero}}(z_j; x)$ and $b^{j,\text{zero}}(z; x)$ for the binary and full-menu probabilities under the parameter value with zero cost. Fix z_j with $p = p_j^+(z_j; x) \in (0,1)$

and apply the preceding construction to the positive-cost parameter value. It gives z_n and Q_n satisfying (C.5.4), with $z_{j,n} \to z_j$. For the parameter value with zero cost at x, choosing the target from all available alternatives implies that its payoff exceeds the default's. If its binary comparison would choose the target but the full menu does not, a rival must be chosen. A6(b) excludes ties, so

$$0 \le p_j^{\text{zero}}(z_{j,n}; x) - b^{j,\text{zero}}(z_n; x) \le \sum_{k \notin \{0,j\}} b^{k,\text{zero}}(z_n; x) = \sum_{k \notin \{0,j\}} Q_{k,n} \longrightarrow 0.$$

The function $z_j \mapsto p_j^{\text{zero}}(z_j; x)$ is continuous by A6(b), and $z_{j,n} \to z_j$. Its value at z_j therefore equals $p = p_j^+(z_j; x)$. Both functions are nondecreasing CDFs with limits zero and one, so the equality extends to points where $p_j^+(z_j; x)$ equals zero or one. Thus $p_j^{\text{zero}}(z_j; x) = p_j^+(z_j; x)$ for every $z_j \in \mathbb{R}$.

If $\lambda(x) = \widetilde{\lambda}(x) = 0$, their choice probabilities with all alternatives available are observed on $\mathbb{R}^J$ by A6(a). For $J = 1$, these are already the probabilities for choice from $\{0, 1\}$, so the conclusion follows directly. For $J \ge 2$, fix z_j and write $a_j = \alpha_j + \delta_j z_j$. Set $z_{j,n} = z_j$ and $z_{k,n} = -n$ for $k \in \{1, \dots, J\} \setminus \{j\}$. Under either parameter value, each rival's payoff tends to $-\infty$ for every finite taste vector because its slope is positive. Assumption A6(b) gives $\mathbb{P}_G(a_j + \Omega_j = 0) = 0$.

Outside this event,

$$\mathbf{1}\{a_j + \Omega_j > \max(0, \max_{k \notin \{0,j\}} (\alpha_k - \delta_k n + \Omega_k))\} \longrightarrow \mathbf{1}\{a_j + \Omega_j > 0\}.$$

Bounded convergence gives the full-information binary curve; equality of the observed choice probabilities gives $p_j(z_j; x) = \widetilde{p}_j(z_j; x)$ for every $z_j \in \mathbb{R}$. □

Lemma C.5.3 (The default share identifies the joint taste distribution)**.** *Suppose the payoff coefficients (α, δ) and the cost function $\lambda : \mathcal{X} \to [0, \infty)$ are known and A2–A3 hold. Fix $x \in \mathcal{X}_+$. On an observed nonempty open set $D_x \subseteq \text{int}(\mathcal{Z}_x)$ with $b^j(z; x) > 0$ for every $j = 0, \dots, J$, the share vector $z \mapsto b(z; x)$ identifies the joint taste distribution G through the default probability $z \mapsto b^0(z; x)$.*

Proof. Fix such an x, write $\lambda = \lambda(x)$, and take independent standard Gumbel variables $\eta_0, \dots, \eta_J$, independent of Ω. At z with all $b^j(z; x) > 0$, the conditional logit formula gives the probability that alternative zero has the largest value among

$$a_j(z) + \Omega_j + \lambda \log b^j(z; x) + \lambda \eta_j \quad (j = 0, \dots, J), \qquad a_0(z) = \Omega_0 = 0.$$

Let $\zeta_j = \eta_j - \eta_0$ and $H = \Omega + \lambda\zeta$. Let D_x be the observed open set in the lemma, on which $b^j(z; x) > 0$ for every $j = 0, \dots, J$. Assumption A3 and continuity ensure that such a nonempty

set exists. Define the known map

$$\begin{aligned} c_x : D_x &\longrightarrow \mathbb{R}^J, \\ z &\longmapsto c_x(z) = (c_{x,1}(z), \ldots, c_{x,J}(z))', \\ c_{x,j}(z) &= -\alpha_j - \delta_j z_j + \lambda \log \frac{b^0(z;x)}{b^j(z;x)}. \end{aligned}$$

The default probability satisfies

$$b^0(z;x) = \mathbb{P}\{H_j \le c_{x,j}(z) \text{ for every } j\} = F_H(c_x(z)).$$

The two threshold maps satisfy $c_x(z) = c(a(z))$. The affine map $z \mapsto a(z)$ has nonsingular Jacobian $\operatorname{diag}(\delta_1, \ldots, \delta_J)$, and Lemma C.5.1 makes $a \mapsto c(a)$ a diffeomorphism. Thus $c_x(D_x)$ is a nonempty open subset of $\mathbb{R}^J$. The values $b^0(z;x)$ therefore determine F_H on $c_x(D_x)$.

By Lemma C.1.7, H has a real-analytic density f_H. Taking one derivative of its CDF in each coordinate determines that density f_H on $c_x(D_x)$. If two taste-shock distributions G and $\widetilde{G}$ give the same observed choice probabilities, their implied densities of H agree on this open set. The identity argument just used in Lemma C.5.2 extends equality to the connected domain $\mathbb{R}^J$. Thus the complete distribution of H is determined. For $t \in \mathbb{R}^J$, write $\varphi_G(t) = \int e^{it'\omega}\, dG(\omega)$ and $\varphi_H(t) = \mathbb{E}e^{it'H}$ for the characteristic functions of Ω and H. Independence and (C.1.9) then give

$$\varphi_H(t) = \varphi_G(t)\varphi_\zeta(\lambda t), \qquad \varphi_G(t) = \frac{\varphi_H(t)}{\varphi_\zeta(\lambda t)} \quad (t \in \mathbb{R}^J).$$

The denominator never vanishes. Uniqueness of multivariate characteristic functions (Billingsley, 1995, Section 29, p. 382) identifies the joint distribution G, including dependence among $\Omega_1, \ldots, \Omega_J$. □

Lemma C.5.4 (Payoff coefficients and joint tastes under full information). *Suppose A1 and A6 hold, $\lambda(x) = 0$ for every $x \in \mathcal{X}$, $\mathbb{E}_G\Omega = 0$, $\mathbb{E}_G\|\Omega\|^2 < \infty$, and $\delta_j > 0$ for $j = 1,\dots,J$. The observed family $\mathcal{B}_{\text{obs}}$ identifies α_j/δ_j for $j = 1,\dots,J$ and δ_j/δ_1 for $j = 2,\dots,J$. Once δ_1 is identified, it therefore identifies (α, δ, G).*

Proof. Fix $x^* \in \mathcal{X}$. A6(a) gives $\mathcal{Z}_{x^*} = \mathbb{R}^J$, so every finite utility-shifter vector used below lies in the observed conditional support. A6(b) excludes ties between any two payoffs. For a fixed $z_j = v$, sending every rival shifter z_k, $k \neq j$, to $-\infty$ gives the binary comparison by bounded convergence, as in the zero-cost case of Lemma C.5.2. For $J = 1$ this comparison is directly observed. Its choice curve is the distribution of the full-information switch point:

$$S^{(j)} = \frac{-\alpha_j - \Omega_j}{\delta_j}, \qquad p_j(v; x^*) = \mathbb{P}_G\{S^{(j)} \le v\}, \qquad v \in \mathbb{R}.$$

Its mean exists by the finite second-moment restriction, and

$$\mu_j := \int_{\mathbb{R}} v \, dp_j(v; x^*) = \mathbb{E}_G S^{(j)} = -\frac{\alpha_j}{\delta_j}.$$

Thus the observed binary curves determine each intercept-to-slope ratio.

To recover the slope ratios when $J \ge 2$, fix $j \in \{2,\dots,J\}$. For $r > 0$, $u \in \mathbb{R}$, and $t > 0$, define the covariate vector $z^{(j)}(t; r, u) \in \mathbb{R}^J$ by

$$z_1^{(j)}(t; r, u) = t, \qquad z_j^{(j)}(t; r, u) = rt + u, \qquad z_k^{(j)}(t; r, u) = 0 \quad (k \notin \{1, j\}).$$

The payoffs of alternatives 1 and j both tend to $+\infty$ for every finite taste vector as $t \to \infty$. Every other payoff, including the default's, stays fixed along this sequence. Consequently, eventually the chosen alternative is either 1 or j, except when their payoffs tie. Their payoff difference is

$$v_j(\Omega; z^{(j)}(t; r, u)) - v_1(\Omega; z^{(j)}(t; r, u)) = (\delta_j r - \delta_1)t + \alpha_j - \alpha_1 + \delta_j u + \Omega_j - \Omega_1.$$

If $\delta_j r > \delta_1$, this difference tends to $+\infty$; if $\delta_j r < \delta_1$, it tends to $-\infty$. If $\delta_j r = \delta_1$, the difference does not depend on t, and A6(b) gives probability zero to its being zero. Bounded convergence therefore establishes the observable limit

$$\begin{aligned} L_j(r, u) &:= \lim_{t\to\infty} b^j(z^{(j)}(t; r, u); x^*) \\ &= \begin{cases} 0, & r < \delta_1/\delta_j, \\ \mathbb{P}_G\{\Omega_1 - \Omega_j < \alpha_j - \alpha_1 + \delta_j u\}, & r = \delta_1/\delta_j, \\ 1, & r > \delta_1/\delta_j. \end{cases} \end{aligned}$$

The distribution of $\Omega_1 - \Omega_j$ is continuous by A6(b). Its CDF has limits zero and one at the two ends of $\mathbb{R}$. Because $\delta_j > 0$, some finite u therefore makes the middle probability strictly between zero and one. At every other value of r, the limiting probability is zero or one for every u. Thus the observed limits determine the unique slope ratio through

$$\{r > 0 : \exists u \in \mathbb{R} \text{ with } 0 < L_j(r,u) < 1\} = \{\delta_1/\delta_j\}.$$

Once δ_1 is known, these ratios give every δ_j. The already identified means give $\alpha_j = -\delta_j \mu_j$. For $J = 1$, no additional slope ratio is needed, and the same mean formula identifies α_1.

It remains to recover G. The default is chosen when all nondefault payoffs are nonpositive; using weak rather than strict inequalities does not affect this probability by A6(b). Therefore

$$b^0(z; x^*) = \mathbb{P}_G\{\Omega_j \le -\alpha_j - \delta_j z_j \text{ for every } j = 1, \ldots, J\}.$$

For any $h = (h_1, \ldots, h_J)' \in \mathbb{R}^J$, set $z_j(h) = (-\alpha_j - h_j)/\delta_j$ and $z(h) = (z_1(h), \ldots, z_J(h))'$. This is a finite vector in the observed support, and

$$G(h) = \mathbb{P}_G\{\Omega_j \le h_j \text{ for every } j\} = b^0(z(h); x^*).$$

Hence the default probabilities identify the joint CDF on all of $\mathbb{R}^J$, including dependence among the taste shocks. □

Proof of Theorem 4. Compare $\theta = (\alpha, \delta, \lambda, G)$ and $\widetilde{\theta} = (\widetilde{\alpha}, \widetilde{\delta}, \widetilde{\lambda}, \widetilde{G})$ satisfying the assumptions and generating the same $\mathcal{B}_{\text{obs}}$. Lemma C.5.2 gives $p_j(z_j; x) = \widetilde{p}_j(z_j; x)$ for every $z_j \in \mathbb{R}$, j, and x. For each j and x, let $S_x^{(j)}$ be the switch point from Lemma C.1.2 for the binary comparison with CDF $p_j(\cdot; x)$. At positive cost, this CDF is continuous by Lemma C.1.1; at zero cost, A6(b), applied to $(j, 0)$, gives continuity. The common binary curves therefore determine

$$\mu_j(x) = \mathbb{E}S_x^{(j)}, \qquad V_j(x) = \operatorname{Var}(S_x^{(j)}), \qquad C_j(x) = \operatorname{Cov}\big(S_x^{(j)}, \ell(p_j(S_x^{(j)}; x))\big).$$

If $\mathcal{X}_+ \neq \varnothing$, A2 rules out a constant Ω_j: otherwise 1 and $e^{\Omega_j/\lambda(x)}$ would be linearly dependent at any $x \in \mathcal{X}_+$. If $\mathcal{X}_+ = \varnothing$, A6(b), applied to the pair $(j, 0)$, gives a continuous distribution for Ω_j, which also rules out a constant. Thus $\sigma_j^2 = \operatorname{Var}_G(\Omega_j) \in (0, \infty)$ in either case. Lemma C.1.2 gives

$$\mu_j(x) = -\frac{\alpha_j}{\delta_j}, \qquad \delta_j^2 V_j(x) + 2\delta_j \lambda(x) C_j(x) = \sigma_j^2, \qquad C_j(x) > 0. \tag{C.5.6}$$

I first recover δ_1, α_1, and λ using the normalization $\sigma_1^2 = \operatorname{Var}_G(\Omega_1) = 1$. A4 gives a sequence of cost-shifter values x_M with $\lambda(x_M) < 1/M$. The cost may equal zero at some of these values. The variance bound in Lemma C.1.2 gives $V_1(x_M) \to 1/\delta_1^2$, while every variance is at most

$1/\delta_1^2$. Hence the common curves identify

$$\delta_1 = (\sup_x V_1(x))^{-1/2}, \qquad \alpha_1 = -\delta_1 \mu_1(x), \qquad \lambda(x) = \frac{1-\delta_1^2 V_1(x)}{2\delta_1 C_1(x)}.$$

Applying the same formulas to the other parameter value gives $\widetilde{\delta}_1 = \delta_1$, $\widetilde{\alpha}_1 = \alpha_1$, and $\widetilde{\lambda}(x) = \lambda(x)$ for every $x \in \mathcal{X}$, including values at which $\lambda(x) = 0$.

The common binary curves have therefore identified λ, including whether it is zero at every x. In particular, an everywhere-zero-cost parameter value cannot generate the same observed shares as a parameter value whose cost is positive at some x. It remains to identify the remaining payoff coefficients and G.

First suppose $\mathcal{X}_+ \neq \varnothing$ and choose $\bar{x} \in \mathcal{X}_+$. For $j > 1$, the variance σ_j^2 is not normalized, but λ is now known. At $\bar{x}$, $V_j(\bar{x}) < \sigma_j^2/\delta_j^2$. The same low-cost sequence satisfies $V_j(x_M) \to \sigma_j^2/\delta_j^2$, so for some x', $V_j(x') > V_j(\bar{x})$. Subtracting (C.5.6) at these two cost-shifter values cancels the unknown taste variance:

$$\delta_j^2\{V_j(\bar{x}) - V_j(x')\} + 2\delta_j\{\lambda(\bar{x})C_j(\bar{x}) - \lambda(x')C_j(x')\} = 0.$$

Since the variance difference is nonzero and $\delta_j > 0$,

$$\delta_j = \frac{2\{\lambda(x')C_j(x') - \lambda(\bar{x})C_j(\bar{x})\}}{V_j(\bar{x}) - V_j(x')}, \qquad \alpha_j = -\delta_j \mu_j(x).$$

These common moments give $\widetilde{\alpha}_j = \alpha_j$ and $\widetilde{\delta}_j = \delta_j$ for every $j > 1$. Substitution also identifies each σ_j^2.

Only the joint taste distribution G remains to be identified. Lemma C.5.3, applied at $\bar{x}$, recovers it from $z \mapsto b(z;\bar{x})$ on the observed open set $D_{\bar{x}} \subseteq \mathrm{int}(\mathcal{Z}_{\bar{x}})$, where $b^j(z;\bar{x}) > 0$ for every $j = 0, \ldots, J$. At this stage (α, δ) and λ are known. The argument does not require the choice probabilities at unobserved covariate values. Thus $G = \widetilde{G}$ and $\theta = \widetilde{\theta}$ when $\mathcal{X}_+ \neq \varnothing$.

If $\mathcal{X}_+ = \varnothing$, the already identified common cost function satisfies $\lambda(x) = \widetilde{\lambda}(x) = 0$ for every $x \in \mathcal{X}$. The value of δ_1 is also already known. Lemma C.5.4 then identifies every remaining slope δ_j, every intercept α_j, and the joint distribution G from the same observed family $\mathcal{B}_{\mathrm{obs}}$. Consequently $\theta = \widetilde{\theta}$ in this case as well. The two cases exhaust the nonnegative cost functions allowed in the model and prove the theorem.

Since $\theta = \widetilde{\theta}$, the choice probabilities satisfy

$$b(z;x) = \widetilde{b}(z;x) \quad \text{for every } z \in \mathbb{R}^J,\ x \in \mathcal{X}.$$

When $\lambda(x) > 0$, both share vectors maximize the same strictly concave objective $q \mapsto \Phi(q; a(z), x)$. When $\lambda(x) = 0$, both are generated by the same payoff comparisons under the same distribution G; A6(b) excludes payoff ties. This equality also holds at z where $b^0(z;x) = 0$. □